

Structural and Numerical Models for the (Visco)elastic Response of Arterial Walls with Residual Stresses

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Structural and Numerical Models for the (Visco)elastic Response of Arterial Walls with Residual Stresses

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Abstract. In this chapter we focus attention on the description of structural and numerical models for the elastic and viscoelastic response of arterial walls with residual stresses. We start by reviewing briefly the arterial histology and describing the mechanical characteristics of arterial components. We also present a fully automatic technique for identifying the orientations of cellular nuclei.

Particular attention is concentrated on multi-layer models for predicting reliably the passive elastic and viscoelastic three-dimensional stress and deformation states of arterial walls under various loading conditions. All the models proposed are well suited for FE implementation. Each arterial layer is treated as a fiber-reinforced material with the fibers corresponding to the collagenous constituent of the material and symmetrically disposed with respect to the cylinder axis. The resulting constitutive law is orthotropic in each layer. A specific form of the law, which requires only three material parameters for each layer, is used to study the response of an artery under combined axial extension, inflation and torsion. The characteristic and very important residual stress in an artery *in vitro* is accounted for by assuming that the unstressed configuration of the material corresponds to an open sector of a tube. The viscoelastic model admits hysteresis loops that are known to be relatively insensitive to strain rate, an essential mechanical feature of muscular arteries. The concept of internal variables is introduced in order to replicate the characteristic dissipative mechanism. We summarize the equations that provide the general continuum description of the deformation and the hyperelastic stress response of arterial walls, which are assumed to behave isochorically. One particular simple mixed FE method is discussed in detail (leading to the $Q1/P0$ -element). This approach circumvents numerical difficulties that arise from the overstiffening of the system associated with the analysis of isochoric deformations. Stiffness matrices and some insights into solution methods for nonlinear dynamic problems are provided.

Three numerical examples are included to show the performance of the structural arterial models and to document FE results that are in good qualitative agreement with experimental data. The first example is concerned with a FE analysis of the mechanical behavior of an artery during clamping. The remaining two examples are concerned with investigation of the characteristic viscoelastic behavior of a healthy young artery under static and dynamic boundary loadings.

In the last section a layer-specific FE-model of balloon angioplasty is described. The model makes use of an *in vitro* MRI of a human stenotic *post-mortem* artery and mechanical tests of the corresponding vascular tissues under supra-physiological loadings. The three-dimensional FE realization considers the balloon-artery interaction and accounts for vessel-specific axial *in situ* pre-stretches. The proposed approach provides a tool that has the potential to improve procedural protocols and the design of interventional instruments on a lesion-specific basis, and to determine post-angioplasty mechanical environments, which may be correlated with restenosis responses.

1 Arterial Histology

This section aims to review briefly the histology of arteries and to describe the mechanical characteristics of arterial components that provide the main *elastic* and *viscoelastic* contributions to the arterial deformation process. In addition, we review a fully automatic technique for identifying the orientations of cellular nuclei, these coinciding with the preferred orientations in the tissue. The distribution of preferred (fiber) orientations provide important histological information for the structural models documented in Sections 2 and 3.

1.1 Histology and Mechanical Characteristics of Arterial Components

Arteries are blood vessels with a wide variety of diameters. They are roughly subdivided into two types: *elastic* and *muscular*. Elastic arteries such as the aorta and the carotid are located close to the heart (proximal arteries), have relatively large diameters and may be regarded as *elastic* structures. Muscular arteries (distal arteries) such as femoral, coeliac and cerebral arteries are smaller, located at the periphery (more distal) and may be regarded as *viscoelastic* structures. Smaller arteries typically display more pronounced viscoelastic behavior than arteries with large diameters; see, for example, Tanaka and Fung (1974). It is generally assumed that the content of smooth muscle cells present in an artery is responsible for its viscosity. For example, according to the study by Learoyd and Taylor (1966), high viscosity is found in the human femoral artery, and this is attributed to its very large muscle content.

Arterial walls are composed of three distinct concentric layers, the *intima* (*tunica intima*), the *media* (*tunica media*) and the *adventitia* (*tunica externa*). The mechanical properties of arterial walls are strongly influenced by the concentration and structural arrangement of constituents such as *collagen* and *elastin*, the hydrated matrix of *proteoglycans*, and the topographical site and respective function in the organism (see, for example, the early work by Roy (1880-82)).

The tensile responses of circumferential and longitudinal specimens vary along the aortic tree (Bergel (1961b), Bergel (1961a), Learoyd and Taylor (1966) and Tanaka and Fung (1974) among many others). Non-diseased arterial walls (with straight vessel axes) behave *cylindrically orthotropically*, which is well-established in the literature; see, for example, the early work by Patel and Fry (1969). For mechanical properties and constitutive equations of arterial walls, see the reviews by, for example, Hayashi (1993), Humphrey (1995), the data book edited by Abè et al. (1996), Section 2, and Holzapfel et al. (2000a) and Humphrey (2002). In the following we discuss the constituents of arterial walls from the mechanical perspective. Figure 1 shows a model of a healthy elastic artery.

Intima. The intima is the innermost layer of the artery. It consists of a single layer of endothelial cells lining the arterial wall and resting on a thin basal membrane (basal lamina). There is also a subendothelial layer whose thickness varies with topography, age and disease. In healthy young muscular arteries, however, the subendothelial layer is almost non-existent. In healthy young individuals the intima is very thin and makes an insignificant contribution to the *solid* mechanical properties of the arterial wall.

However, the mechanical contribution of the intima may become significant for aged arteries (*arteriosclerosis*) (the intima becomes thicker and stiffer); see for example, Schulze-Bauer et al. (2001). In addition, it is important to note that pathological changes of the intimal components

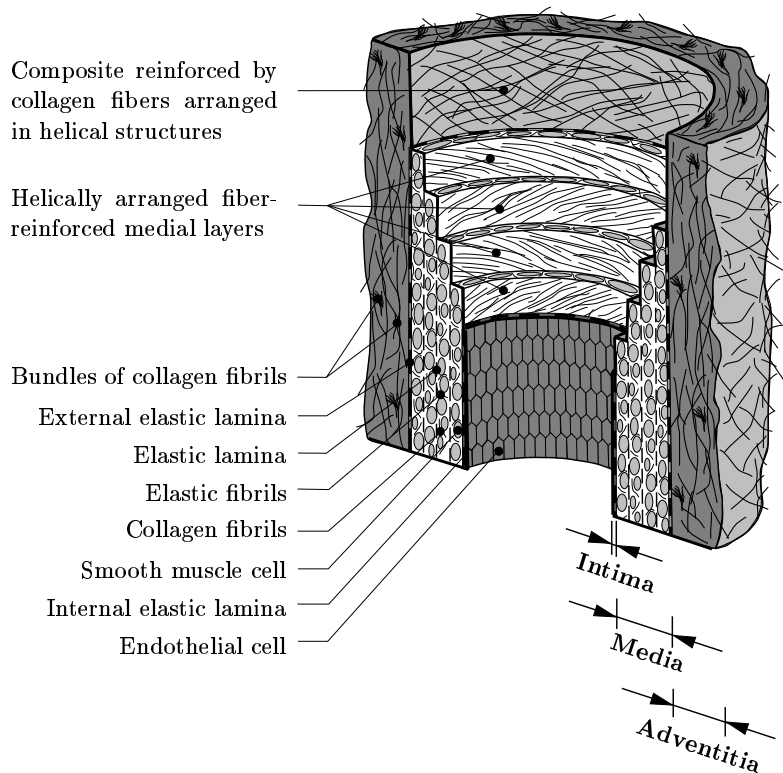


Figure 1. Diagrammatic model of the major components of a healthy *elastic* artery composed of three layers: intima (I), media (M), adventitia (A). I is the innermost layer consisting of a single layer of endothelial cells that rests on a thin basal membrane and a subendothelial layer whose thickness varies with topography, age and disease. M is composed of smooth muscle cells, a network of elastic and collagen fibrils and elastic laminae which separate M into a number of fiber-reinforced layers. The primary constituents of A are thick bundles of collagen fibrils arranged in helical structures; A is the outermost layer surrounded by loose connective tissue.

(*atherosclerosis*) are associated with significant alterations in the mechanical properties of arterial walls, differing significantly from those of healthy arteries (Learoyd and Taylor (1966) and Langewouters et al. (1984)).

Media. The media is the middle layer of the artery and consists of a complex three-dimensional network of smooth muscle cells, elastin and collagen fibrils. According to Rhodin (1980) the fenestrated elastic laminae separate the media into a varying number of well-defined concentrically fiber-reinforced medial layers. The number of elastic laminae decreases toward the periphery (as the size of the vessels decreases) so that elastic laminae are hardly present in muscular arteries.

The media is separated from the intima and adventitia by the so-called internal elastic lamina and external elastic lamina (absent in cerebral blood vessels), respectively. In muscular arteries these laminae appear as prominent structures, whereas in elastic arteries they are hardly dis-

tinguishable from the regular elastic laminae. The close interconnection between the elastic and collagen fibrils, elastic laminae, and smooth muscle cells together constitute a continuous fibrous helix (Faserschraube), see Schultze-Jena (1939) and Staubesand (1959). The helix has a small pitch so that the smooth muscle cells in the media are almost circumferentially oriented. This structured arrangement gives the media high strength, resilience and the ability to resist loads in both the longitudinal and circumferential directions. From the mechanical perspective, the media is the most significant layer in a young healthy artery.

Due to the high content of smooth muscle cells, it is the media that is believed to be mainly responsible for the viscoelastic behavior of an arterial segment.

Adventitia. The adventitia is the outermost layer of the artery and consists mainly of fibroblasts and fibrocytes (cells that produce collagen and elastin), histological ground substance and thick bundles of collagen fibrils forming a fibrous tissue. The adventitia is surrounded continuously by loose connective tissue. The thickness of the adventitia depends strongly on the type (elastic or muscular) and the physiological function of the blood vessel and its topographical site. For example, in cerebral blood vessels there is virtually no adventitia.

The wavy collagen fibrils are arranged in helical structures and serve to reinforce the wall. They contribute significantly to the stability and strength of the arterial wall. The adventitia is much less stiff in the load-free configuration and at low pressures than the media. However, at higher levels of pressure the collagen fibers reach their straightened lengths and the adventitia changes to a stiff 'jacket-like' tube which prevents the artery from overstretch and rupture (Schulze-Bauer et al. (2002)).

Collagen. Collagen is a protein which is very important for vertebrate physiology. It is a macromolecule with length of about 280 nm. The rod-like shape of the collagen molecule comes from three polypeptide chains which are composed in a right-handed triple-helical conformation. Most of the collagen molecule consists of three amino acids; glycine (33%), which enhances the stability of the molecule, proline (15%) and hydroxyproline (15%). Collagen molecules are linked to each other by covalent bonds building collagen fibrils. Depending on the primary function and the requirement of strength of the tissue the diameter of collagen fibrils varies (the order of magnitude is 1.5 nm; see Nimni and Harkness (1988)). Collagen appears as concentrically arranged fibers in the structure of blood vessels. More than 12 types of collagen have been identified (for comprehensive data see Nimni (1988)). The most common collagen is type I, which can be isolated from any tissue. It is the major constituent in blood vessels.

The intramolecular crosslinks of collagen gives the tissues the strength which varies with age, pathology, etc. The function and integrity of organs are maintained by the *tension* in collagen fibers. They shrink upon heating due to breakdown of the crystalline structure (at 65°C, for example, mammalian collagen shrinks to about one-third of its initial length, Fung (1993), p. 263). Collagen fibers represent the main load carrying elements of arterial walls that render the material properties *anisotropic*.

Elastin. Elastin is a protein which is present as thin strands in soft tissues. The ratio of elastin to collagen in the aorta decreases away from the heart (Wolinsky and Glagov (1967)). The long flexible elastin molecules build up a three-dimensional (rubber-like) network, which may be stretched to about 2.5 of the initial length of the unloaded configuration. In contrast to collagen fibers, this network does not exhibit a pronounced hierarchical organization. As for collagen, 33%

of the total amino acids of elastin consists of glycine. However, the proline and hydroxyproline contents are much lower than in collagen molecules. Elastin behaves like a rubber band and can sustain extremely large strains without rupturing (it fractures at low stresses). It is essentially a linear elastic material (tested for the ligamentum nuchae of cattle – Fung and Sobin (1981)) and displays very small relaxation effects (they are larger for collagen).

Further discussion of arterial histology is contained in Chapter 1 by HUMPHREY and McCULLOCH in this volume. For a more detailed account of the different mechanical characteristics, the structure (distribution and orientation) of the interrelated arterial components, the morphological structure and the overall functioning of the blood vessel the reader is referred to, for example, Rhodin (1980), Silver et al. (1989), Humphrey (1995), Holzapfel et al. (2000a), Section 2, Holzapfel (2001) and Humphrey (2002).

1.2 Identification of Preferred Orientations in Arterial Layers

The quantitative knowledge of preferred orientations in arterial layers enhances the understanding of the general mechanical characteristics of arterial walls significantly. It is important to note that realistic structural models rely strongly on this knowledge. Collagen fibers are those components of arterial walls that render the material properties anisotropic. To describe the anisotropic feature, appropriate geometrical data (fiber angles) are required. They serve as an essential set of input data for numerical models. In this section we review an automatic technique for obtaining information about preferred orientations in isolated arterial tissues, and, additionally, the concentration of nuclei recently proposed by Holzapfel et al. (2002a).

Automatic technique for identifying preferred orientations in arterial layers. High directional correlations between the long axes of cellular shapes and nuclei of, for example, a stained patch of a media may be studied under a microscope. Since a stained patch reveals the orientations of smooth muscle cells, in particular of the associated nuclei (see Figure 2), it is possible to identify the statistical distributions of nuclei in the patch plane. In particular, we detect nuclei on an algorithmic basis by scanning each line of the histological section (image) consecutively. All connected pixels that make up a nucleus are copied to a stack. The stack is then transferred to a program, which determines the orientation of the shape of the nucleus from its associated second-order moment. To continue, the nucleus so detected is then deleted from the original image, and the procedure starts again at the location where the first pixel of the preceding nucleus was detected.

The above mentioned program starts with the computation of the centroid $\mathbf{x}_c$ of a nucleus according to $\mathbf{x}_c = \sum_{i=1}^N \mathbf{x}_i / N$, where N refers to the total number of pixels making up the nucleus. A representative shape of a nucleus is illustrated in Figure 2. The position vector of a typical pixel i relative to a fixed origin is denoted by $\mathbf{x}_i$ and the components of the vectors $\mathbf{x}_c$ and $\mathbf{x}_i$ are x_c, y_c and x_i, y_i , respectively.

Knowing the centroid of the nucleus we may compute the second-order moment, $\mathbf{G}$ say, i.e.

$$\mathbf{G} = \sum_{i=1}^N [(\mathbf{r}_i \cdot \mathbf{r}_i) \mathbf{I} - \mathbf{r}_i \otimes \mathbf{r}_i], \quad (1)$$

where we introduced the definition $\mathbf{r}_i = \mathbf{x}_i - \mathbf{x}_c$ of the position vectors $\mathbf{r}_i$, and $\mathbf{I}$ denotes the second-order unit tensor which, in index notation, has the form $(\mathbf{I})_{ij} = \delta_{ij}$ with δ_{ij} being the

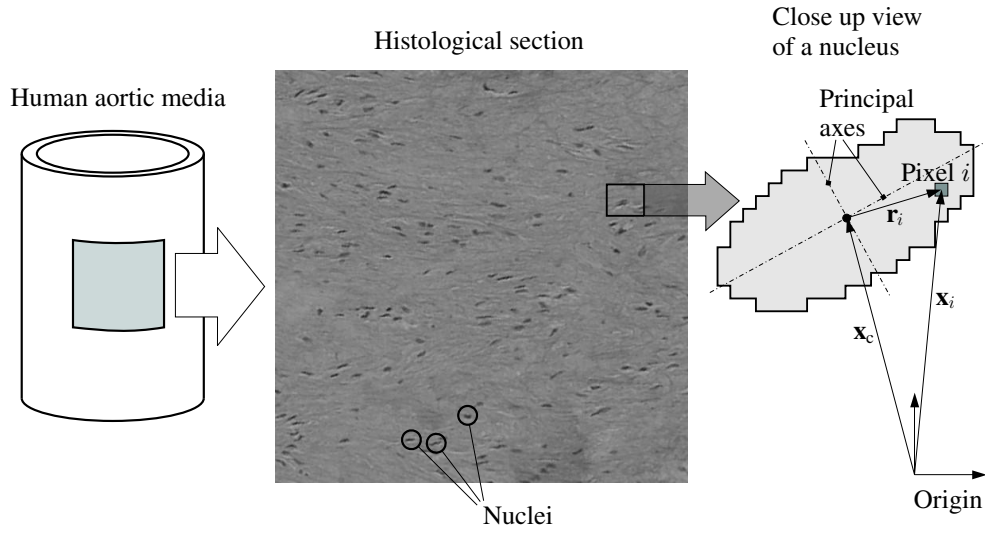


Figure 2. Histological section of a stained patch of a human aortic media. The circumferential orientation of the patch was aligned with the horizontal axis. The algorithm detected 179 nuclei within the image representing a histological size of $0.5 \times 0.5 \text{mm}^2$ ($p_{\min} = 15$, $r_{\min} = 2$).

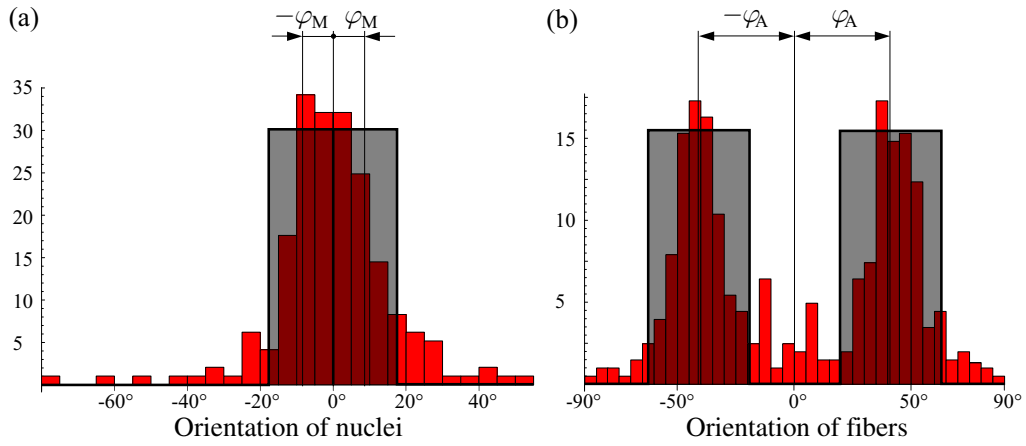


Figure 3. Histograms show statistical distributions of the orientation of nuclei in a human aortic media (a), and of collagen fibers in a human aortic adventitia (b). Statistical analyses led to mean angles $\varphi_M = \pm 8.4^\circ$ in the media and $\varphi_A = \pm 41.9^\circ$ in the adventitia.

Kronecker delta. To write eq. (1) in a more convenient *matrix* representation, which is useful for computational purposes, we have

$$\mathbf{G} = \sum_{i=1}^N \begin{bmatrix} (y_i - y_c)^2 & -(x_i - x_c)(y_i - y_c) \\ -(x_i - x_c)(y_i - y_c) & (x_i - x_c)^2 \end{bmatrix}. \quad (2)$$

If we imagine that every pixel, for example, possesses mass, then $\mathbf{G}$ may be seen as the *inertia tensor* relative to a fixed origin. It is a symmetric second-order tensor with components forming the entries of the inertia matrix, denoted by $\mathbf{G}$ and expressed through eq. (2). The diagonal components $(x_i - x_c)^2$, $(y_i - y_c)^2$ and the off-diagonal components $-(x_i - x_c)(y_i - y_c)$ of $\mathbf{G}$ are associated with the *moments of inertia* and the *product of inertia*, respectively. Considering the eigenvalue problem for the matrix $\mathbf{G}$ we may finally find the eigenvalues (which are the principal moments of inertia) and the associated eigenvectors which define the corresponding principal axes. Hence, the orientation of the nucleus is clearly determined by its principal axes, relative to which the product of inertia is zero. To this end it is immaterial whether only perimeter-pixels or all the pixels making up the nucleus are considered. Subsequently, the (mean) angle that occurs between the collagen fibers and the circumferential direction of the respective arterial layer (media, adventitia) is denoted by φ .

Finally, note that special attention is focused on the type of data stored in a stack. In order to avoid artifacts such as scratches, which may occur in a histological section, or nuclei that are represented by almost circular shapes, we require (i) a minimum number of pixels $p_{\min}$ that make up a nucleus and (ii) a minimum ratio $r_{\min}$ of larger to smaller eigenvalues (principal moments of inertia). If both criteria are satisfied the nucleus is accepted and its orientation is considered as relevant data.

Example. A histological section of a stained patch of a human aortic *media* with size $0.5 \times 0.5 \text{ mm}^2$ is considered. The image, as illustrated in Figure 2, was digitized and prepared for scanning by the software tool developed. The horizontal direction of the patch was aligned with the circumferential direction of the vessel. The criteria which ensure that the orientation of a nucleus is considered as relevant data were set to the values $p_{\min} = 15$ and $r_{\min} = 2$.

The algorithm detected 179 nuclei within the image and determined the orientations of all nuclei. A typical statistical distribution of the orientation of nuclei is illustrated in Figure 3(a) in the form of a histogram. The two (light grey) peaks in the figure fill the same area as the histogram. Statistical analyses led to mean angles φ_M of the orientations of nuclei (collagen fibers) in the media of $\pm 8.4^\circ$. The result that the smooth muscle cell orientations in the media are almost circumferentially oriented is in agreement with general histological data of a human aortic media (see, for example, Rhodin (1980)).

Figure 3(b) shows the representative results of a statistical analysis performed for a human aortic *adventitia*, which is characterized by predominant orientations of collagen fibers. The mean angles between the collagen fibers and the circumferential direction in the adventitia are, for this sample, determined as $\varphi_A = \pm 41.9^\circ$. The automatic technique for identifying fiber directions in the adventitia is very similar to that described above. Alternatively, for low numbers of nuclei the intra-spatial voids between collagen fiber bundles may be used as indicators for preferred orientations of the tissue.

2 A Structural Model for the Elastic Behavior of Arterial Walls

In this section we propose a multi-layer model for the passive (unstimulated) behavior of healthy young arterial walls in the *elastic* deformation domain. The structural model is formulated within the framework of nonlinear continuum mechanics and is well-suited for a finite element implementation.

We summarize the equations that provide the general continuum description of the deformation and the hyperelastic stress response of the material. From experimental observations of arterial walls it is known that their deformation response is almost isochoric in nature (the bulk modulus considerably exceeds the shear modulus). For this physical reason it seems to be most beneficial to choose a *decoupled* representation of the free-energy function. The decoupled representation turns out to be advantageous in avoiding numerical complications within the finite element analysis of incompressible materials (see Section 4). As a basis for reporting the performance of the constitutive model we consider the mechanical response of a thick-walled circular cylindrical tube under various boundary loads. We specify the stretches of the deformation and the amount of shear to be used, and discuss the equilibrium equation which arises in the considered problem. We also give expressions for the torsional couple and the reduced axial force acting on the tube, these being crucial for the numerical study of the constitutive model performed on a carotid artery. The numerical example in Section 5.1 of this chapter demonstrates one (clinical) application of the proposed multi-layer model. It documents a large scale finite element analysis of the elastic behavior of a healthy and young arterial segment during clamping.

2.1 Hyperelasticity

In the following we describe the (finite) deformation and the stress response of hyperelastic materials.

Description of the deformation. Let Ω_0 be a region of a continuum body $\mathcal{B}$ positioned in $\mathbb{R}^3$, referred to as the fixed *reference* configuration (assumed to be stress-free). We use the notation $\chi : \Omega_0 \rightarrow \mathbb{R}^3$ for the deformation, which transforms a typical material point $\mathbf{X} \in \Omega_0$ (with material coordinates X_A , $A = 1, 2, 3$) to a position $\mathbf{x} = \chi(\mathbf{X}) \in \Omega$ (with spatial coordinates x_a , $a = 1, 2, 3$) in the *current* configuration, denoted Ω . Further, let $\mathbf{F}(\mathbf{X}) = \partial\chi(\mathbf{X})/\partial\mathbf{X}$ be the deformation gradient and $J(\mathbf{X}) = \det\mathbf{F} > 0$ the local volume ratio.

Following Flory (1961) and Ogden (1978), we consider the multiplicative decomposition

$$\mathbf{F} = (J^{1/3}\mathbf{I})\bar{\mathbf{F}} \quad (3)$$

of the deformation gradient into a spherical (dilatational) part $J^{1/3}\mathbf{I}$, and into a unimodular (distortional) part, denoted by the modified deformation gradient $\bar{\mathbf{F}}$ ($\det\bar{\mathbf{F}} = 1$). We use the symmetric right and left Cauchy-Green tensors, denoted $\mathbf{C}$ and $\mathbf{b}$ respectively, and their modified counterparts, denoted $\bar{\mathbf{C}}$ and $\bar{\mathbf{b}}$ respectively, associated with $\bar{\mathbf{F}}$. From eq. (3) we then have

$$\mathbf{C} = \mathbf{F}^T \mathbf{F} = J^{2/3} \bar{\mathbf{C}}, \quad \bar{\mathbf{C}} = \bar{\mathbf{F}}^T \bar{\mathbf{F}}, \quad (4)$$

$$\mathbf{b} = \mathbf{F} \mathbf{F}^T = J^{2/3} \bar{\mathbf{b}}, \quad \bar{\mathbf{b}} = \bar{\mathbf{F}} \bar{\mathbf{F}}^T. \quad (5)$$

Hyperelastic stress response. In order to describe the hyperelastic stress response of arterial walls, we employ a set $\{\mathbf{A}_\alpha \mid \alpha = 1, \dots, n\}$ of (second-order) tensors which characterize the wall structure, and we postulate the existence of a Helmholtz free-energy function $\Psi(\mathbf{C}, \mathbf{A}_1, \dots, \mathbf{A}_n)$ defined per unit reference volume rather than per unit mass. Note that the free energy depends on $\mathbf{X}$ in addition to $\mathbf{C}, \mathbf{A}_1, \dots, \mathbf{A}_n$, but, for simplicity, we do not indicate subsequently this dependency.

Subsequently, we assume the *decoupled* form

$$\Psi(\mathbf{C}, \mathbf{A}_1, \dots, \mathbf{A}_n) = U(J) + \bar{\Psi}(\bar{\mathbf{C}}, \mathbf{A}_1, \dots, \mathbf{A}_n), \quad (6)$$

where the given strictly convex function U , taking on its unique minimum at $J = 1$, is motivated mathematically, and the given convex function $\bar{\Psi}$ is responsible for the isochoric elastic response of the material.

From the Clausius-Planck inequality, standard arguments lead to the well-established expression $\mathbf{S} = 2\partial\Psi(\mathbf{C}, \mathbf{A}_1, \dots, \mathbf{A}_n)/\partial\mathbf{C}$ for the second Piola-Kirchhoff stress. Equation (6) then gives

$$\mathbf{S} = \mathbf{S}_{\text{vol}} + \bar{\mathbf{S}} \quad \text{with} \quad \mathbf{S}_{\text{vol}} = 2\frac{\partial U(J)}{\partial \mathbf{C}}, \quad \bar{\mathbf{S}} = 2\frac{\partial \bar{\Psi}(\bar{\mathbf{C}}, \mathbf{A}_1, \dots, \mathbf{A}_n)}{\partial \bar{\mathbf{C}}}, \quad (7)$$

where $\mathbf{S}_{\text{vol}}$ and $\bar{\mathbf{S}}$ represent the purely volumetric and isochoric contributions to the second Piola-Kirchhoff stress $\mathbf{S}$. We shall also require the standard results

$$\frac{\partial J}{\partial \mathbf{C}} = \frac{1}{2}J\mathbf{C}^{-1} \quad \text{and} \quad \frac{\partial \bar{\mathbf{C}}}{\partial \mathbf{C}} = J^{-2/3} \left(\mathbb{I} - \frac{1}{3}\bar{\mathbf{C}} \otimes \bar{\mathbf{C}}^{-1} \right) \quad (8)$$

from tensor analysis (see, for example, Holzapfel (2000)), where $\mathbf{C}^{-1} = \mathbf{F}^{-1}\mathbf{F}^{-T}$ is the inverse of $\mathbf{C}$. The symbol $\mathbb{I}$ has been introduced to denote the fourth-order unit tensor which, in index notation, has the form $(\mathbb{I})_{IJKL} = (\delta_{IK}\delta_{JL} + \delta_{IL}\delta_{JK})/2$. With these results, eqs. (7)₂ and (7)₃ become, after some straightforward tensor manipulations and the introduction of the arbitrary hydrostatic pressure $p = dU/dJ$ as in Holzapfel (2000),

$$\mathbf{S}_{\text{vol}} = Jp\mathbf{C}^{-1}, \quad \bar{\mathbf{S}} = J^{-2/3}\mathbb{P} : \mathbf{S}^*. \quad (9)$$

We used the definitions

$$\mathbf{S}^* = 2\partial\bar{\Psi}/\partial\bar{\mathbf{C}}, \quad \mathbb{P} = \mathbb{I} - \frac{1}{3}\mathbf{C}^{-1} \otimes \mathbf{C} \quad (10)$$

for the fictitious second Piola-Kirchhoff stress and the fourth-order projection tensor, respectively. The kinematic quantity $\mathbb{P}$ is introduced with respect to the reference configuration (therefore expressed through $\mathbf{C}$), so that $(\mathbb{P} : \mathbf{S}^*) : \mathbf{C} = 0$ (see Holzapfel (2000)). It is important to emphasize that in the description of an incompressible material (which an artery is assumed to be) the function U has no physical meaning any more and the hydrostatic pressure p becomes an indeterminate Lagrange multiplier.

A Piola transformation of eqs. (9) enables the Cauchy stress tensor $\boldsymbol{\sigma} = J^{-1}\mathbf{F}\mathbf{S}\mathbf{F}^T$ to be put in the *decoupled* form

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}_{\text{vol}} + \bar{\boldsymbol{\sigma}} \quad \text{with} \quad \boldsymbol{\sigma}_{\text{vol}} = p\mathbf{I}, \quad \bar{\boldsymbol{\sigma}} = \mathbb{P} : \boldsymbol{\sigma}^*, \quad (11)$$

analogously to eq. (9). Here we used the definition $\boldsymbol{\sigma}^* = 2J^{-1}\bar{\mathbf{F}}(\partial\bar{\Psi}/\partial\bar{\mathbf{C}})\bar{\mathbf{F}}^T$ for the fictitious Cauchy stress and $\mathbb{P} = \mathbb{I} - \frac{1}{3}\mathbf{I} \otimes \mathbf{I}$ for the fourth-order projection tensor introduced with respect to the current configuration, so that $(\mathbb{P} : \boldsymbol{\sigma}^*) : \mathbf{I} = 0$.

2.2 Combined Bending, Inflation, Extension and Torsion of a Tube

Basic kinematics. We consider the artery as an incompressible thick-walled cylindrical tube subjected to various loads. It is known that the load-free configuration, Ω_{res} say, in which the artery is excised from the body and *not* subjected to any loads is not a stress-free (or strain-free) reference configuration Ω_0 . Thus, the arterial ring springs open when cut in a radial direction. It appears that Vaishnav and Vossoughi (1983) were the first to publish this finding. The load-free configuration Ω_{res} of an artery arises from certain growth mechanisms of the different layers. For discussion of growth and remodeling we refer to, for example, Rodriguez et al. (1994), Skalak et al. (1996) and Rachev (1997); for a more recent theoretical work on residual stresses see Ogden and Schulze-Bauer (2000) and also the lecture notes by Ogden in this course. We *assume* that the open sector is the undeformed (stress-free and fixed) reference configuration Ω_0 of a circular cylindrical tube, as depicted in Figure 4. Note that, in general, the residually stressed state is more complex than considered here. There are also residual stresses in the axial direction of an arterial segment. For a striking example of this phenomenon the reader is referred to the paper by Schulze-Bauer et al. (2002).

Thus, in terms of cylindrical polar coordinates (R, Θ, Z) , the geometrical region Ω_0 of the tube is defined by

$$R_i \leq R \leq R_o, \quad 0 \leq \Theta \leq (2\pi - \alpha), \quad 0 \leq Z \leq L, \quad (12)$$

where R_i , R_o , α and L denote the inner and outer radii, the opening angle and length of the undeformed (split) tube, respectively.

The deformation χ takes Ω_0 into the current configuration Ω . For the considered problem $\chi = \chi_p \circ \chi_{\text{res}}$ is the composition of the deformations χ_{res} and χ_p , as indicated in Figure 4, where χ_{res} generates the load-free configuration Ω_{res} associated with residual stresses, while χ_p is associated with inflation, axial elongation and torsion of the tube, and leads to the final configuration Ω .

In terms of cylindrical polar coordinates (r, θ, z) , the geometry of the deformed configuration Ω is given by

$$r_i \leq r \leq r_o, \quad 0 \leq \theta \leq 2\pi, \quad 0 \leq z \leq l, \quad (13)$$

where r_i , r_o and l denote the inner and outer radii and the length of the deformed tube, respectively.

The deformation χ , which is taken to be isochoric, may then be written in the form $\chi = r\mathbf{e}_r + z\mathbf{e}_z$ with reference to the (unit) basis vectors $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_z\}$ associated with the cylindrical polar coordinates (r, θ, z) , where

$$r = \sqrt{\frac{R^2 - R_i^2}{k\lambda_z} + r_i^2}, \quad \theta = k\Theta + Z\frac{\Phi}{L}, \quad z = \lambda_z Z, \quad (14)$$

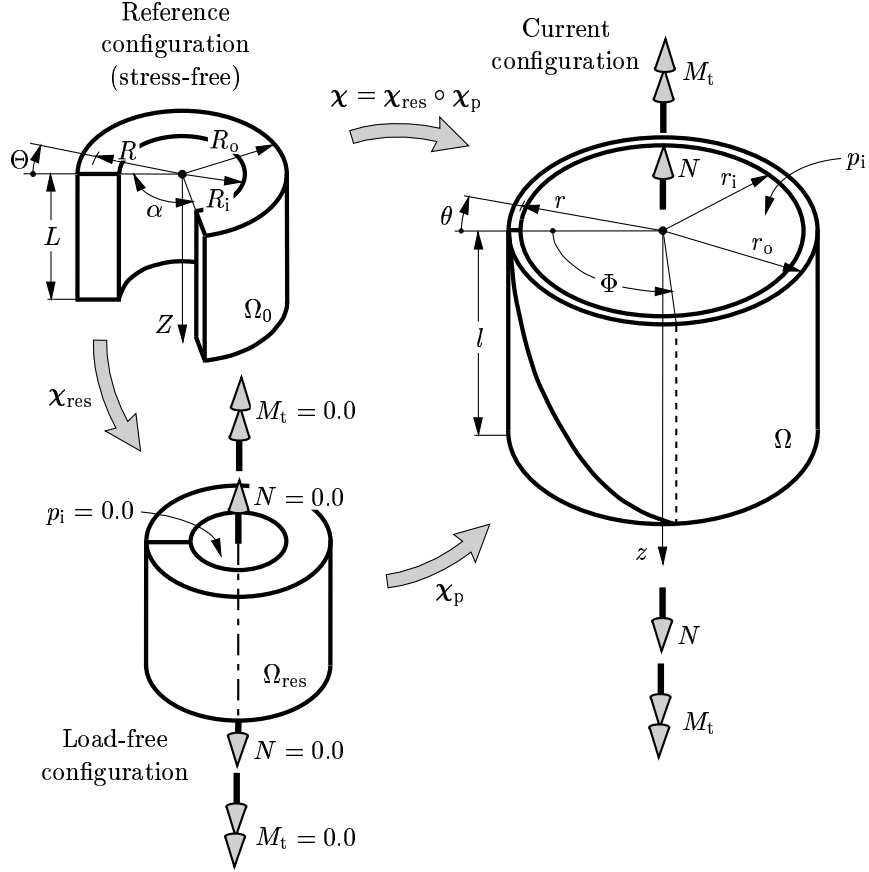


Figure 4. Arterial ring in the (stress-free) reference configuration Ω_0 , the load-free configuration Ω_{res} and the current configuration Ω .

λ_z is the (constant) axial stretch, the parameter k , defined by $k = 2\pi/(2\pi - \alpha)$, is a convenient measure of the tube opening angle in the unstressed configuration, r_i is the inner radius in the deformed configuration and Φ is the angle of twist of the tube arising from the torsion. Hence, the first term $k\Theta$ in $(14)_2$ represents the deformation from configuration Ω_0 to Ω_{res} while the second term $Z\Phi/L$ describes the influence of the torsion.

In addition to λ_z , it is convenient to introduce the notations defined by

$$\lambda_r(R) = \frac{\partial r}{\partial R} = \frac{R}{rk\lambda_z}, \quad \lambda_\theta(R) = \frac{r}{R} \frac{\partial \theta}{\partial \Theta} = \frac{kr}{R}, \quad \gamma(R) = r \frac{\partial \theta}{\partial z} = r \frac{\Phi}{l}. \quad (15)$$

Here $\lambda_r(R)$, $\lambda_\theta(R)$ and λ_z are the principal stretches of the deformation associated with the radial, circumferential and axial directions *when there is no twist*, while $\gamma(R)$, which is associated with the twist, represents locally the amount of shear in a (θ, z) -plane. Since each of these quan-

ties depends only on the radius R , the one-dimensional character of the problem is apparent. When $\gamma \neq 0$, λ_r is the principal stretch in the radial direction but λ_θ and λ_z are not then principal stretches. The condition that the volume is preserved during the deformation is independent of γ and requires simply that

$$\lambda_r \lambda_\theta \lambda_z = 1. \quad (16)$$

Note that

$$r_i = \lambda_{\theta i} \frac{R_i}{k}, \quad (17)$$

where $\lambda_{\theta i}$ denotes the value of λ_θ at the inner surface of the tube. Use of incompressibility constraint (16) and eqs. (4), (5) enables the deformation gradient $\mathbf{F} = \bar{\mathbf{F}}$ and the Cauchy-Green tensors $\mathbf{C} = \bar{\mathbf{C}}$ and $\mathbf{b} = \bar{\mathbf{b}}$ to be given in terms of cylindrical polar coordinates (see Holzapfel et al. (2000a) for more details).

In terms of the parameters k , $\lambda_{\theta i}$, λ_z and Φ , equations (14), (17) define the combined bending, inflation, axial extension and torsion of a thick-walled tube.

Equilibrium equations. In the absence of body forces the equilibrium equations are

$$\text{div } \boldsymbol{\sigma} = \mathbf{0}, \quad (18)$$

where $\text{div}(\bullet)$ denotes the spatial divergence of the spatial tensor field $(\bullet)$. Note that in cylindrical polar coordinates (r, θ, z) , because of the geometrical and constitutive symmetry, the only non-trivial component of (18) is

$$\frac{d\sigma_{rr}}{dr} + \frac{(\sigma_{rr} - \sigma_{\theta\theta})}{r} = 0 \quad (19)$$

(see, for example, Ogden (1997)). From this equation and the boundary condition $\sigma_{rr}|_{r=r_o} = 0$ on the outer surface of the tube, the radial Cauchy stress σ_{rr} may be calculated as

$$\sigma_{rr}(\xi) = \int_{\xi}^{r_o} (\sigma_{rr} - \sigma_{\theta\theta}) \frac{dr}{r}, \quad r_i \leq \xi \leq r_o. \quad (20)$$

The internal pressure $p_i = -\sigma_{rr}|_{r=r_i}$ is then obtained in the form

$$p_i = \int_{r_i}^{r_o} (\sigma_{\theta\theta} - \sigma_{rr}) \frac{dr}{r}. \quad (21)$$

This equation will play an important role in the numerical solution of the problem considered.

When the state of deformation is known, expressions for the axial force N and the torsional couple M_t can be calculated via the definitions

$$N = 2\pi \int_{r_i}^{r_o} \sigma_{zz} r dr, \quad M_t = 2\pi \int_{r_i}^{r_o} \sigma_{\theta z} r^2 dr. \quad (22)$$

In view of the decoupled representation of the Cauchy stress tensor σ according to (11)₁, we may recast equations (21) and (22)₂ by using the decompositions $\sigma_{\theta\theta} = p + \bar{\sigma}_{\theta\theta}$ and $\sigma_{rr} = p + \bar{\sigma}_{rr}$ to obtain

$$p_i = \int_{r_i}^{r_o} (\bar{\sigma}_{\theta\theta} - \bar{\sigma}_{rr}) \frac{dr}{r}, \quad M_t = 2\pi \int_{r_i}^{r_o} \bar{\sigma}_{\theta z} r^2 dr, \quad (23)$$

where $\bar{\sigma}_{\theta\theta}$, $\bar{\sigma}_{rr}$ denote the isochoric parts of the normal components of (Cauchy) stress in the circumferential and radial directions, while $\bar{\sigma}_{\theta z} = \sigma_{\theta z}$ is the shear component of (Cauchy) stress acting tangentially to the cross-section of the tube.

Use of the additive split (11) and eq. (20) enables the axial force N in eq. (22)₁ to be expressed as

$$N = 2\pi \int_{r_i}^{r_o} \left[\int_{\mathcal{U}_o}^{\xi} (\bar{\sigma}_{\theta\theta} - \bar{\sigma}_{rr}) \frac{dr}{r} - \bar{\sigma}_{rr} + \bar{\sigma}_{zz} \right] \xi d\xi. \quad (24)$$

Reversal of the order of integration in (24) and use of the expression (23)₁ leads to the general formula

$$F = \pi \int_{r_i}^{r_o} (2\bar{\sigma}_{zz} - \bar{\sigma}_{\theta\theta} - \bar{\sigma}_{rr}) r dr \quad (25)$$

for the *reduced axial force* $F = N - r_i^2 \pi p_i$. This expression for F is very important since it gives precisely the force that is measured during inflation tests on arteries. A specific form of (25) is given in eq. (15) of the paper by Chuong and Fung (1983).

The theory described above is designed to capture the deformation behavior in the central part of a tube so as to exclude end effects. Therefore, axial dependence of the deformation is not considered. This reflects the typical setting used in experiments (see, for example, Fung et al. (1979) or Schulze-Bauer et al. (2001), amongst others).

Numerical technique to solve the nonlinear boundary-value problem. Here we describe briefly the numerical technique to solve the problem of bending, axial extension, inflation and torsion of a thick-walled cylindrical tube.

By assuming a particular state of residual strain (characterized by the parameter k), the fixed axial stretch λ_z and fixed angle of twist Φ of the tube, the isochoric part of the strain (and hence the stress) can always be expressed in terms of the two variables $\lambda_{\theta i}$ and r , i.e. the circumferential stretch at the inner surface of the tube and the radius, respectively. Hence, the equation of equilibrium (23)₁ may be written in the general form

$$p_i = \int_{r_i}^{r_o} \mathcal{F}(\lambda_{\theta i}, r) \frac{dr}{r}, \quad (26)$$

where r_i is given in terms of $\lambda_{\theta i}$ by (17). Since closed-form evaluation of eq. (26) is only possible for very simple constitutive equations, we employ a Gaussian integration scheme (Hughes

(2000)), i.e.

$$p_i \approx \sum_{j=1}^n \mathcal{F}(\lambda_{\theta i}, r_j) \frac{w_j}{r_j}, \quad (27)$$

where w_j and r_j , ($j = 1, \dots, n$), denote the weights and the Gaussian points, and n is the order of integration. Equation (27) is, in general, nonlinear in the single unknown $\lambda_{\theta i}$, and, for given p_i , can be solved for $\lambda_{\theta i}$ using, for example, a standard *Newton iteration* with the initial value $\lambda_{\theta i} = 1.0$.

Since the deformation is now determined, the torsional couple M_t and the reduced axial force F follow directly from eqs. (23)₂ and (25), respectively. This computation is carried out by employing another Gaussian integration. It turns out that for the considered range of deformations a three-point integration ($n = 3$) with the accuracy of order five gives sufficiently accurate solutions.

2.3 A Multi-layer Model for Arterial Walls

In this section we propose a potential that models each layer of the artery as a fiber-reinforced composite. The basic idea is to formulate a constitutive model which incorporates some histological information. Hence, the material parameters involved may be associated with the histological structure of arterial walls (i.e. fiber directions), a feature which is not possible with phenomenological models. The underlying physical background of the described constitutive model leads to a formulation that is consistent with convexity requirements ensuring mechanically and mathematically reliable behavior. The comparative study by Holzapfel et al. (2000a) reports about the general lack of convexity detected in several well-established potentials used to describe the pronounced mechanical behavior of arterial walls. The multi-layer model is based on the theory of the mechanics of fiber-reinforced composites (Spencer (1984)) and embodies the symmetries of a cylindrically orthotropic material.

Constitutive model for the arterial layers. Since arteries are composed of (thick-walled) layers we model each of these layers with a separate strain-energy function. From the engineering point of view each layer may be considered as a composite reinforced by two families of (collagen) fibers which are arranged in symmetrical spirals.

We assume that each layer responds with similar mechanical characteristics and we therefore use the same form of strain-energy function (but a different set of material parameters) for each layer. We suggest an additive split of the isochoric strain-energy function $\bar{\Psi}$ into a part $\bar{\Psi}_{\text{iso}}$ associated with isotropic deformations and a part $\bar{\Psi}_{\text{aniso}}$ associated with anisotropic deformations (Holzapfel and Weizsäcker (1998)). Since the (wavy) collagen fibers of arterial walls are not active at low pressures (they do not store strain energy) we associate $\bar{\Psi}_{\text{iso}}$ with the mechanical response of the non-collagenous matrix material, which we assume to be isotropic. The resistance to stretch at high pressures is almost entirely due to collagenous fibers (Roach and Burton (1957)) and this mechanical response is therefore taken to be governed by the anisotropic function $\bar{\Psi}_{\text{aniso}}$. Hence, for a representative arterial layer the (two-term) potential is written as

$$\bar{\Psi}(\bar{\mathbf{C}}, \mathbf{a}_{01}, \mathbf{a}_{02}) = \bar{\Psi}_{\text{iso}}(\bar{\mathbf{C}}) + \bar{\Psi}_{\text{aniso}}(\bar{\mathbf{C}}, \mathbf{a}_{01}, \mathbf{a}_{02}), \quad (28)$$

where the families of collagenous fibers are characterized by the two (reference) direction vectors $\mathbf{a}_{0i}$, $i = 1, 2$, with $|\mathbf{a}_{0i}| = 1$. For a possible identification of collagen fiber orientations recall Section 1.2.

We now include structure tensors in accordance with the formulation in Section 2.1. Specifically, we incorporate two such tensors, $\mathbf{A}_i$, $i = 1, 2$, defined as the tensor products $\mathbf{a}_{0i} \otimes \mathbf{a}_{0i}$. The integrity basis for the three symmetric second-order tensors $\bar{\mathbf{C}}$, $\mathbf{A}_1$, $\mathbf{A}_2$ then consists of the invariants

$$\bar{I}_1(\bar{\mathbf{C}}) = \text{tr} \bar{\mathbf{C}}, \quad \bar{I}_2(\bar{\mathbf{C}}) = \frac{1}{2}[(\text{tr} \bar{\mathbf{C}})^2 - \text{tr} \bar{\mathbf{C}}^2], \quad \bar{I}_3(\bar{\mathbf{C}}) = \det \bar{\mathbf{C}} = 1, \quad (29)$$

$$\bar{I}_4(\bar{\mathbf{C}}, \mathbf{a}_{01}) = \bar{\mathbf{C}} : \mathbf{A}_1, \quad \bar{I}_5(\bar{\mathbf{C}}, \mathbf{a}_{01}) = \bar{\mathbf{C}}^2 : \mathbf{A}_1, \quad (30)$$

$$\bar{I}_6(\bar{\mathbf{C}}, \mathbf{a}_{02}) = \bar{\mathbf{C}} : \mathbf{A}_2, \quad \bar{I}_7(\bar{\mathbf{C}}, \mathbf{a}_{02}) = \bar{\mathbf{C}}^2 : \mathbf{A}_2, \quad (31)$$

$$\bar{I}_8(\bar{\mathbf{C}}, \mathbf{a}_{01}, \mathbf{a}_{02}) = (\mathbf{a}_{01} \cdot \mathbf{a}_{02}) \mathbf{a}_{01} \cdot \bar{\mathbf{C}} \mathbf{a}_{02}, \quad \bar{I}_9(\mathbf{a}_{01}, \mathbf{a}_{02}) = (\mathbf{a}_{01} \cdot \mathbf{a}_{02})^2; \quad (32)$$

see, for example, Spencer (1984), Holzapfel (2000) and the lecture notes by Ogden in this course. Since the invariants $\bar{I}_3$, $\bar{I}_9$ are constants we may express eq. (28) in the reduced form

$$\bar{\Psi}(\bar{\mathbf{C}}, \mathbf{A}_1, \mathbf{A}_2) = \bar{\Psi}_{\text{iso}}(\bar{I}_1, \bar{I}_2) + \bar{\Psi}_{\text{aniso}}(\bar{I}_1, \bar{I}_2, \bar{I}_4, \dots, \bar{I}_8). \quad (33)$$

Note that the invariants $\bar{I}_4$ and $\bar{I}_6$ are the squares of the stretches in the directions of $\mathbf{a}_{01}$ and $\mathbf{a}_{02}$, respectively, so that they are stretch measures for the two families of (collagen) fibers and therefore have a clear physical interpretation. For simplicity, in order to minimize the number of material parameters, we consider the reduced form of (33) given by

$$\bar{\Psi}(\bar{\mathbf{C}}, \mathbf{A}_1, \mathbf{A}_2) = \bar{\Psi}_{\text{iso}}(\bar{I}_1) + \bar{\Psi}_{\text{aniso}}(\bar{I}_4, \bar{I}_6). \quad (34)$$

The anisotropy then arises only through the invariants $\bar{I}_4$ and $\bar{I}_6$, but this is sufficiently general to capture the typical features of arterial response.

Finally, the two contributions $\bar{\Psi}_{\text{iso}}$ and $\bar{\Psi}_{\text{aniso}}$ to the function $\bar{\Psi}$ must be particularized so as to fit the material parameters to the experimentally observed response of the arterial layers. We use the (classical) neo-Hookean model to determine the isotropic response in each layer, and we write

$$\bar{\Psi}_{\text{iso}}(\bar{I}_1) = \frac{c}{2}(\bar{I}_1 - 3), \quad (35)$$

where $c > 0$ is a stress-like material parameter. The strong stiffening effect of each layer observed at high pressures motivates the use of an exponential function for the description of the strain energy stored in the collagen fibers, and for this we propose

$$\bar{\Psi}_{\text{aniso}}(\bar{I}_4, \bar{I}_6) = \frac{k_1}{2k_2} \sum_{i=4,6} \{ \exp[k_2(\bar{I}_i - 1)^2] - 1 \}, \quad (36)$$

where $k_1 > 0$ is a stress-like material parameter and $k_2 > 0$ is a dimensionless parameter. An appropriate choice of k_1 and k_2 enables the histologically-based assumption that the collagen fibers do not influence the mechanical response of the artery in the low pressure domain (Roach and Burton (1957)) to be modeled.

All that remains is to determine an expression for the stress, which we provide here in the Eulerian description. Using (11)₃ and the proposed particularizations (35) and (36), we obtain, after some straightforward manipulations, the explicit isochoric contribution $\bar{\boldsymbol{\sigma}}$ to the Cauchy stress tensor, namely

$$\bar{\boldsymbol{\sigma}} = c \operatorname{dev} \bar{\mathbf{b}} + \sum_{i=4,6} 2\bar{\Psi}_i \operatorname{dev}(\mathbf{a}_i \otimes \mathbf{a}_i), \quad (37)$$

where $\bar{\Psi}_4 = \partial \bar{\Psi}_{\text{aniso}} / \partial \bar{I}_4$, $\bar{\Psi}_6 = \partial \bar{\Psi}_{\text{aniso}} / \partial \bar{I}_6$ denote (scalar) response functions and $\mathbf{a}_i = \bar{\mathbf{F}} \mathbf{a}_{0i}$, $i = 1, 2$, the Eulerian counterparts of $\mathbf{a}_{0i}$. For a detailed derivation of equation (37) the reader is referred to the more general constitutive framework described in Gasser and Holzapfel (2001). For other applications of the proposed constitutive framework to soft tissue biomechanics, see, for example, Eberlein et al. (2001) and Holzapfel (2001).

Constitutive representation of a healthy young artery. For a *healthy young* artery, which has no pathological intimal changes, the innermost layer is not of (solid) mechanical interest, and we therefore focus attention on modeling the two remaining layers, i.e. the media and the adventitia. It is then appropriate to model the artery as a two-layer thick-walled tube with residual strains, as illustrated in Figure 5.

This model uses 6 material parameters, i.e. c_M, k_{1M}, k_{2M} for the media and c_A, k_{1A}, k_{2A} for the adventitia. In respect of eqs. (34)–(36) the free-energy functions for the considered two-layer problem may be written as

$$\bar{\Psi}_M = \frac{c_M}{2}(\bar{I}_1 - 3) + \frac{k_{1M}}{2k_{2M}} \sum_{i=4,6} \{ \exp[k_{2M}(\bar{I}_i - 1)^2] - 1 \}, \quad R_i \leq R \leq R_i + H_M, \quad (38)$$

$$\bar{\Psi}_A = \frac{c_A}{2}(\bar{I}_1 - 3) + \frac{k_{1A}}{2k_{2A}} \sum_{i=4,6} \{ \exp[k_{2A}(\bar{I}_i - 1)^2] - 1 \}, \quad R_i + H_M \leq R \leq R_o \quad (39)$$

for the media and adventitia, respectively. The constants c_M and c_A are associated with the non-collagenous matrix of the material, which describes the *isotropic* part of the overall response of the tissue. Note, however, that the matrix material is significantly less stiff than its elastin fiber constituent. The constants k_{1M}, k_{2M} and k_{1A}, k_{2A} are associated with the *anisotropic* contribution of collagen to the overall response. The material parameters are constants and do not depend on the geometry, opening angle or fiber angle.

The invariants, associated with the media M and the adventitia A, are defined by $\bar{I}_{4j} = \mathbf{A}_{1j} : \bar{\mathbf{C}}$ and $\bar{I}_{6j} = \mathbf{A}_{2j} : \bar{\mathbf{C}}$, $j = M, A$, and H_M is the reference thickness of the media, as illustrated in Figure 5. The tensors $\mathbf{A}_{1j}, \mathbf{A}_{2j}$, characterizing the structure of the media and adventitia, are given by

$$\mathbf{A}_{1j} = \mathbf{a}_{01j} \otimes \mathbf{a}_{01j}, \quad \mathbf{A}_{2j} = \mathbf{a}_{02j} \otimes \mathbf{a}_{02j}, \quad j = M, A, \quad (40)$$

where, in a cylindrical polar coordinate system, the components of the direction vectors $\mathbf{a}_{01j}$ and $\mathbf{a}_{02j}$ have, in matrix notation, the forms

$$\mathbf{a}_{01j} = \begin{bmatrix} 0 \\ \cos \varphi_j \\ \sin \varphi_j \end{bmatrix}, \quad \mathbf{a}_{02j} = \begin{bmatrix} 0 \\ \cos \varphi_j \\ -\sin \varphi_j \end{bmatrix}, \quad j = M, A, \quad (41)$$

and φ_j , $j = M, A$, are the (mean) angles between the collagen fibers (arranged in symmetrical spirals) and the circumferential direction in the media and adventitia, as indicated in Figure 5 (compare also with Section 1.2).

Because of the wavy structure of collagen it is regarded as not being able to support compressive stresses. We therefore assume that the fibers contribute to the strain energy in extension and do not contribute in compression. Hence, in the proposed model the anisotropic terms in the free-energy functions (38) and (39) should only contribute when the fibers are extended, that is when $\bar{I}_{4j} > 1$ or $\bar{I}_{6j} > 1$, $j = M, A$. If one or more of these conditions is not satisfied then the relevant part of the anisotropic function is omitted from the expressions (38) and (39). If, for example, $\bar{I}_{4A}$ and $\bar{I}_{6A}$ are less than or equal to 1, then the response of the adventitia is purely isotropic.

When these conditions are taken into account, it can be shown that strict local convexity is *guaranteed* by the form of the free-energy functions (38) and (39). Strict local convexity means that the second derivative of Ψ with respect to the Green-Lagrange strain tensor is positive definite, with appropriate modifications to account for incompressibility. This fundamental physical requirement in hyperelasticity ensures that undesirable material instabilities are precluded. It also induces desirable mathematical features in the governing equations, which are important from the point of view of numerical computations. It is worthwhile mentioning that some potentials which have been proposed in the literature are not convex for all possible sets (or *any* set) of material parameters. For investigations of convexity for a small number of constitutive models used in the literature to describe the mechanical response of arteries see the paper by Holzapfel et al. (2000a).

2.4 Elastic Behavior of a Carotid Artery. Performance of the Model

This section aims to report the performance of the fully three-dimensional constitutive model. In particular, the anisotropic behavior of a healthy carotid artery under combined bending, inflation, axial extension and torsion is investigated in more detail.

We use geometrical data from Chuong and Fung (1983) for a carotid artery from a rabbit (experiment 71 in Fung et al. (1979)) and make the assumptions that the media occupies 2/3 of the arterial wall thickness and that the wall thickness of each layer in the unloaded configuration ($\alpha = 0.0^\circ$) is the same as for the case without residual stress ($\alpha = 160.0^\circ$). The undeformed length L of the arterial tube was taken to be equal to the value of the inner radius R_i corresponding to $\alpha = 0.0^\circ$. According to Section 2.2 we make the assumption that the open sector shown in Figure 5 is stress-free. This simplifying assumption must be regarded as an approximation. In practice, the opening angles and the stress-free configurations for the separate layers would be different.

In order to identify the material parameters of the two-layer model for healthy arterial walls, we fitted the parameters to the experimental data from experiment 71 in Fung et al. (1979) and used the standard nonlinear *Levenberg-Marquardt* algorithm. The material parameters obtained are summarized in Figure 5.

Experimental tests performed by Maltzahn and Warriyar (1984), Yu et al. (1993) and Xie et al. (1995) indicate that the elastic properties of the media and adventitia are different. Their results show that the media is much stiffer than the adventitia. In particular, it was found that in the neighborhood of the reference configuration the mean value of Young's modulus for the

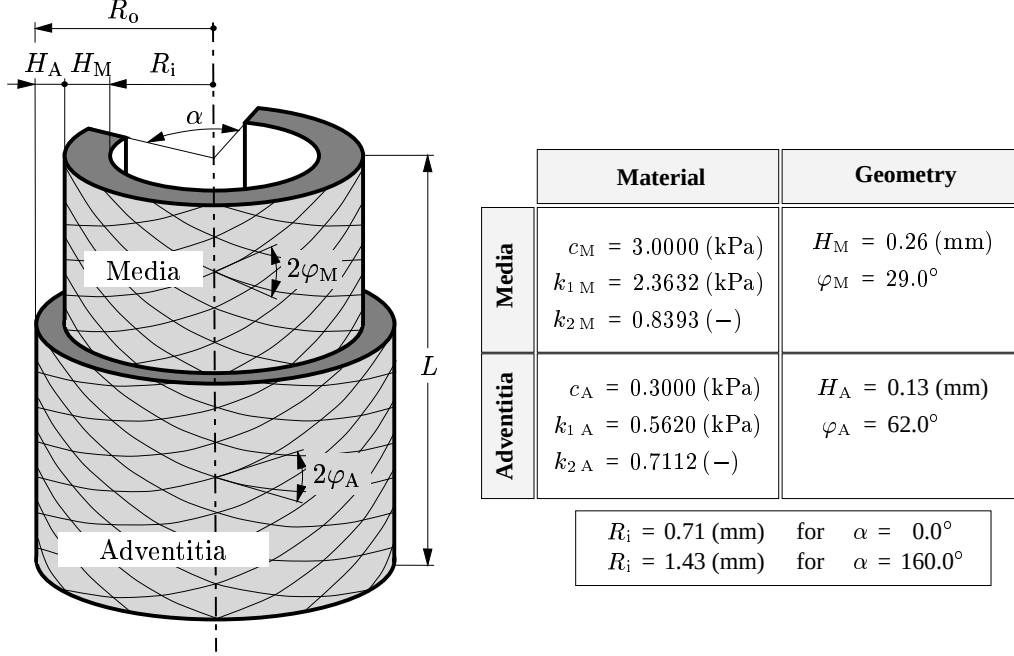


Figure 5. Material and geometrical data for a carotid artery from a rabbit in respect of (38) and (39) (see experiment 71 in Chuong and Fung (1983)).

media, for several pig thoracic aortas, is about an *order of magnitude higher* than that of the adventitia (Yu et al. (1993)). For our proposed constitutive model this observation implies that for these materials the neo-Hookean parameters are such that the ratio c_M/c_A is typically in the range of 6 to 14. This effectively reduces the number of material parameters, and for definiteness we therefore set $c_M = 10c_A$ for purposes of numerical calculation. In general, however, this ratio depends on the topographical site.

For the functions (38) and (39) the isochoric contribution $\bar{\sigma}$ to the Cauchy stress tensor σ is obtained from (37). Hence, with the definition (5)₂ of the modified left Cauchy-Green tensor $\bar{\mathbf{b}}$, the isochoric Cauchy stress components, which are used in eqs. (23)₁, (23)₂ and (25), are given.

The mechanical response of the carotid artery during bending, inflation, axial extension and torsion is shown in Figure 6. The internal pressure p_i and the angle of twist Φ are varied within the ranges

$$0 \leq p_i \leq 21.33 \text{ (kPa)} \quad \text{and} \quad -0.10 \leq \Phi \leq 0.10 \text{ (rad)}. \quad (42)$$

These loadings are applied at fixed axial stretches of the artery varying between $\lambda_z = 1.5$ and $\lambda_z = 1.9$. The solid lines in Figure 6 show numerical results based on a load-free, but not stress-free, configuration ($\alpha = 160.0^\circ$), while the dashed lines are based on a load-free and stress-free configuration ($\alpha = 0.0^\circ$).

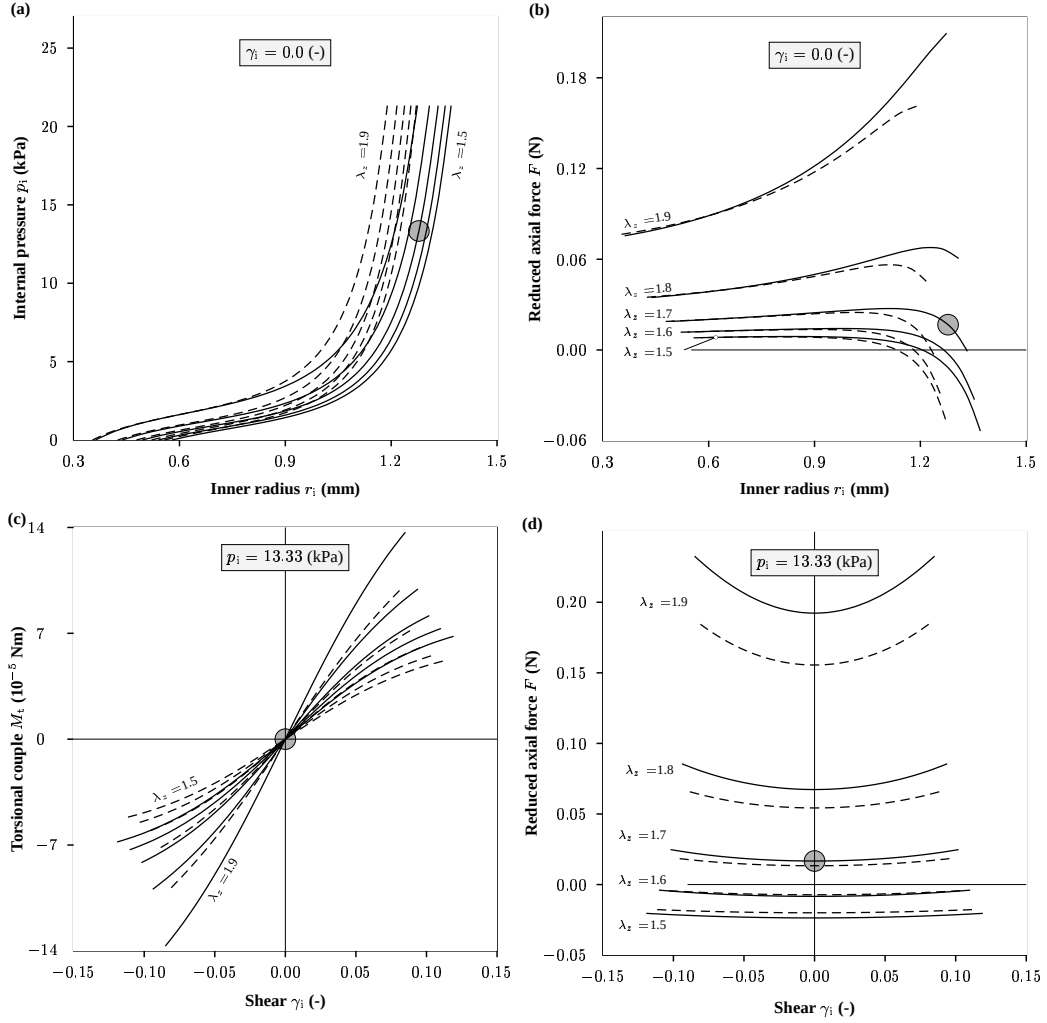


Figure 6. Deformation behavior of a carotid artery during inflation and torsion using the constitutive model (38)–(39). Solid lines are numerical results with residual strains included ($\alpha = 160.0^\circ$) and the dashed lines are results without residual strains ($\alpha = 0.0^\circ$). Dependence of (a) the internal pressure p_i and (b) the reduced axial force F on inner radius r_i , without shear deformation ($\gamma_i = 0$). Dependence of (c) the torsional couple M_t and (d) the reduced axial force F on the shear γ_i at fixed internal pressure $p_i = 13.33$ (kPa). The shaded circles indicate the approximate central region of the physiological state.

The internal pressure versus radius behavior is shown in Figure 6(a). The proposed model (38)–(39) is able to describe the salient features of arterial elasticity, such as the experimentally observed ‘sigma-shaped’ form of the pressure/radius relationship; see Figure 2(a) in the paper by Weizsäcker and Pinto (1988). Note that residual strains have a strong influence on the global pressure/radius response of the artery. The internal pressure/radius response, of course, does depend on geometry, opening angle and fiber angle, but we have not included here an analysis of the effect of changes in these quantities. However, our studies have found that in the high-pressure regime the stress-strain response depends significantly on the fiber angles (as should be expected). The fiber angles are associated with the stress-free configuration, as indicated in Figure 5, and we have here assumed that they are the same in the load-free configuration. The difference in angle between the unstressed and unloaded configurations for the case we considered goes from (approximately) -3.0° on the inner boundary to $+2.7^\circ$ on the outer boundary (mean value 0.2°). This approximation has a negligible influence on the analysis.

Figure 6(b) shows that the proposed potential (38)–(39) is also able to model the typical evolution of the reduced axial force F with inflation (increase of the inner radius) of the artery; see Figure 2(b) in Weizsäcker and Pinto (1988). This means that F is a decreasing function of r_i at axial stretches λ_z less than some value above the physiological stretch and an increasing function for λ_z greater than this value.

The response of the artery during torsion at the internal pressure $p_i = 13.33$ (kPa) is plotted in Figure 6(c)(d). As can be seen from Figure 6(c), the torsional couple M_t increases more slowly than the shear $\gamma_i = \Phi r_i / l = \Phi r_i / \lambda_z L$ on the inner boundary increases (i.e. the slope of the curve decreases). One possible explanation of this interesting phenomenon is as follows: since the artery is inflated with the internal physiological pressure, the (collagen) fiber reinforcement is activated and the fibers are much stiffer than the matrix material. During torsion from this state of deformation the nearly inextensible fibers cause the arterial diameter to decrease, which leads to a reduction in the torsional couple M_t given by eq. (22)₂. This realistic diameter-shrinking behavior of the artery during torsion seems to be a consequence of the considered fiber reinforcement (orthotropy). However, this effect may also be predicted by a non-convex isotropic strain-energy function.

In Figure 6(d) the reduced axial force F during torsion is plotted against the shear γ_i . For an axial pre-stretch $\lambda_z = 1.5$ the reduction in the inner radius r_i due to torsion is about 5.8% ($\gamma_i = 0.119$) and 7.8% for $\lambda_z = 1.9$ ($\gamma_i = 0.085$). This behavior is in qualitative agreement with experimental observations presented in Deng et al. (1994).

3 A Structural Model for the Viscoelastic Behavior of Arterial Walls

In this section we present a structural model suitable for predicting reliably the passive *time-dependent* three-dimensional stress and deformation states of arterial walls under various loading conditions. It extends the constitutive framework for the elastic strain response of arterial walls, as developed in Section 2.3, to the viscoelastic regime (for more details see Holzapfel et al. (2002a)).

The reviewed viscoelastic model admits hysteresis loops that are known to be relatively insensitive to strain rate, an essential mechanical feature of arteries of the muscular type. Instead of modeling the loading and unloading curves by two different laws of elasticity (pseudo-elasticity),

the concept of internal variables is introduced in order to replicate the characteristic dissipative mechanism of muscular arteries. Evolution equations and associated closed form solutions in the form of convolution integrals are provided for the internal variables. An efficient update algorithm for the stresses and the associated closed-form expression for the algorithmic elasticity tensor are briefly discussed. For a theoretical basis of the viscoelastic model at finite strains and for applications to elastomeric structures see Holzapfel (1996) and Holzapfel (2000), Section 6.10. A more general constitutive framework for fiber-reinforced composites and associated computational aspects is provided in Holzapfel and Gasser (2001). In Section 5.2, two representative numerical examples are designed to demonstrate the good performance and mechanism of the structural model.

3.1 Constitutive Equations and Internal Dissipation

In order to derive a structural model for the viscoelastic behavior of healthy young arterial walls, we define a Helmholtz free-energy function Ψ , which describes the viscoelastic deformation of a material point from the reference configuration Ω_0 to some current configuration Ω . The free energy is assumed to be based on the kinematic decomposition (3) and expressed by the unique *decoupled* representation

$$\Psi = U(J) + \bar{\Psi}(\bar{\mathbf{C}}, \mathbf{A}_1, \mathbf{A}_2) + \sum_{\alpha=1}^m \bar{T}_\alpha(\bar{\mathbf{C}}, \mathbf{A}_1, \mathbf{A}_2, \bar{\mathbf{T}}_\alpha), \quad (43)$$

valid over some closed time interval $t \in [0, T]$. The history of the deformation is described by a set of *internal (strain-like) variables*, denoted by the second-order tensors $\bar{\mathbf{T}}_\alpha$, $\alpha = 1, \dots, m$. For the basic idea of internal variables that we use for the description of inelastic processes, see, for example, Valanis (1972), Lubliner (1990), Simo and Hughes (1998) or Holzapfel (2000), Section 6.9. Note that all viscoelasticity is assumed to occur purely by *isochoric* deformations and all *volume changing* deformations are forced to be reversible. Hence, the tensorial (history) variables $\bar{\mathbf{T}}_\alpha$ are akin to $\bar{\mathbf{C}}$. They are not accessible to direct observation, characterize the current departure from equilibrium and contribute to the total strain (stress). The viscoelastic behavior is modeled by $\alpha = 1, \dots, m$ viscoelastic processes with corresponding relaxation (or retardation) times $\tau_\alpha \in (0, \infty)$.

The first two terms on the right hand side of eq. (43) characterize the *equilibrium state* of the viscoelastic solid at fixed $\mathbf{F}$ as $t \rightarrow \infty$, which is a state of balance, while the third term, the ‘dissipative’ potential $\sum_{\alpha=1}^m \bar{T}_\alpha$, characterizes the *non-equilibrium state*, i.e. the relaxation and creep behavior. As already mentioned on p. 9, in the description of an incompressible material (which an artery is assumed to be) the function U is motivated mathematically and serves as a penalty function within a numerical analysis. For a discussion of the underlying issue the reader is referred to p. 29 in this volume. Representation (43) is well-suited for numerical implementation (see, for example, Simo et al. (1985), Simo and Taylor (1991) and Holzapfel and Gasser (2001) amongst others).

Now we particularize the second law of thermodynamics through the Clausius-Planck inequality, i.e. $\mathcal{D}_{\text{int}} = \mathbf{S} : \dot{\mathbf{C}}/2 - \dot{\Psi} \geq 0$, where $\mathcal{D}_{\text{int}}$ is the internal dissipation (local entropy

production). By computing the rate of change of Ψ and using the chain rule, we find that

$$\left(\mathbf{s} - J \frac{dU}{dJ} \mathbf{C}^{-1} - 2 \frac{\partial \bar{\Psi}}{\partial \mathbf{C}} - \sum_{\alpha=1}^m 2 \frac{\partial \bar{T}_\alpha}{\partial \mathbf{C}} \right) : \frac{1}{2} \dot{\mathbf{C}} - \sum_{\alpha=1}^m \frac{\partial \bar{T}_\alpha}{\partial \bar{\mathbf{\Gamma}}_\alpha} : \dot{\bar{\mathbf{\Gamma}}}_\alpha \geq 0. \quad (44)$$

In deriving (44) we used the properties $\dot{J} = \partial J / \partial \mathbf{C} : \dot{\mathbf{C}} = J \mathbf{C}^{-1} : \dot{\mathbf{C}} / 2$ and $\dot{\bar{\mathbf{C}}} = 2(\partial \bar{\mathbf{C}} / \partial \mathbf{C}) : \dot{\mathbf{C}} / 2$.

In order to satisfy $\mathcal{D}_{\text{int}} \geq 0$ for all admissible processes we apply the standard Coleman-Noll procedure (see Coleman and Noll (1963) and Coleman and Gurtin (1967)). For arbitrary choices of $\dot{\mathbf{C}}$, we deduce the constitutive equations for compressible hyperelasticity and a remainder inequality governing the non-negativeness of the internal dissipation. In particular, the stress response constitutes an additive split of the second Piola-Kirchhoff stress tensor. We write

$$\mathbf{s} = 2 \frac{\partial \Psi}{\partial \mathbf{C}} = \mathbf{s}_{\text{vol}} + \bar{\mathbf{s}} + \sum_{\alpha=1}^m \mathbf{Q}_\alpha. \quad (45)$$

This split is based on the definition

$$\mathbf{Q}_\alpha = 2 \frac{\partial \bar{T}_\alpha(\bar{\mathbf{C}}, \mathbf{A}_1, \mathbf{A}_2, \bar{\mathbf{\Gamma}}_\alpha)}{\partial \mathbf{C}} \quad (46)$$

of the isochoric *viscoelastic* stress contributions $\mathbf{Q}_\alpha$, $\alpha = 1, \dots, m$. The *elastic* stress contributions $\mathbf{s}_{\text{vol}}$ and $\bar{\mathbf{s}}$ are according to (9). The stresses $\mathbf{Q}_\alpha$ may be interpreted as *non-equilibrium stresses* in the sense of non-equilibrium thermodynamics, so that the ‘dissipative’ potential takes on the form

$$\bar{T}_\alpha = \int_{\mathbf{C}} \mathbf{Q}_\alpha(\bar{\mathbf{C}}_*, \mathbf{A}_1, \mathbf{A}_2, \bar{\mathbf{\Gamma}}_\alpha) : \frac{1}{2} d\mathbf{C}_*, \quad \alpha = 1, \dots, m, \quad (47)$$

(for a detailed exposition of the thermodynamic background see the book by Holzapfel (2000), which contains further references). The tensor quantities $\mathbf{Q}_\alpha$ describe purely isochoric stresses, so that $\mathbf{Q}_\alpha : \mathbf{C} = 0$, $\alpha = 1, \dots, m$. The internal dissipation is given by

$$\mathcal{D}_{\text{int}} = - \sum_{\alpha=1}^m \frac{\partial \bar{T}_\alpha}{\partial \bar{\mathbf{\Gamma}}_\alpha} : \dot{\bar{\mathbf{\Gamma}}}_\alpha \geq 0. \quad (48)$$

Note that for arbitrary elastic processes $\dot{\bar{\mathbf{\Gamma}}}_\alpha = \mathbf{0}$, and hence the internal dissipation $\mathcal{D}_{\text{int}}$ is zero (the material is considered to be elastic).

This constitutive framework, which is based on the concept of internal variables, is now used for the representation of viscoelastic stress responses of arterial layers.

3.2 Decoupled Stress Response

In this section we are concerned with the viscoelastic stress response of *healthy young* arteries of the muscular type at finite strains.

Viscoelastic stress response in the material description. As for healthy young arteries of the elastic type (recall pp. 16, 17) we model the artery as a two-layer thick-walled tube, i.e.

the media and the adventitia, which make the significant contributions to the solid mechanical properties of the wall. Hence, for each arterial layer the stress response is derived from the associated Helmholtz free-energy function. According to the constitutive framework discussed in Section 3.1 we arrive at the (second Piola-Kirchhoff) stress response

$$\left. \begin{aligned} \mathbf{S}_M &= (Jp\mathbf{C}^{-1})_M + \sum_{a=1,4,6} (\bar{\mathbf{S}}_a + \sum_{\alpha=1}^m \mathbf{Q}_{\alpha a})_M, \\ \mathbf{S}_A &= (Jp\mathbf{C}^{-1})_A + \sum_{a=1,4,6} (\bar{\mathbf{S}}_a + \sum_{\alpha=1}^m \mathbf{Q}_{\alpha a})_A \end{aligned} \right\} \quad (49)$$

for the media and adventitia, respectively (compare with the mathematical structure (45)₂). However, $\bar{\mathbf{S}}$, $\mathbf{Q}_\alpha$ are replaced by $\bar{\mathbf{S}}_a$, $\mathbf{Q}_{\alpha a}$ signifying the a -th constituent of the arterial layer ($a = 1$ is associated with the non-collagenous matrix material and $a = 4, 6$ with the two families of collagen fibers). We recall that the index α stands for the number of viscoelastic processes. Note that for the case $a = 1$ isotropy is recovered as a special case.

Considering the particularization (38) and (39) of the free energy and the decoupled representation of the kinematics according to Section 2.1, we may express the isochoric elastic stress contributions (eq. (9)₂ with definition (10)₁ in the convenient form

$$\bar{\mathbf{S}}_a = J^{-2/3} \mathbb{P} : \mathbf{S}_a^*, \quad \mathbf{S}_a^* = 2\psi_a \mathbf{D}_a, \quad (a = 1, 4, 6; \text{no summation}), \quad (50)$$

which is used in relations (49). We introduced the useful definition

$$\psi_a = \frac{\partial \bar{\Psi}(\bar{I}_1, \bar{I}_4, \bar{I}_6)}{\partial \bar{I}_a}, \quad a = 1, 4, 6 \quad (51)$$

of the stress functions ψ_a , which are affected by the special choice of $\bar{\Psi}$ (the material). In addition, in eq. (50) we have introduced the definitions

$$\mathbf{D}_a = \frac{\partial \bar{\mathbf{I}}_a}{\partial \bar{\mathbf{C}}}, \quad a = 1, 4, 6 \quad (52)$$

of kinematic quantities, i.e. the second-order tensors $\mathbf{D}_a$, $a = 1, 4, 6$. From eq. (52) we may conclude, using definitions (29)₁, (30)₁, (31)₁, that $\mathbf{D}_1 = \mathbf{I}$, $\mathbf{D}_4 = \mathbf{A}_1$ and $\mathbf{D}_6 = \mathbf{A}_2$. With the given functions (38) and (39) it is also straightforward to particularize eq. (51) in order to obtain

$$\psi_1 = c/2, \quad (53)$$

$$\psi_4 = k_1(\bar{I}_4 - 1) \exp[k_2(\bar{I}_4 - 1)^2], \quad (54)$$

$$\psi_6 = k_1(\bar{I}_6 - 1) \exp[k_2(\bar{I}_6 - 1)^2]. \quad (55)$$

As seen from definition (50) (with eqs. (53)–(55)) the stress response consists of purely *isotropic* contributions $\bar{\mathbf{S}}_1$ due to the matrix material, and *anisotropic* contributions $\bar{\mathbf{S}}_4$, $\bar{\mathbf{S}}_6$ due to the two families of fibers, which characterize decoupled stresses (associated only with the fibers).

The non-equilibrium states of the arterial layers are associated with the additional tensor variables $\mathbf{Q}_{\alpha a}$, for $\alpha = 1, \dots, m$ and $a = 1, 4, 6$. They are zero at a state of thermodynamic

equilibrium, which implies that the anisotropic material responds perfectly elastically. The non-equilibrium stresses are governed by complementary equations of evolution, as discussed in the following section.

Evolution equations for the non-equilibrium stresses. In order to determine how a viscoelastic process in an artery evolves, we have to postulate additional equations governing the non-equilibrium stresses.

Hysteresis loops of arterial tissues are known to be not very sensitive to strain rates over several decades (see, for example, the study by Tanaka and Fung (1974) performed for various arteries of dogs). This is also true for other types of biological soft tissues such as articular cartilage (Woo et al. (1979)) or the mesentery (Chen and Fung (1973)). Hence, we have to select a rheological model which considers this characteristic feature. Classical mechanical devices such as the *Maxwell model* (spring in series with a dashpot), the *Kelvin-Voigt model* (spring in parallel with a dashpot) or a device of the ‘*standard solid*’ type, which is a free spring on one end and one Maxwell element arranged in parallel, are not able to represent the typical viscoelastic behavior of soft tissues. The damping mechanisms of these devices are strongly frequency dependent and are not suitable candidates for formulating meaningful evolution equations. However, a mechanical device which is composed of a number of springs and dashpots gives the required viscoelastic behavior (see, for example, Fung (1993), Section 7.6, for more details and references).

For this reason we extend the attractive one-dimensional *generalized Maxwell model* to the three-dimensional region. The generalized Maxwell model may be seen as a mechanical device with a *free spring* on one end and an arbitrary number m of *Maxwell elements* arranged in parallel (see Holzapfel (2000), Section 6.10; in particular, see Example 6.10). The more Maxwell elements and associated (different) relaxation times used the nearer is the response to constant damping over a wide frequency spectrum.

Hence, for each of the (isochoric) non-equilibrium stresses, *separately* for each α and constituent a of the arterial layer, we formulate an evolution equation. We assume the set of linear differential equations

$$\dot{\mathbf{Q}}_{\alpha a} + \frac{\mathbf{Q}_{\alpha a}}{\tau_{\alpha a}} = \beta_{\alpha a}^{\infty} \dot{\tilde{\mathbf{S}}}_a, \quad \mathbf{Q}_{\alpha a}|_{t=0} = \mathbf{0}, \quad (a = 1, 4, 6; \text{no summation}),$$

and $\alpha = 1, \dots, m$) (56)

valid for some semi-open time interval $t \in (0, T]$. The initial conditions (56)₂ ensure that the reference configuration has no viscoelastic stress contribution. The constants $\beta_{\alpha a}^{\infty} \in [0, \infty)$ introduced are given so-called *free-energy factors*, which are non-dimensional and associated with the relaxation times $\tau_{\alpha a} \in (0, \infty)$, which describe the rate of decay of the stress and strain in a viscoelastic process.

Closed-form solutions of the linear equations (56)₁ may be represented by the simple convolution integrals

$$\mathbf{Q}_{\alpha a} = \int_{t=0^+}^{t=T} \exp[-(T-t)/\tau_{\alpha a}] \beta_{\alpha a}^{\infty} \dot{\tilde{\mathbf{S}}}_a(t) dt, \quad (57)$$

for $a = 1, 4, 6$ and $\alpha = 1, \dots, m$. The typical features of anisotropic arterial response in the large strain domain are now described completely by the constitutive equations (49), with expressions (50)–(55) and (57).

Algorithmic stress tensor at time t_{n+1} . In the following we review an efficient update algorithm for the stresses suitable for the numerical implementation in a finite element program. The update algorithm, which is basically concerned with the numerical integration of the convolution integrals (57), goes back to Herrmann and Peterson (1968) and Taylor et al. (1970). For a detailed exploitation of the time integration algorithm (for $a = 1$) and for additional references see Holzapfel (2000), pp. 290-293.

We consider some closed time interval $t \in [0^+, T]$ of interest and focus attention on a typical closed *time sub-interval* $[t_n, t_{n+1}]$, with $\Delta t = t_{n+1} - t_n$ characterizing the associated *time increment*. Assume now that at certain times t_n and t_{n+1} all relevant kinematic quantities are given. In addition to that, the stress $\mathbf{S}_n$ at t_n is also specified uniquely via the associated constitutive equation serving as an ‘initial’ data base. All that remains is to compute the so-called algorithmic stress tensor $\mathbf{S}_{n+1}$ at t_{n+1} according to

$$\mathbf{S}_{n+1} = (\mathbf{S}_{\text{vol}} + \bar{\mathbf{S}} + \sum_{\alpha=1}^m \mathbf{Q}_{\alpha})_{n+1}. \quad (58)$$

The first two elastic stress contributions $\mathbf{S}_{\text{vol } n+1}$ and $\bar{\mathbf{S}}_{n+1}$ can be computed simply from the given strain measures at t_{n+1} . The third term $\mathbf{Q}_{\alpha n+1}$ in (58), which is based on convolution integrals (57), is responsible for the viscoelastic stress contribution and remains to be evaluated. Using the mid-point rule we arrive at a second-order accurate *recurrence update formula* for the non-equilibrium stresses, namely

$$\mathbf{Q}_{\alpha n+1} = \mathcal{H}_{\alpha n} + \sum_{a=1,4,6} (\mathbf{Q}_{\alpha a})_{n+1}, \quad \alpha = 1, \dots, m, \quad (59)$$

with the definition

$$\mathcal{H}_{\alpha n} = \sum_{a=1,4,6} \exp(-\Delta t/2\tau_{\alpha a}) [\exp(-\Delta t/2\tau_{\alpha a}) (\mathbf{Q}_{\alpha a})_n - \beta_{\alpha a}^{\infty} (\bar{\mathbf{S}}_a)_n] \quad (60)$$

of the (algorithmic) history term $\mathcal{H}_{\alpha n}$, $\alpha = 1, \dots, m$, at t_n . In addition to the history term we have used the definitions

$$(\mathbf{Q}_{\alpha a})_{n+1} = \delta_{\alpha a} (\bar{\mathbf{S}}_a)_{n+1}, \quad \delta_{\alpha a} = \beta_{\alpha a}^{\infty} \exp(-\Delta t/2\tau_{\alpha a}) \quad (61)$$

of the non-equilibrium stresses $\mathbf{Q}_{\alpha a}$ at t_{n+1} and the so-called *viscoelastic factors* $\delta_{\alpha a}$, for each α and a . Note that the update algorithm is formulated in terms of material coordinates, and hence the objectivity requirement based on an Euclidean transformation is trivially satisfied.

3.3 Elasticity Tensor in the Material Description

The use of the viscoelastic model described above within a finite element context, with which it is desired to combined a Newton-type solution method, requires knowledge of the linearized

constitutive equation (for specific solution methods see Section 4.4 in this volume). In general, $\mathbb{C} = 2\partial\mathbf{S}/\partial\mathbf{C}$ is defined to be the fourth-order *elasticity tensor (tangent moduli)* in the material description. It is the gradient of the nonlinear tensor-valued (tensor) function $\mathbf{S}$, which possesses minor and major symmetries (Holzapfel (2000)).

We now restrict our attention to the purely isochoric (elastic and viscoelastic) contribution to $\mathbb{C}$, which we wish to denote by $\bar{\mathbb{C}}^v$. Given the structure of the decoupled stress relation (45), the associated isochoric part of the elasticity tensor at time t_{n+1} may be written in the form

$$\bar{\mathbb{C}}_{n+1}^v = \left[\sum_{a=1,4,6} (1 + \delta_a) \bar{\mathbb{C}}_a \right]_{n+1}, \quad \delta_a = \sum_{\alpha=1}^m \delta_{\alpha a}, \quad (62)$$

where $\bar{\mathbb{C}}_{n+1}^v$ is referred to as the *algorithmic elasticity tensor* in the material description at t_{n+1} . In eq. (62) each factor δ_a is determined as the sum of the associated m viscoelastic factors $\delta_{\alpha a}$, $\alpha = 1, \dots, m$, given through eq. (61)₂. Note that the viscoelastic contribution to the algorithmic elasticity tensor $\bar{\mathbb{C}}_{n+1}^v$ at t_{n+1} may be expressed in terms of the viscoelastic factors $\delta_{\alpha a}$, governing the time-dependent part, and $(\bar{\mathbb{C}}_a)_{n+1}$ at t_{n+1} , which is associated with the isochoric elastic response as $t \rightarrow \infty$. For a detailed derivation of (62), in particular of $\bar{\mathbb{C}}_a$, the reader is referred to the Appendix of the paper by Holzapfel and Gasser (2001). Here, only the explicit result for the isochoric part $\bar{\mathbb{C}}_a$ is presented, i.e.

$$\bar{\mathbb{C}}_a = 2 \frac{\partial \bar{\mathbf{S}}_a}{\partial \mathbf{C}} = \mathbb{P} : \mathbb{C}_a^* : \mathbb{P}^T + \frac{2}{3} \text{Tr}(J^{-2/3} \mathbf{S}_a^*) \tilde{\mathbb{P}} - \frac{2}{3} (\mathbf{C}^{-1} \otimes \bar{\mathbf{S}}_a + \bar{\mathbf{S}}_a \otimes \mathbf{C}^{-1}), \quad (63)$$

with the abbreviations

$$\tilde{\mathbb{P}} = \mathbf{C}^{-1} \odot \mathbf{C}^{-1} - \frac{1}{3} (\mathbf{C}^{-1} \otimes \mathbf{C}^{-1}), \quad \text{Tr}(\bullet) = (\bullet) : \mathbf{C} \quad (64)$$

of the fourth-order *modified projection tensor* $\tilde{\mathbb{P}}$ and the trace $\text{Tr}(\bullet)$, respectively. In eq. (64)₁ we used the definition $-\mathbf{C}^{-1} \odot \mathbf{C}^{-1}$ of the fourth-order tensor $\partial \mathbf{C}^{-1} / \partial \mathbf{C}$, for convenience. The symbol $\odot$ has been introduced to denote the tensor product according to the rule

$$-(\mathbf{C}^{-1} \odot \mathbf{C}^{-1})_{ABCD} = -\frac{1}{2} (C_{AC}^{-1} C_{BD}^{-1} + C_{AD}^{-1} C_{BC}^{-1}) = \frac{\partial C_{AB}^{-1}}{\partial C_{CD}}. \quad (65)$$

Recall that the kinematic quantity $\mathbb{P}$ is given by eq. (10)₂, and the stress contributions $\bar{\mathbf{S}}_a$ and $\mathbf{S}_a^*$ by eq. (50). In addition, in eq. (63)₂ we have used the definition $\mathbb{C}_a^* = 2J^{-4/3} \partial \mathbf{S}_a^* / \partial \bar{\mathbf{C}}$, $a = 1, 4, 6$ of the fourth-order *fictitious elasticity tensors* in the material description. Their closed form expressions are

$$\mathbb{C}_a^* = 4J^{-4/3} \sum_{a=1,4,6} \psi_{ab} \mathbf{D}_a \otimes \mathbf{D}_b, \quad (a = 1, 4, 6; \text{no summation}), \quad (66)$$

in which we have introduced the useful definitions

$$\psi_{ab} = \psi_{ba} = \frac{\partial^2 \bar{\Psi}(\bar{I}_1, \bar{I}_4, \bar{I}_6)}{\partial \bar{I}_a \partial \bar{I}_b}, \quad a, b = 1, 4, 6 \quad (67)$$

of the elasticity functions ψ_{ab} , which depend on the special choice of $\bar{\Psi}$. The second-order tensors $\mathbf{D}_a$ are introduced in eq. (52). With the given functions (38) and (39) it is straightforward to particularize eq. (67). The non-vanishing scalars are

$$\psi_{44} = k_1 [1 + 2k_2(\bar{I}_4 - 1)^2] \exp [k_2(\bar{I}_4 - 1)^2] , \quad (68)$$

$$\psi_{66} = k_1 [1 + 2k_2(\bar{I}_6 - 1)^2] \exp [k_2(\bar{I}_6 - 1)^2] . \quad (69)$$

Note that for the case of thermodynamic equilibrium in which $t \rightarrow \infty$, the non-equilibrium stresses vanish and the isochoric part $\bar{\mathbb{C}}_{n+1}^v$ of the elasticity tensor given by eq. (62)₁ reduces to $\bar{\mathbb{C}}$. General finite anisotropic elasticity is recovered with

$$\bar{\mathbb{C}} = \sum_{a=1,4,6} \bar{\mathbb{C}}_a, \quad (70)$$

and the constitutive framework for the description of the *elastic* behavior of arterial walls holds, as presented in Section 2.

4 Mixed Finite Element Formulation for Incompressible Finite Elasticity

Various types of biological soft tissues show a very high resistance to volumetric changes compared with that to isochoric changes. Hence, due to the nearly incompressible or incompressible deformation behavior of soft tissues a very careful numerical treatment is required. A conventional displacement-based method cannot be applied directly to these types of kinematic constraint problems, because ‘locking’ and ‘checkerboard’ phenomena will occur. A mixed finite element formulation is to be used.

In this section we describe one particular simple and well-established mixed finite element method in detail, which turns out to be very efficient within the framework of incompressible finite elasticity. We prefer to use the Eulerian description and define the tensor *and* matrix problems for fully nonlinear elastodynamics. Without loss of generality, we choose a Lagrangian function which does not depend on structural tensors as introduced in Section 2.1. We start with Hamilton’s principle, as the counterpart of the principle of stationary potential energy in dynamics, and continue with a brief review of the three-field variational principle by Simo et al. (1985), which takes account of nearly incompressible response. Spatial discretization of the continuous problem gives three Euler-Lagrange equations in weak form. Thereby, the volumetric variables are eliminated at the element level leading to the quite robust $Q1/P0$ -element, which circumvents numerical difficulties that arise from the overstiffening of the system associated with the analysis of isochoric constitutive responses of arterial walls. By a consistent linearization process we arrive at the stiffness matrices suitable for implementation in a finite element program. The section concludes by providing some insights into solution methods for nonlinear dynamic problems.

4.1 Variational Principles

Hamilton’s variational principle. The important *Hamilton’s variational principle* represents the laws of dynamics of (discrete and) continuous systems in concise form. It is equivalent to

the principle of virtual work and the Euler-Lagrange equations. As a particularization for static problems, the principle of stationary potential energy may be obtained. Hamilton's principle is presented by the stationary condition

$$\delta \int_{t_0}^{t_1} L(\mathbf{u}, \dot{\mathbf{u}}) dt = 0, \quad L(\mathbf{u}, \dot{\mathbf{u}}) = \Pi(\mathbf{u}) - \mathcal{K}(\dot{\mathbf{u}}), \quad (71)$$

where $\mathbf{u}(\mathbf{x}, t) = \mathbf{u}(\mathbf{X}, t)$ denotes the displacement vector field of a typical particle of a continuum body $\mathcal{B}$ positioned in $\mathbb{R}^3$. It relates its position $\mathbf{X}$ in the undeformed configuration to its position $\mathbf{x}$ in the deformed configuration at time t . The displacement vector field is regarded as continuously differentiable with respect to *position* and *time*. The velocity field, denoted by $\dot{\mathbf{u}}(\mathbf{x}, t) = \dot{\mathbf{u}}(\mathbf{X}, t)$, is the material time derivative of $\mathbf{u}$. The scalar-valued functionals Π and $\mathcal{K}$ denote the total potential energy and the kinetic energy of the moving body, respectively. The difference between Π and $\mathcal{K}$, called the *Lagrangian function* L (in the literature sometimes introduced as $-L$), is integrated with respect to time t over a closed time interval $t \in [t_0, t_1]$ (t_0 and t_1 are two arbitrary instants of time). We now assume the restriction that at the times t_0 and t_1 , the variation of the displacement vector field, denoted by $\delta \mathbf{u}$, vanishes at all points of the continuum body, i.e.

$$\delta \mathbf{u}|_{t=t_0} = \delta \mathbf{u}|_{t=t_1} = \mathbf{0}. \quad (72)$$

Note that $\delta \mathbf{u}$ is regarded as a function of position and time.

For further considerations we assume that the reference configuration Ω_0 of the continuum body $\mathcal{B}$ is bounded by a reference boundary surface $\partial \Omega_0$. We agree that this boundary surface is partitioned into two parts, $\partial \Omega_{0\mathbf{u}}$ and $\partial \Omega_{0\sigma}$, on which *Dirichlet* and *von Neumann* boundary conditions will be specified. The two parts are non-overlapping so that

$$\partial \Omega_0 = \partial \Omega_{0\mathbf{u}} \cup \partial \Omega_{0\sigma} \quad \text{with} \quad \partial \Omega_{0\mathbf{u}} \cap \partial \Omega_{0\sigma} = \emptyset. \quad (73)$$

Subsequently, we characterize the total potential energy Π as the sum of the *internal* and *external potential energies*, Π^{int} and Π^{ext} , i.e.

$$\Pi(\mathbf{u}) = \Pi^{\text{int}}(\mathbf{u}) + \Pi^{\text{ext}}(\mathbf{u}), \quad (74)$$

$$\Pi^{\text{int}}(\mathbf{u}) = \int_{\Omega_0} \Psi(\mathbf{C}(\mathbf{u})) dV, \quad \Pi^{\text{ext}}(\mathbf{u}) = - \int_{\Omega_0} \mathbf{B} \cdot \mathbf{u} dV - \int_{\partial \Omega_{0\sigma}} \bar{\mathbf{T}} \cdot \mathbf{u} dS, \quad (75)$$

where Ψ represents the strain-energy function defined per unit reference volume. It is expressed as a function of the six components C_{AB} of the right Cauchy-Green tensor $\mathbf{C}$. The infinitesimal volume element dV and the infinitesimal surface element dS are defined in the reference configuration. In addition, we assume here that the loads do not depend on the motion of the body (Π^{ext} is a linear function of $\mathbf{u}$). The *reference body force* $\mathbf{B} = \mathbf{B}(\mathbf{X}, t)$ is referred to the referential position $\mathbf{X}$ (force measured per unit reference volume), while $\bar{\mathbf{T}}$ denotes the prescribed (given) first Piola-Kirchhoff traction vector (force measured per unit reference surface area) specified on the part $\partial \Omega_{0\sigma} \subset \partial \Omega_0$ of the boundary surface.

The kinetic energy of a moving continuum body is defined to be

$$\mathcal{K}(\dot{\mathbf{u}}) = \int_{\Omega_0} \frac{1}{2} \rho_0 \dot{\mathbf{u}}^2 dV, \quad (76)$$

where $\rho_0(\mathbf{X}) \in \Omega_0$ is the (time-independent) reference mass density.

As we known from the stationary energy principle, the vanishing variation of L with the imposed restrictions (72) gives the *principle of virtual work* for all $\delta \mathbf{u}$ which are zero on the part $\partial \Omega_{0u} \subset \partial \Omega_0$ of the boundary surface where the displacement field is prescribed throughout the entire closed time interval $t \in [t_0, t_1]$.

Simo-Taylor-Pister variational principle. The principle of virtual work is *not* the appropriate variational approach for invoking kinematic constraint conditions such as incompressibility ($J \equiv 1$) frequently arise in soft tissue biomechanics. Within variational methods, constraint conditions may be considered by introducing Lagrange multipliers in the associated functional (see Washizu (1982)).

Here we recall the well-established variational principle proposed by Simo et al. (1985), which will result in a simple but efficient finite element formulation. It takes account of nearly incompressible response and is based on the kinematic assumption

$$\tilde{\mathbf{F}} = \tilde{J}^{1/3} \bar{\mathbf{F}}. \quad (77)$$

Here, $\tilde{J}$ is a kinematic field variable which enforces the kinematic constraint $\tilde{J} = J(\mathbf{u})$ ($J > 0$ is the local volume ratio), and $\bar{\mathbf{F}} = J^{-1/3} \mathbf{F}$ is the modified deformation gradient. The deformation measure is given by $\tilde{\mathbf{C}} = \tilde{\mathbf{F}}^T \tilde{\mathbf{F}} = \tilde{J}^{2/3} \bar{\mathbf{C}}(\mathbf{u})$, where $\bar{\mathbf{C}} = J^{-2/3} \mathbf{C}$ is the modified right Cauchy-Green tensor as introduced in Section 2.1. Thus, for the strain-energy function Ψ we may write

$$\Psi = U(\tilde{J}) + \bar{\Psi}(\bar{\mathbf{C}}(\mathbf{u})). \quad (78)$$

The three-field variational principle of Simo et al. (1985), which is of *Hu-Washizu's* type, treats the displacement and pressure fields $\mathbf{u}$ and p , and the kinematic field variable $\tilde{J}$ as *independent* variables. Hence, we extend the Lagrangian function (71)₂ to the Lagrangian function L_{STP} , which takes on the form

$$L_{\text{STP}}(\mathbf{u}, p, \tilde{J}) = \int_{\Omega_0} [U(\tilde{J}) + p(J(\mathbf{u}) - \tilde{J}) + \bar{\Psi}(\bar{\mathbf{C}}(\mathbf{u}))] dV + \Pi^{\text{ext}}(\mathbf{u}) - \mathcal{K}(\dot{\mathbf{u}}), \quad (79)$$

where eqs. (74), (75)₁ and (78) have been used. The constraint condition $J = \tilde{J}$ is enforced by p . A suitable candidate for the function U is

$$U(\tilde{J}) = \frac{\kappa}{2} (\tilde{J} - 1)^2, \quad (80)$$

with $U(1) = 0$. Note that for the incompressible case the (positive) parameter $\kappa > 0$ serves as a user-specified *penalty* parameter which has no physical relevance. Since $\kappa > 0$, $dU^2(\tilde{J})/d^2\tilde{J}$ is positive, and hence U is convex. It is worthwhile mentioning that the Lagrangian function (79) is often *augmented* by a term of the form $\Pi^{\text{aug}}(\tilde{J}; \lambda) = \int_{\Omega_0} \lambda h(\tilde{J}) dV$, where $h(\tilde{J})$ is a

continuously differentiable function that satisfies $h(1) = 0$ (for example, $h = (\tilde{J} - 1)$), and λ is a Lagrange multiplier. According to Simo and Taylor (1991), λ may be determined by a standard update procedure of the form $\lambda \leftarrow \lambda + \kappa h(\tilde{J})$ until the magnitude of h is less than a given tolerance of accuracy. The augmented term enforces the incompressibility constraint and prevents the global stiffness matrix from becoming increasingly ill-conditioned for increasing κ , a problem known from the penalty method.

In addition to the virtual displacement field $\delta \mathbf{u}$ we now introduce arbitrary smooth functions δp and $\delta \tilde{J}$, which may be interpreted as the *virtual pressure field* and the *virtual volume change*, respectively. Hence, the necessary stationary conditions for the function L_{STP} with respect to the three field variables $(\mathbf{u}, p, \tilde{J})$ are evaluated separately. We require the stationary conditions

$$D_{\delta \mathbf{u}} \int_{t_0}^{t_1} L_{\text{STP}} dt = 0, \quad D_{\delta p} \int_{t_0}^{t_1} L_{\text{STP}} dt = 0, \quad D_{\delta \tilde{J}} \int_{t_0}^{t_1} L_{\text{STP}} dt = 0, \quad (81)$$

for all $\delta \mathbf{u}$ satisfying $\delta \mathbf{u} = \mathbf{0}$ on the part $\partial \Omega_{0,u}$ of the boundary surface $\partial \Omega_0$ throughout the entire time interval $t \in [t_0, t_1]$, for all $\delta \mathbf{u}$ that vanish at times t_0 and t_1 at all points of the body (compare with eq. (72)), and all $\delta p, \delta \tilde{J}$.

In eqs. (81) we have introduced the concept of *directional derivative* and used the equivalent representation of the first variation and the directional derivative of a functional, Φ say, according to

$$\delta \Phi(\mathbf{x}, \delta \mathbf{u}) = D_{\delta \mathbf{u}} \Phi(\mathbf{x}) = \frac{d}{d\varepsilon} \Phi(\mathbf{x} + \varepsilon \delta \mathbf{u})|_{\varepsilon=0}, \quad (82)$$

where $D(\bullet)$ is the *Gâteaux operator* and ε is a scalar parameter. We say that $D_{\delta \mathbf{u}} \Phi(\mathbf{x})$ is the directional derivative of Φ at given $\mathbf{x}$ (fixed) in the direction of the virtual displacement field $\delta \mathbf{u}$. By means of the chain rule we immediately find from definition (82) that $D_{\delta \mathbf{u}} \Phi = (\partial \Phi / \partial \mathbf{x}) \cdot \delta \mathbf{u}$.

Next, we differentiate L_{STP} , as provided in eq. (79), with respect to changes in $\mathbf{u}$. By using the commutative property that the order of the directional derivative and the definite integral is interchangeable, then eq. (81)₁ and the analogue of eq. (7)₃, we find from (81)₁ that

$$\int_{t_0}^{t_1} \int_{\Omega_0} (Jp\mathbf{C}^{-1} + \bar{\mathbf{S}}) : \frac{1}{2} D_{\delta \mathbf{u}} \mathbf{C} dV dt + \int_{t_0}^{t_1} D_{\delta \mathbf{u}} II^{\text{ext}}(\mathbf{u}) dt - \int_{t_0}^{t_1} D_{\delta \mathbf{u}} \mathcal{K}(\dot{\mathbf{u}}) dt = 0, \quad (83)$$

where $D_{\delta \mathbf{u}} \mathbf{C}$ characterizes the directional derivative of $\mathbf{C}$ in the direction of $\delta \mathbf{u}$, and $\bar{\mathbf{S}}$ represents the purely isochoric contribution to the second Piola-Kirchhoff stress $\mathbf{S}$. From expression (75)₂ we find that the second term in eq. (83) has the form

$$\int_{t_0}^{t_1} D_{\delta \mathbf{u}} II^{\text{ext}}(\mathbf{u}) dt = - \int_{t_0}^{t_1} \int_{\Omega_0} \mathbf{B} \cdot \delta \mathbf{u} dV dt - \int_{t_0}^{t_1} \int_{\partial \Omega_{0,\sigma}} \bar{\mathbf{T}} \cdot \delta \mathbf{u} dS dt. \quad (84)$$

Next, we specify the third term in eq. (83) by using the kinetic energy (76) and interchanging the order of the directional derivative and the definite integral. Subsequently, integration by parts

gives

$$\begin{aligned}
-\int_{t_0}^{t_1} D_{\delta \mathbf{u}} \mathcal{K}(\dot{\mathbf{u}}) dt &= -\int_{t_0}^{t_1} D_{\delta \mathbf{u}} \int_{\Omega_0} \frac{1}{2} \rho_0 \dot{\mathbf{u}}^2 dV dt = -\int_{t_0}^{t_1} \int_{\Omega_0} \rho_0 \dot{\mathbf{u}} \cdot D_{\delta \mathbf{u}} \dot{\mathbf{u}} dV dt \\
&= -\int_{\Omega_0} \rho_0 \dot{\mathbf{u}} \cdot D_{\delta \mathbf{u}} \mathbf{u} dV \Big|_{t_0}^{t_1} + \int_{t_0}^{t_1} \int_{\Omega_0} \rho_0 \ddot{\mathbf{u}} \cdot D_{\delta \mathbf{u}} \mathbf{u} dV dt. \tag{85}
\end{aligned}$$

With the equivalent notation $D_{\delta \mathbf{u}} \mathbf{u} = \delta \mathbf{u}$ and restriction (72) we deduce from (85)₃ that

$$-\int_{t_0}^{t_1} D_{\delta \mathbf{u}} \mathcal{K}(\dot{\mathbf{u}}) dt = \int_{t_0}^{t_1} \int_{\Omega_0} \rho_0 \ddot{\mathbf{u}} \cdot \delta \mathbf{u} dV dt. \tag{86}$$

Finally, with eqs. (84), (86) and the equivalent notation $D_{\delta \mathbf{u}} \mathbf{C} = \delta \mathbf{C}$ we may rewrite eq. (83) in the variational form

$$\int_{\Omega_0} [(Jp\mathbf{C}^{-1} + \bar{\mathbf{S}}) : \frac{1}{2} \delta \mathbf{C} - (\mathbf{B} - \rho_0 \ddot{\mathbf{u}}) \cdot \delta \mathbf{u}] dV - \int_{\partial \Omega_0 \sigma} \bar{\mathbf{T}} \cdot \delta \mathbf{u} dS = 0, \tag{87}$$

with the condition $\delta \mathbf{u} = \mathbf{0}$ imposed on the boundary surface $\partial \Omega_{0u} \subset \partial \Omega_0$. Note that eq. (87) holds for each $t \in [t_0, t_1]$.

A straightforward differentiation of L_{STP} with respect to changes in the field variables p and $\tilde{J}$ gives the weak enforcement of the equivalence between J and $\tilde{J}$, and the expression for the volumetric changes. Hence, from eqs. (81)₂ and (81)₃ we find the remaining two Euler-Lagrange equations in weak form

$$\int_{\Omega_0} (J(\mathbf{u}) - \tilde{J}) \delta p dV = 0, \tag{88}$$

$$\int_{\Omega_0} \left(\frac{dU(\tilde{J})}{d\tilde{J}} - p \right) \delta \tilde{J} dV = 0, \tag{89}$$

which hold for each $t \in [t_0, t_1]$. Note that for arbitrary δp , the variational equation (88) results in the local form $J = \tilde{J}$ of the kinematic constraint condition. For arbitrary $\delta \tilde{J}$, eq. (89) gives the standard constitutive equation $p = dU/d\tilde{J}$.

For subsequent use it is beneficial to express the stationary condition (87) in terms of spatial coordinates. Before proceeding we assume that region Ω_0 moves to a new region Ω occupied by $\mathcal{B}$ at a subsequent time $t > 0$. It is bounded by a current boundary surface $\partial \Omega$, which is decomposed into disjoint parts so that $\partial \Omega = \partial \Omega_u \cup \partial \Omega_\sigma$ with $\partial \Omega_u \cap \partial \Omega_\sigma = \emptyset$. According to, for example, Holzapfel (2000), Sections 8.2, 4.3, we transform now the material fields $\delta \mathbf{C}$ and $\mathbf{B}$ into the associated spatial fields $\delta \mathbf{e}$ and $\mathbf{b}$. This yields,

$$\frac{1}{2} \delta \mathbf{C} = \mathbf{F}^T \delta \mathbf{e} \mathbf{F}, \quad \mathbf{B} = J \mathbf{b}, \tag{90}$$

where $\mathbf{b}$ denotes the body force with respect to the current position $\mathbf{x}$ (force measured per unit current volume), and

$$\delta \mathbf{e} = \frac{1}{2}(\text{grad}^T \delta \mathbf{u} + \text{grad} \delta \mathbf{u}) = \text{sym}(\text{grad} \delta \mathbf{u}), \quad (91)$$

is the first variation of the Euler-Almansi strain tensor $\mathbf{e}$. Note that the symbol $\mathbf{b}$ should not be confused with the strain tensor introduced in eq. (5)₁. In relation (91), $\text{grad}(\bullet)$ denotes the *spatial gradient* of $(\bullet)$ (lowercase g), which is the derivative of $(\bullet)$ with respect to the current position $\mathbf{x}$, at a fixed time t , and the notation $\text{sym}(\bullet)$ is used to indicate the symmetric part of $(\bullet)$.

The change in volume between the current and the reference configuration and the relation between the reference mass density and the spatial mass density, denoted by $\rho(\mathbf{x}, t)$, at time t , is given by

$$dv = J(\mathbf{X}, t)dV, \quad \rho_0(\mathbf{X}) = J(\mathbf{X}, t)\rho(\mathbf{x}, t). \quad (92)$$

Here, dv defines the infinitesimal volume element in the current configuration.

With relations (90)-(92) we deduce from (87) the corresponding function in terms of spatial quantities, i.e.

$$\int_{\Omega} [(p\mathbf{I} + \bar{\boldsymbol{\sigma}}) : \delta \mathbf{e} - (\mathbf{b} - \rho \ddot{\mathbf{u}}) \cdot \delta \mathbf{u}] dv - \int_{\partial \Omega_{\sigma}} \bar{\mathbf{t}} \cdot \delta \mathbf{u} ds = 0, \quad (93)$$

where $\bar{\boldsymbol{\sigma}} = J^{-1} \mathbf{F} \bar{\mathbf{S}} \mathbf{F}^T$ defines the purely isochoric stress contribution to the Cauchy stress tensor $\boldsymbol{\sigma}$. The virtual displacement field $\delta \mathbf{u}$ is here defined on the *current* configuration (we require that $\delta \mathbf{u}$ vanishes on $\partial \Omega_u$). The prescribed Cauchy traction vector (force measured per unit current surface area s), denoted by $\bar{\mathbf{t}}$, is specified on the portion $\partial \Omega_{\sigma} \subset \partial \Omega$.

For subsequent use it is useful to rephrase eqs. (87) and (93) in the more convenient form

$$\delta \tilde{W} = \delta \tilde{W}^{\text{int}} - \delta W^{\text{ext}} + \delta W^{\text{kin}} = 0, \quad (94)$$

in which we have introduced the definitions (in the Lagrangian and Eulerian descriptions)

$$\delta \tilde{W}^{\text{int}} = \int_{\Omega} (p\mathbf{I} + \bar{\boldsymbol{\sigma}}) : \delta \mathbf{e} dv = \int_{\Omega_0} (Jp\mathbf{C}^{-1} + \bar{\mathbf{S}}) : \frac{1}{2} \delta \mathbf{C} dV, \quad (95)$$

$$\delta W^{\text{ext}} = \delta W_b^{\text{ext}} + \delta W_{\sigma}^{\text{ext}}, \quad \delta W_b^{\text{ext}} = \int_{\Omega} \mathbf{b} \cdot \delta \mathbf{u} dv = \int_{\Omega_0} \mathbf{B} \cdot \delta \mathbf{u} dV, \quad (96)$$

$$\delta W_{\sigma}^{\text{ext}} = \int_{\partial \Omega_{\sigma}} \bar{\mathbf{t}} \cdot \delta \mathbf{u} ds = \int_{\partial \Omega_0_{\sigma}} \bar{\mathbf{T}} \cdot \delta \mathbf{u} dS, \quad (97)$$

$$\delta W^{\text{kin}} = \int_{\Omega} \rho \ddot{\mathbf{u}} \cdot \delta \mathbf{u} dv = \int_{\Omega_0} \rho_0 \ddot{\mathbf{u}} \cdot \delta \mathbf{u} dV, \quad (98)$$

of the *modified internal virtual work* $\delta \tilde{W}_{\text{int}}$, the *external virtual work* $\delta W^{\text{ext}} = -D_{\delta \mathbf{u}} \Pi^{\text{ext}}(\mathbf{u})$ and the *kinetic virtual work* δW^{kin} . We call the nonlinear variational equation (93), or (94), the *modified virtual work equation*.

It is important to note that the second-order differential equations themselves require additional data in the form of *initial conditions*. We agree that the initial configuration, i.e. a region at *initial time* $t = 0$, coincides with the reference configuration; hence, the reference time is at $t = 0$. The displacement field $\mathbf{u}|_{t=0}$ and the velocity field $\dot{\mathbf{u}}|_{t=0}$ at the initial time $t = 0$ are specified in the weak forms

$$\left. \begin{aligned} \int_{\Omega} \mathbf{u}(\mathbf{x}, t)|_{t=0} \cdot \delta \mathbf{u} dv &= \int_{\Omega} \mathbf{u}_0(\mathbf{X}) \cdot \delta \mathbf{u} dv, \\ \int_{\Omega} \dot{\mathbf{u}}(\mathbf{x}, t)|_{t=0} \cdot \delta \mathbf{u} dv &= \int_{\Omega} \dot{\mathbf{u}}_0(\mathbf{X}) \cdot \delta \mathbf{u} dv, \end{aligned} \right\} \quad (99)$$

where $(\bullet)_0$ denotes a prescribed function in Ω_0 . This set of initial conditions together with eqs. (88), (89) and (93) characterizes the weak form (or variational form) of the nonlinear *initial boundary-value problem*.

If the local forms $\tilde{J} = J$ and $p = dU/d\tilde{J}$ are satisfied (compare with eqs. (88) and (89)), then p is identified as the hydrostatic pressure and eq. (93) characterizes precisely the virtual work equation in the spatial description. Thereby, the term $p\mathbf{I}$ determines the purely *volumetric* contribution σ_{vol} to the Cauchy stress tensor σ . Hence, for vanishing acceleration $\ddot{\mathbf{u}}$, the internal virtual work, then governed by $\delta W^{\text{int}} = \int_{\Omega} \sigma : \delta \mathbf{e} dv$, with $\sigma = (dU/dJ)\mathbf{I} + \bar{\sigma}$, equals the external virtual work δW^{ext} , so that $\delta W = 0$.

4.2 Finite Element Discretization

The aim of this section is to *spatially* discretize (approximate) the described continuous problem of solid mechanics (the *time* remains continuous at this point). In particular, we derive the discrete versions of the three nonlinear Euler-Lagrange equations (88), (89) and (93) in weak form. For establishing the discrete quantities we prefer subsequently to use the Eulerian description.

We subdivide the current domain Ω and the portion $\partial\Omega_{\sigma} \subset \partial\Omega$ of the boundary surface $\partial\Omega$ of a continuum body $\mathcal{B}$ into n_e finite elements with domain Ω_e and n_b element surfaces with domain $\partial\Omega_{\sigma b}$ on which traction vectors are specified. We write

$$\Omega \approx \mathbf{A}_{e=1}^{n_e} \Omega_e, \quad \partial\Omega_{\sigma} \approx \mathbf{A}_{b=1}^{n_b} \partial\Omega_{\sigma b}, \quad (100)$$

where $\mathbf{A}$ denotes an operator which assembles the elements. Subsequently, the subscript $(\bullet)_e$ is an index (running between 1 and n_e) which will refer to a typical finite element Ω_e , and the subscript $(\bullet)_b$ (running between 1 and n_b) which will refer to a typical element surface $\partial\Omega_{\sigma b}$.

Discrete kinematics. We prefer to use isoparametric elements and interpolate the geometry of the reference and current configurations according to

$$\mathbf{X}_h = \sum_{I=1}^{n_{\text{node}}} N_I(\xi) \mathbf{X}_I, \quad \mathbf{x}_h = \sum_{I=1}^{n_{\text{node}}} N_I(\xi) \mathbf{x}_I(t), \quad (101)$$

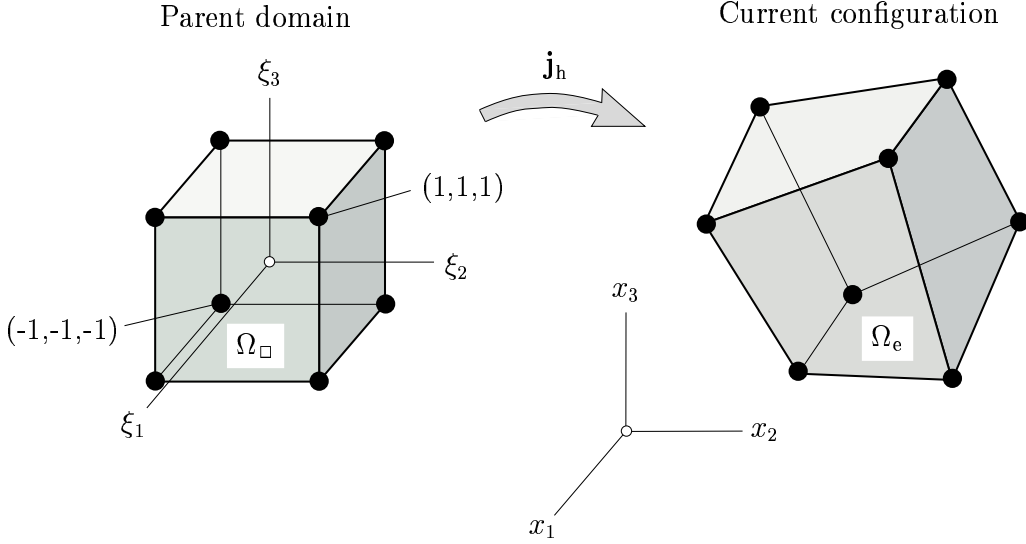


Figure 7. Three-dimensional element in the current configuration and the parent domain.

where the subscript $(\bullet)_h$ indicates the discrete (finite-dimensional) counterpart to quantity $(\bullet)$. The reference and current positions of the element node I are denoted by $\mathbf{X}_I$ and $\mathbf{x}_I$, respectively. The subscript $(\bullet)_I$ is an index running between 1 and the total number of element nodes, denoted by n_{node} . In eqs. (101), $\boldsymbol{\xi} = \{\xi_1, \xi_2, \xi_3\} \in \Omega_\square$ are the local element coordinates (the natural coordinates), where $\Omega_\square = \{\boldsymbol{\xi} \in \mathbb{R}^3 | (-1, 1) \times (-1, 1) \times (-1, 1)\}$ characterize the domain of the *parent element*, i.e. a biunit cube, as illustrated in Figure 7. The isoparametric *interpolation* (or *shape*) function associated with node I , is denoted by $N_I(\boldsymbol{\xi})$ and defined in $\Omega_\square$ (for a more detailed explanation of these standard concepts, see, for example, Hughes (2000)). Now we shall assume that the interpolation functions are *tri-linear* and expressed in the form $N_I(\boldsymbol{\xi}) = \frac{1}{8}(1 + \xi_{1I}\xi_1)(1 + \xi_{2I}\xi_2)(1 + \xi_{3I}\xi_3)$, $I = 1, \dots, n_{\text{node}} = 8$.

With eqs. (101) it is straightforward to compute the discrete displacement $\mathbf{u}_h = \mathbf{x}_h - \mathbf{X}_h$ and the acceleration $\ddot{\mathbf{u}}_h$, i.e.

$$\mathbf{u}_h = \sum_{I=1}^{n_{\text{node}}} N_I(\boldsymbol{\xi}) \mathbf{u}_I(t), \quad \ddot{\mathbf{u}}_h = \sum_{I=1}^{n_{\text{node}}} N_I(\boldsymbol{\xi}) \ddot{\mathbf{u}}_I(t), \quad (102)$$

where the nodal displacements and the nodal accelerations are denoted by $\mathbf{u}_I(t) = \mathbf{x}_I(t) - \mathbf{X}_I$ and $\ddot{\mathbf{u}}_I(t) = \ddot{\mathbf{x}}_I(t)$, respectively. As can be seen from eqs. (101) and (102), a specific element is characterized by the positions and the degrees of freedom associated with the element nodes I attached to it.

In addition, the interpolations of the virtual displacement $\delta \mathbf{u}$, and the related spatial gradient $\text{grad} \delta \mathbf{u}$ and spatial divergence $\text{div} \delta \mathbf{u}$ give the discrete forms

$$\delta \mathbf{u}_h = \sum_{I=1}^{n_{\text{node}}} N_I(\boldsymbol{\xi}) \delta \mathbf{u}_I, \quad (103)$$

$$\text{grad} \delta \mathbf{u}_h = \sum_{I=1}^{n_{\text{node}}} \delta \mathbf{u}_I \otimes \nabla_{\mathbf{x}} N_I(\boldsymbol{\xi}), \quad (104)$$

$$\text{div} \delta \mathbf{u}_h = \sum_{I=1}^{n_{\text{node}}} \delta \mathbf{u}_I \cdot \nabla_{\mathbf{x}} N_I(\boldsymbol{\xi}), \quad (105)$$

where the standard expressions

$$\nabla_{\mathbf{x}} N_I(\boldsymbol{\xi}) = \frac{\partial N_I(\boldsymbol{\xi})}{\partial \mathbf{x}_h} = \mathbf{j}_h^{-T} \frac{\partial N_I}{\partial \boldsymbol{\xi}}, \quad \mathbf{j}_h = \frac{\partial \mathbf{x}_h}{\partial \boldsymbol{\xi}} = \sum_{I=1}^{n_{\text{node}}} \mathbf{x}_I \otimes \nabla_{\boldsymbol{\xi}} N_I \quad (106)$$

are to be used. Therein, $\nabla_{(\bullet)} N_I$ denotes the gradient of the scalar function N_I with respect to the coordinates $(\bullet)$ and $\mathbf{j}_h$ is the *Jacobian operator*, which transforms the gradient $\nabla_{\boldsymbol{\xi}} N_I$ to the gradient $\nabla_{\mathbf{x}} N_I$. The utilization of the *same* interpolation functions to discretize $\mathbf{u}$ and $\delta \mathbf{u}$ is called the *Bubnov-Galerkin* (finite element) *method*.

According to (91), the discretization of the variation of the Euler-Almansi strain tensor $\delta \mathbf{e} = \text{sym}(\text{grad} \delta \mathbf{u})$ within a typical finite element domain may be found with representation (104) as

$$\delta \mathbf{e}_h = \text{sym}(\text{grad} \delta \mathbf{u}_h) = \frac{1}{2} \sum_{I=1}^{n_{\text{node}}} (\delta \mathbf{u}_I \otimes \nabla_{\mathbf{x}} N_I + \nabla_{\mathbf{x}} N_I \otimes \delta \mathbf{u}_I). \quad (107)$$

In addition, for the independent volumetric variables $\tilde{J}$, p , and their variations $\delta \tilde{J}$, δp , we use the same *constant* (discontinuous) interpolation functions over a given element domain without having to satisfy continuity across the element boundaries (Simo et al. (1985)). Thus, we write

$$p_h = \bar{p}, \quad \delta p_h = \delta \bar{p}, \quad \tilde{J}_h = \tilde{J}, \quad \delta \tilde{J}_h = \delta \tilde{J}, \quad (108)$$

where the symbol $(\bullet)$ denotes the constant interpolation function. This type of mixed formulation is known as the mean dilatation technique, leading to the $Q1/P0$ -element, a procedure which goes back to Nagtegaal et al. (1974). This approach may be regarded as the nonlinear extension of the B -bar method, as proposed by Hughes (2000). It turns out that this approach leads to a finite element formulation which prevents ‘locking’ phenomena associated with the volume-preserving condition on the motion. For practical use the $Q1/P0$ -element is quite robust and is therefore implemented in several commercially available finite element programs.

Discrete variational equations. We refer to eq. (94) and study the three terms $\delta \tilde{W}^{\text{int}}$, δW^{ext} and δW^{kin} , i.e. (95)-(98), in more detail. For the three contributions to the modified virtual work $\delta \tilde{W}$ we have the discrete forms

$$\delta \tilde{W}^{\text{int}} \approx \sum_{e=1}^{n_e} \delta \tilde{W}_e^{\text{int}}, \quad (109)$$

$$\delta W^{\text{ext}} \approx \sum_{e=1}^{n_e} \delta W_{b_e}^{\text{ext}} + \sum_{b=1}^{n_b} \delta W_{\sigma_e}^{\text{ext}}, \quad (110)$$

$$\delta W^{\text{kin}} \approx \sum_{e=1}^{n_e} \delta W_e^{\text{kin}}, \quad (111)$$

which constitute the basis for the finite element formulation.

Just for clarity, we consider subsequently the virtual work contributions caused by a typical finite element with domain Ω_e and surface $\partial\Omega_{\sigma_b}$ (on which traction vectors are specified). We start with the modified internal virtual work $\delta\tilde{W}_e^{\text{int}}$, which may be derived from definition (95)₁. By means of the interpolations (107)₂ and (108)₁, the basic properties of double contraction and the symmetry of the Cauchy stress tensor, we obtain

$$\begin{aligned} \delta\tilde{W}_e^{\text{int}} &= \int_{\Omega_e} (p_h \mathbf{I} + \bar{\boldsymbol{\sigma}}_h) : \delta \mathbf{e}_h dv = \sum_{I=1}^{n_{\text{node}}} \int_{\Omega_e} (\bar{p} \mathbf{I} + \bar{\boldsymbol{\sigma}}_h) : (\delta \mathbf{u}_I \otimes \nabla_{\mathbf{x}} N_I) dv \\ &= \sum_{I=1}^{n_{\text{node}}} \delta \mathbf{u}_I \cdot \int_{\Omega_e} (\bar{p} \mathbf{I} + \bar{\boldsymbol{\sigma}}_h) \nabla_{\mathbf{x}} N_I dv. \end{aligned} \quad (112)$$

The Cauchy stress tensor $\bar{\boldsymbol{\sigma}}_h$ is a function of $\mathbf{u}_h$ through the respective constitutive relation.

By means of the discrete equation (103), the external virtual work for a finite element Ω_e with surface $\partial\Omega_{\sigma_b}$ results from definitions (96) and (97) as

$$\delta W_{b_e}^{\text{ext}} = - \int_{\Omega_e} \mathbf{b} \cdot \delta \mathbf{u}_h dv = - \sum_{I=1}^{n_{\text{node}}} \delta \mathbf{u}_I \cdot \int_{\Omega_e} N_I \mathbf{b} dv, \quad (113)$$

$$\delta W_{\sigma_e}^{\text{ext}} = - \int_{\partial\Omega_{\sigma_b}} \bar{\mathbf{t}} \cdot \delta \mathbf{u}_h ds = - \sum_{I=1}^{b_{\text{node}}} \delta \mathbf{u}_I \cdot \int_{\partial\Omega_{\sigma_b}} N_I \bar{\mathbf{t}} ds, \quad (114)$$

with the given boundary conditions $\mathbf{b}$ and $\bar{\mathbf{t}}$. Note that for the term $\delta W_{\sigma_e}^{\text{ext}}$ the interpolation functions are to be used for an element, which is reduced by one space dimension. Thereby, the interpolation functions are denoted by N_I , $I = 1, \dots, b_{\text{node}}$.

We now generate the kinetic virtual work δW_e^{kin} within a typical finite element domain by substitution of the interpolations (102)₂ and (103) into definition (98)₁, causing

$$\delta W_e^{\text{kin}} = \int_{\Omega_e} \rho \ddot{\mathbf{u}}_h \cdot \delta \mathbf{u}_h dv = \sum_{I=1}^{n_{\text{node}}} \sum_{K=1}^{n_{\text{node}}} \delta \mathbf{u}_I \cdot \mathbf{I} \int_{\Omega_e} N_I \rho N_K dv \ddot{\mathbf{u}}_K. \quad (115)$$

Finally, the initial conditions, which are the *given* displacement and velocity fields, are expressed in the discrete forms as

$$\left. \begin{aligned} \mathbf{u}_I|_{t=0} &= \mathbf{u}_0(\mathbf{X}_I), & I &= 1, \dots, n_{\text{node}}, \\ \dot{\mathbf{u}}_I|_{t=0} &= \dot{\mathbf{u}}_0(\mathbf{X}_I), & I &= 1, \dots, n_{\text{node}} \end{aligned} \right\} \quad (116)$$

(rather than enforcing (99)), where $\mathbf{X}_I$ characterizes the referential position of element node I .

The two equations (88) and (89) have to be satisfied locally, since the interpolation functions used for the (independent) volumetric variables $\tilde{J}$ and p are constant (discontinuous). Therefore, with the four quantities provided in eq. (108), we find that the discretizations for the volumetric variables have the forms

$$\tilde{J} = \frac{1}{\int_{\Omega_{0e}} dV} \int_{\Omega_{0e}} J_h dV = \frac{v_e}{V_e}, \quad (117)$$

$$\bar{p} = \frac{1}{\int_{\Omega_{0e}} dV} \int_{\Omega_{0e}} \frac{dU(\tilde{J})}{d\tilde{J}} dV = \left. \frac{dU(\tilde{J})}{d\tilde{J}} \right|_{\tilde{J}=v_e/V_e}, \quad (118)$$

where v_e and V_e are element volumes in the current and reference configurations, respectively. They are defined by $\int_{\Omega_{0e}} J_h dV = \int_{\Omega_e} dv = v_e$ and $\int_{\Omega_{0e}} dV = V_e$. Hence, both volumetric variables can be eliminated at the element level (static condensation), and the expression (118)₂ for the constant pressure $\bar{p}$ (with $\tilde{J}$ determined in (117)₂) is therefore used in eq. (112)₃.

Finite element discretization in matrix notation; Semidiscrete nonlinear equations of motion. The discrete equations (112)₃, (113)₂, (114)₂, (115)₂, which are given in tensorial forms, are now recast in the more standard *matrix notation*.

The analogues of eqs. (102)₂ and (103) in matrix form are $\ddot{\mathbf{u}}_h = \sum_{I=1}^{n_{\text{node}}} N_I(\boldsymbol{\xi}) \ddot{\mathbf{u}}_I(t)$ and $\delta \mathbf{u}_h = \sum_{I=1}^{n_{\text{node}}} N_I(\boldsymbol{\xi}) \delta \mathbf{u}_I(t)$, where $\ddot{\mathbf{u}}_I$ and $\delta \mathbf{u}_I$ denote 3×1 *column matrices* of the form

$$\ddot{\mathbf{u}}_I = [\ddot{u}_1, \ddot{u}_2, \ddot{u}_3]_I^T, \quad \delta \mathbf{u}_I = [\delta u_1, \delta u_2, \delta u_3]_I^T. \quad (119)$$

They contain the components $\{\ddot{u}_1, \ddot{u}_2, \ddot{u}_3\}_I$ of the acceleration vector $\ddot{\mathbf{u}}_I$ and the components $\{\delta u_1, \delta u_2, \delta u_3\}_I$ of the virtual displacement vector $\delta \mathbf{u}_I$ for node I .

Subsequently, we represent the first variation of the *symmetric* Euler-Almansi strain tensor $\delta \mathbf{e}_h$ within a typical finite element domain by the 6×1 column matrix, denoted by $\delta \mathbf{e}_h$. It is of the form

$$\delta \mathbf{e}_h = [\delta e_{11}, \delta e_{22}, \delta e_{33}, 2\delta e_{12}, 2\delta e_{23}, 2\delta e_{13}]_h^T, \quad (120)$$

where, for the three-dimensional case, the (six) independent components of the tensor $\delta \mathbf{e}_h$ form the entries of the column matrix $\delta \mathbf{e}_h$. With this convention we may express the discrete equation

(107)₂ in the standard matrix form

$$\delta \mathbf{e}_h = \sum_{I=1}^{n_{\text{node}}} \mathbf{B}_I \delta \mathbf{u}_I, \quad \mathbf{B}_I = \begin{bmatrix} \frac{\partial N_I}{\partial x_1} & 0 & 0 \\ 0 & \frac{\partial N_I}{\partial x_2} & 0 \\ 0 & 0 & \frac{\partial N_I}{\partial x_3} \\ \frac{\partial N_I}{\partial x_2} & \frac{\partial N_I}{\partial x_1} & 0 \\ 0 & \frac{\partial N_I}{\partial x_3} & \frac{\partial N_I}{\partial x_2} \\ \frac{\partial N_I}{\partial x_3} & 0 & \frac{\partial N_I}{\partial x_1} \end{bmatrix}. \quad (121)$$

The 6×3 matrix $\mathbf{B}_I$ is associated with node I , consists of the derivatives of the interpolation functions N_I with respect to the *spatial* (current) coordinates x_a , and is, therefore, configuration dependent.

By analogy with eq. (120) the isochoric contribution $\bar{\boldsymbol{\sigma}}_h$ to the symmetric Cauchy stress tensor of a typical element is given by the 6×1 column matrix $\bar{\boldsymbol{\sigma}}_h$ of the form

$$\bar{\boldsymbol{\sigma}}_h = [\bar{\sigma}_{11}, \bar{\sigma}_{22}, \bar{\sigma}_{33}, \bar{\sigma}_{12}, \bar{\sigma}_{23}, \bar{\sigma}_{13}]_h^T, \quad (122)$$

where the (six) independent components of the tensor $\bar{\boldsymbol{\sigma}}_h$ form the entries of the column matrix $\bar{\boldsymbol{\sigma}}_h$. The second-order unit tensor $\mathbf{I}$ is represented by the matrix $\mathbf{I}$.

By means of (112)₁ and the interpolations (108)₁ and (121)₁ we find from the approximation (109) the modified internal virtual work $\delta \tilde{W}^{\text{int}}$ for the region $\Omega \approx \mathbf{A}_{e=1}^{n_e} \Omega_e$, i.e.

$$\delta \tilde{W}^{\text{int}} \approx \mathbf{A}_{e=1}^{n_e} \sum_{I=1}^{n_{\text{node}}} \delta \mathbf{u}_I^T \mathbf{F}_I^{\text{int}}(\mathbf{u}_e) = \delta \mathbf{u}^T \mathbf{F}^{\text{int}}(\mathbf{u}), \quad (123)$$

$$\text{with} \quad \mathbf{F}_I^{\text{int}}(\mathbf{u}_e) = \int_{\Omega_e} \mathbf{B}_I^T (\bar{p} \mathbf{I} + \bar{\boldsymbol{\sigma}}_h) dv, \quad (124)$$

where $\mathbf{u}_e$ contains the components of the displacement vector associated with element e . The constant pressure $\bar{p}$ is given explicitly by (118)₂, while the matrices $\bar{\boldsymbol{\sigma}}_h$, $\mathbf{B}_I$ and $\delta \mathbf{u}_I$ have the forms given in (122), (121)₂ and (119)₂, respectively. Note that the abbreviation $\mathbf{F}_I^{\text{int}}$ defines the internal nodal forces, while the global column matrices $\mathbf{F}^{\text{int}}$ and $\delta \mathbf{u}$ define the internal forces and the virtual displacements for all nodes of the finite element mesh, respectively.

By means of the expressions (113)₁ and (114)₁ we find from the approximation (110) the external virtual work $\delta \tilde{W}^{\text{ext}}$, i.e.

$$\delta \tilde{W}^{\text{ext}} \approx - \mathbf{A}_{e=1}^{n_e} \sum_{I=1}^{n_{\text{node}}} \delta \mathbf{u}_I^T \mathbf{F}_I^{\text{ext}} - \mathbf{A}_{b=1}^{n_b} \sum_{I=1}^{b_{\text{node}}} \delta \mathbf{u}_I^T \mathbf{F}_{\sigma I}^{\text{ext}} = \delta \mathbf{u}^T \mathbf{F}^{\text{ext}}, \quad (125)$$

$$\text{with} \quad \mathbf{F}_I^{\text{ext}} = \int_{\Omega_e} N_I \mathbf{b} dv, \quad \mathbf{F}_{\sigma I}^{\text{ext}} = \int_{\partial\Omega_{\sigma b}} N_I \bar{\mathbf{t}} ds, \quad (126)$$

where the column matrices $\mathbf{b}$ and $\bar{\mathbf{t}}$ contain components of the body force and the surface traction, respectively. In eq. (125), the abbreviations $\mathbf{F}_I^{\text{ext}}$ and $\mathbf{F}_{\sigma I}^{\text{ext}}$ define external nodal forces, associated with $\mathbf{b}$ and $\bar{\mathbf{t}}$, while the global column matrix $\mathbf{F}^{\text{ext}}$ captures the loads which act on the continuum body occupying the region Ω with boundary surface $\partial\Omega$.

By means of the expression (115)₁ we find finally from the approximation (111) the kinetic virtual work $\delta\tilde{W}^{\text{kin}}$ for the region Ω in the form

$$\delta W^{\text{kin}} \approx \mathbf{A} \sum_{e=1}^{n_e} \sum_{I=1}^{n_{\text{node}}} \sum_{K=1}^{n_{\text{node}}} \delta \mathbf{u}_I^T \mathbf{M}_{IK} \ddot{\mathbf{u}}_K = \delta \mathbf{u}^T \mathbf{M} \ddot{\mathbf{u}}, \quad (127)$$

$$\text{with} \quad \mathbf{M}_{IK} = \mathbf{I} \int_{\Omega_e} N_I \rho N_K dv, \quad (128)$$

where $\ddot{\mathbf{u}}_K$ and $\delta \mathbf{u}_I$ are given by (119), and $\mathbf{M}_{IK}$ defines the *consistent mass matrix* related to the element domain Ω_e . The global mass matrix $\mathbf{M}$ and the global column matrix $\ddot{\mathbf{u}}$, which consists of accelerations, are calculated as an assembly of contributions arising from all finite elements n_e in the mesh. Note that, in general, the mass matrix will not be diagonal; however, for practical purposes $\mathbf{M}$ is often diagonalized, which leads to the *lumped mass matrix* (see, for example, Hinton et al. (1976) and Hughes (2000)). Note the equivalence of the tensor equations (112)-(115) and the matrix equations (123)-(128).

With the definitions of the internal and external nodal forces and the mass matrix (see the expressions (124), (126), (128)), and by means of the virtual work expressions (123)₂, (125)₂ and (127)₂ we obtain the associated form of the modified virtual work equation (94). Thus,

$$\delta \mathbf{u}^T \mathbf{R}(\mathbf{u}, t) = 0, \quad \mathbf{R}(\mathbf{u}, t) = \mathbf{M} \ddot{\mathbf{u}}(t) + \mathbf{F}^{\text{int}}(\mathbf{u}(t)) - \mathbf{F}^{\text{ext}}(t), \quad (129)$$

where residual forces form the entries of the column matrix $\mathbf{R}$, which is referred to briefly as a *residual*. It is important to note that some nodal displacements of the finite element mesh are known. Hence, for nodes I lying on the discrete boundary surface $\partial\Omega_u \subset \partial\Omega$ the vector $\mathbf{u}_I$ is given, while the corresponding values $\delta \mathbf{u}_I$ at these nodes are zero.

Since eq. (129) must hold for all vectors $\delta \mathbf{u}$, we obtain

$$\mathbf{R}(\mathbf{u}, t) = \mathbf{O}, \quad \mathbf{M} \ddot{\mathbf{u}}(t) + \mathbf{F}^{\text{int}}(\mathbf{u}(t)) = \mathbf{F}^{\text{ext}}(t), \quad (130)$$

which are the *semidiscrete equations of motion* (discrete in space and continuous in time). Here $\mathbf{F}^{\text{int}}$ is a nonlinear function of the displacement $\mathbf{u}$. Equations (130) form a system of N *non-linear* ordinary differential equations in the nodal displacements, where N is the number of *unprescribed* degrees of freedom. Those equations which correspond to the *prescribed* degrees of freedom are ignored. The second-order differential equations are supplemented by initial conditions for the given displacement and velocity fields at the instant of time $t = 0$, i.e.

$$\mathbf{u}|_{t=0} = \mathbf{u}_0, \quad \dot{\mathbf{u}}|_{t=0} = \dot{\mathbf{u}}_0. \quad (131)$$

The matrix equations (130) and (131) define the discrete problem for fully nonlinear elastodynamics.

If the data depend on time and the contributions due to dynamical quantities are negligible, we have a *quasi-static* problem and (130)₂ takes on the form

$$\mathbf{F}^{\text{int}}(\mathbf{u}(t)) = \mathbf{F}^{\text{ext}}(t), \quad (132)$$

with the initial condition $\mathbf{u}|_{t=0} = \mathbf{u}_0$. Hence, at each time t we seek a solution $\mathbf{u}(t)$ that makes the internal forces equal external forces. In contrast to the dynamic (transient) problem, the variable t in eq. (132) may not correspond to the actual time. This is the case, for example, when rate-independent problems are considered. Finally, note that if the data are independent of time the body is considered to be in static equilibrium and the set of equations reduces to the associated nonlinear *boundary-value problem* of elastostatics.

4.3 Consistent Linearization

In order to obtain solutions of the nonlinear initial boundary-value problem, as introduced above, *incremental/iterative solution techniques of Newton's type* are often applied to solve a sequence of linearized problems. This strategy implies a *consistent* linearization of the nonlinear equations with respect to the positions of the nodal points, which yields the corresponding geometrical and material stiffness matrices. The notion ‘consistent linearization’ means a linearization of *all* quantities associated with the nonlinear problem. The procedure of linearization is a very important task and a key element in computational mechanics. The underlying technique was first introduced in the mechanics of solids and structures by Hughes and Pister (1978).

We recall that we started our discussion with the *Simo-Taylor-Pister* variational principle, which treats the displacement, pressure and kinematic field variables $(\mathbf{u}, p, \tilde{J})$ as *independent* variables. In the next two steps we derived the necessary stationary conditions for the function L_{STP} and provided the associated discretizations. Based on the chosen discretization it turned out, however, that the volumetric variables are now *dependent* variables. In particular, the pressure $\bar{p}$ depends on $\tilde{J}$ via eq. (118)₂ and $\tilde{J}$ depends on the deformation via eq. (117)₂. Hence, both variables are subject to linearization, whose implementation is an important component. Consequently, due to the interpolation functions used it is crucial to start with the spatial discretization of the Euler-Lagrange equations in the weak form and to continue with the linearization of the discrete equations with respect to the positions of the nodal points. To linearize the Euler-Lagrange equations in the weak form first and then to discretize them would not be appropriate; the operations of spatial discretization and linearization do not commute in the theory of interest here.

Linearization in the Lagrangian description. In the following the linearization procedure is presented at the element level. Since we have considered the loads as independent of the deformation of the continuum body, the linearization only affects the modified internal virtual work $\delta \tilde{W}_e^{\text{int}}$. Rather than linearizing the discrete relation (112)₃, which is expressed in terms of *spatial* quantities, we perform the linearization of its *material* version and then do a push-forward operation on the material tensors. By recalling (95)₂ we have the expression $\delta \tilde{W}_e^{\text{int}} = \int_{\Omega_0} (J_h p_h \mathbf{C}_h^{-1} + \bar{\mathbf{S}}_h) : \delta \mathbf{C}_h / 2 dV$ in terms of *material* quantities for a typical finite element, on which we will focus subsequently.

It is important to emphasize that variation and linearization of functions are equivalent operations, which are based on the concept of directional derivative. Hence, in eq. (82), we just use the linearization operator $\Delta(\bullet)$ instead of $\delta(\bullet)$, and have the similar relation $\Delta\Phi(\mathbf{x}, \Delta\mathbf{u}) = D_{\Delta\mathbf{u}}\Phi(\mathbf{x}) = \frac{d}{d\varepsilon}\Phi(\mathbf{x} + \varepsilon\Delta\mathbf{u})|_{\varepsilon=0}$. For a detailed description of the concept of linearization the reader is referred to Holzapfel (2000), Section 8.4. By adopting rule (82) and by using the product rule, we arrive at the form

$$D_{\Delta\mathbf{u}}\delta\tilde{W}_e^{\text{int}} = \int_{\Omega_{0e}} [(Jp\mathbf{C}^{-1} + \bar{\mathbf{S}}) : \frac{1}{2}D_{\Delta\mathbf{u}}\delta\mathbf{C} + \frac{1}{2}\delta\mathbf{C} : D_{\Delta\mathbf{u}}(Jp\mathbf{C}^{-1} + \bar{\mathbf{S}})]dV \quad (133)$$

of the linearization, where $D_{\Delta\mathbf{u}}\delta\mathbf{C}$ characterizes the directional derivative of $\delta\mathbf{C}$ in the direction of the *increment* $\Delta\mathbf{u}$ of the displacement field. Note that the domain of integration in (133) remains fixed (it does not depend on the deformation), which is particularly useful for the linearization procedure. For convenience, we omit subsequently the subscript $(\bullet)_h$ indicating the discretization.

We may then proceed to specify the directional derivatives of the two quantities $Jp\mathbf{C}^{-1}$ and $\bar{\mathbf{S}}$. We use the chain rule to obtain

$$\begin{aligned} D_{\Delta\mathbf{u}}(Jp\mathbf{C}^{-1} + \bar{\mathbf{S}}) &= \left(2p\frac{\partial(J\mathbf{C}^{-1})}{\partial\mathbf{C}} + 2\frac{\partial\bar{\mathbf{S}}}{\partial\mathbf{C}} \right) : \frac{1}{2}D_{\Delta\mathbf{u}}\mathbf{C} + D_{\Delta\mathbf{u}}pJ\mathbf{C}^{-1} \\ &= Jp(\mathbf{C}^{-1} \otimes \mathbf{C}^{-1} - 2\mathbf{C}^{-1} \odot \mathbf{C}^{-1}) : \frac{1}{2}D_{\Delta\mathbf{u}}\mathbf{C} \\ &\quad + \bar{\mathbb{C}} : \frac{1}{2}D_{\Delta\mathbf{u}}\mathbf{C} + D_{\Delta\mathbf{u}}pJ\mathbf{C}^{-1}, \end{aligned} \quad (134)$$

where the tensor product $-\mathbf{C}^{-1} \odot \mathbf{C}^{-1}$ is given by eq. (65). We introduced the definition $\bar{\mathbb{C}} = 2(\partial\bar{\mathbf{S}}/\partial\mathbf{C})$ of the fourth-order elasticity tensor, which depends upon the material used. The explicit expression for $\bar{\mathbb{C}}$ has the form

$$\bar{\mathbb{C}} = 2\frac{\partial\bar{\mathbf{S}}}{\partial\mathbf{C}} = \mathbb{P} : \mathbb{C}^* : \mathbb{P}^T + \frac{2}{3}\text{Tr}(J^{-2/3}\mathbf{S}^*)\tilde{\mathbb{P}} - \frac{2}{3}(\mathbf{C}^{-1} \otimes \bar{\mathbf{S}} + \bar{\mathbf{S}} \otimes \mathbf{C}^{-1}), \quad (135)$$

with the abbreviation

$$\mathbb{C}^* = 2J^{-4/3}\frac{\partial\mathbf{S}^*}{\partial\bar{\mathbf{C}}} = 4J^{-4/3}\frac{\partial^2\bar{\Psi}(\bar{\mathbf{C}})}{\partial\bar{\mathbf{C}}\partial\bar{\mathbf{C}}} \quad (136)$$

for the fictitious elasticity tensor $\mathbb{C}^*$ in the material description. The modified projection tensor $\tilde{\mathbb{P}}$ and the trace $\text{Tr}(\bullet)$ are defined by (64). For a detailed account of these results the reader is referred to Holzapfel (2000), Example 6.8. A constitutive particularization of eqs. (135)₂ and (136)₂ for arterial wall mechanics is presented in Section 3.3.

By means of (134)₂, the linearized relation (133) may be re-expressed as

$$\begin{aligned} D_{\Delta\mathbf{u}}\delta\tilde{W}_e^{\text{int}} &= \int_{\Omega_{0e}} \{ (Jp\mathbf{C}^{-1} + \bar{\mathbf{S}}) : \frac{1}{2}D_{\Delta\mathbf{u}}\delta\mathbf{C} + D_{\Delta\mathbf{u}}pJ\mathbf{C}^{-1} : \frac{1}{2}\delta\mathbf{C} \\ &\quad + \frac{1}{2}\delta\mathbf{C} : [(\mathbf{C}^{-1} \otimes \mathbf{C}^{-1} - 2\mathbf{C}^{-1} \odot \mathbf{C}^{-1})Jp + \bar{\mathbb{C}}] : \frac{1}{2}D_{\Delta\mathbf{u}}\mathbf{C} \} dV. \end{aligned} \quad (137)$$

For notational simplicity we now use Δp for $D_{\Delta \mathbf{u}} p$ and $\Delta \mathbf{C}$ for $D_{\Delta \mathbf{u}} \mathbf{C}$. In addition, we use the explicit expression $D_{\Delta \mathbf{u}} \delta \mathbf{C} / 2 = \text{sym}(\text{Grad}^T \Delta \mathbf{u} \text{Grad} \delta \mathbf{u})$ for the first term of the integrand in eq. (137) (see Holzapfel (2000)). Herein, $\text{Grad}(\bullet)$ (with uppercase G), denotes the *material gradient* of the material field $(\bullet)$, which is obtained by differentiating $(\bullet)$ with respect to the referential position $\mathbf{X}$, at a fixed time t . By taking advantage of symmetries associated with $\mathbf{C}$ and $\bar{\mathbf{S}}$ the linearization of $D_{\Delta \mathbf{u}} \delta \tilde{W}_e^{\text{int}}$ in the *material* description leads to the set of linear increments

$$\begin{aligned} D_{\Delta \mathbf{u}} \delta \tilde{W}_e^{\text{int}} = \int_{\Omega_0} \{ & \text{Grad} \delta \mathbf{u} : \text{Grad} \Delta \mathbf{u} (J p \mathbf{C}^{-1} + \bar{\mathbf{S}}) + \Delta p J \mathbf{C}^{-1} : \frac{1}{2} \delta \mathbf{C} \\ & + \frac{1}{2} \delta \mathbf{C} : [(\mathbf{C}^{-1} \otimes \mathbf{C}^{-1} - 2 \mathbf{C}^{-1} \odot \mathbf{C}^{-1}) J p + \bar{\mathbb{C}}] : \frac{1}{2} \Delta \mathbf{C} \} dV. \end{aligned} \quad (138)$$

Linearization in the Eulerian description. The next goal is to re-express the result (138) in the *spatial* description. This is done by means of appropriate push-forward operations on the material tensors. The details, which can be found by Holzapfel (2000), Section 8.4, are left to be supplied by the reader. Here we summarize the result for the linearization of $D_{\Delta \mathbf{u}} \delta \tilde{W}_e^{\text{int}}$ in the *spatial* description, i.e.

$$\begin{aligned} D_{\Delta \mathbf{u}} \delta \tilde{W}_e^{\text{int}} = \int_{\Omega_e} \{ & \text{grad} \delta \mathbf{u}_h : \text{grad} \Delta \mathbf{u}_h (p_h \mathbf{I} + \bar{\boldsymbol{\sigma}}_h) + \Delta p_h \mathbf{I} : \delta \mathbf{e}_h \\ & + \delta \mathbf{e}_h : [(\mathbf{I} \otimes \mathbf{I} - 2 \mathbb{I}) p_h + \bar{\boldsymbol{\tau}}_h] : \Delta \mathbf{e}_h \} dv, \end{aligned} \quad (139)$$

where $\delta \mathbf{e}_h$ is given by (107), and the discretization

$$\Delta \mathbf{e}_h = \text{sym}(\text{grad} \Delta \mathbf{u}_h) = \frac{1}{2} \sum_{I=1}^{n_{\text{node}}} (\Delta \mathbf{u}_I \otimes \nabla_{\mathbf{x}} N_I + \nabla_{\mathbf{x}} N_I \otimes \Delta \mathbf{u}_I) \quad (140)$$

of the increment of the Euler-Almansi strain tensor is the analogue of eq. (107). The superscript $(\bullet)_h$ in (139) is to be used again to indicate discrete quantities. We introduced the notation $\bar{\boldsymbol{\tau}}$ for the *elasticity tensor* in the *spatial description*, which is defined as the push-forward operation of $\bar{\mathbb{C}}$ times a factor of J^{-1} . It is the *Piola transformation* of $\bar{\mathbb{C}}$ on each large index so that

$$\bar{\tau}_{abcd} = J^{-1} F_{aA} F_{bB} F_{cC} F_{dD} \bar{\mathbb{C}}_{ABCD}. \quad (141)$$

Equations (138) and (139) have a similar *symmetric* structure. They are linear with respect to the terms $\delta \mathbf{u}$ and $\Delta \mathbf{u}$, depending on $\mathbf{X}$ and $\mathbf{x}$, respectively, and lead to a *symmetric* (tangent) stiffness matrix. Formulations according to (138) and (139) are sometimes in the literature called *total-Lagrangian* and *updated-Lagrangian*, respectively. This really means that integrals are calculated over the respective regions of the reference and current configurations. It is important to emphasize that the material representation (138) of the linearized modified virtual internal work is **equivalent** to the spatial version (139). The two representations are based on the use of change of variables, and the results are the same in both cases.

For convenience, we may write the linearization of the discrete modified internal virtual work (139) in the decoupled form

$$D_{\Delta \mathbf{u}} \delta \tilde{W}_e^{\text{int}} = D_{\Delta \mathbf{u}} \delta W_e^{\text{geo}} + D_{\Delta \mathbf{u}} \delta W_e^{\text{mat}} + D_{\Delta \mathbf{u}} \delta W_e^{\text{STP}}, \quad (142)$$

with the definitions

$$D_{\Delta \mathbf{u}} \delta W_e^{\text{geo}} = \int_{\Omega_e} \text{grad} \delta \mathbf{u}_h : \text{grad} \Delta \mathbf{u}_h (p_h \mathbf{I} + \bar{\boldsymbol{\sigma}}_h) dv, \quad (143)$$

$$D_{\Delta \mathbf{u}} \delta W_e^{\text{mat}} = \int_{\Omega_e} \delta \mathbf{e}_h : [(\mathbf{I} \otimes \mathbf{I} - 2\mathbb{I})p_h + \bar{\mathbb{C}}_h] : \Delta \mathbf{e}_h dv, \quad (144)$$

$$D_{\Delta \mathbf{u}} \delta W_e^{\text{STP}} = \int_{\Omega_e} \Delta p_h \mathbf{I} : \delta \mathbf{e}_h dv \quad (145)$$

of the three contributions. The first term in eq. (142) comes from the current state of stress and represents the **geometrical stress contribution** to the linearization, while the second term represents the **material contribution**. The last contribution $D_{\Delta \mathbf{u}} \delta W_e^{\text{STP}}$ governs the change of p_h and is due to the mean dilation approach, which is based on the three-field variational principle of **Simo, Taylor** and **Pister**.

Derivations of the stiffness matrices. The aim now is to compute explicitly the three contributions (143)-(145) and to provide the associated (*tangent*) *stiffness matrices*. Let us start with the geometrical stress contribution $D_{\Delta \mathbf{u}} \delta W_e^{\text{geo}}$ to the linearization, as given in (143), and study the explicit discrete form for a typical element Ω_e . By means of the interpolations (104) and (108)₁ we obtain from (143) that

$$\begin{aligned} D_{\Delta \mathbf{u}} \delta W_e^{\text{geo}} &= \int_{\Omega_e} \text{grad} \delta \mathbf{u}_h : \text{grad} \Delta \mathbf{u}_h (p_h \mathbf{I} + \bar{\boldsymbol{\sigma}}_h) dv \\ &= \sum_{I=1}^{n_{\text{node}}} \sum_{K=1}^{n_{\text{node}}} \int_{\Omega_e} (\delta \mathbf{u}_I \otimes \nabla_{\mathbf{x}} N_I) : (\Delta \mathbf{u}_K \otimes \nabla_{\mathbf{x}} N_K) (\bar{p} \mathbf{I} + \bar{\boldsymbol{\sigma}}_h) dv. \end{aligned} \quad (146)$$

For later use it is now beneficial to rearrange the integrand of this equation. The straightforward rearrangement is shown by means of the analysis of the term $(\mathbf{a} \otimes \mathbf{b}) : (\mathbf{c} \otimes \mathbf{d}) \mathbf{A}$, which has the same structure as the integrand in eq. (146)₂, where $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ are vectors and $\mathbf{A}$ is a *symmetric* second-order tensor. Basic tensor algebra gives

$$\begin{aligned} (\mathbf{a} \otimes \mathbf{b}) : (\mathbf{c} \otimes \mathbf{d}) \mathbf{A} &= (\mathbf{c} \otimes \mathbf{d})^T (\mathbf{a} \otimes \mathbf{b}) : \mathbf{A} \\ &= (\mathbf{d} \otimes \mathbf{c}) (\mathbf{a} \otimes \mathbf{b}) : \mathbf{A} \\ &= (\mathbf{a} \cdot \mathbf{c}) (\mathbf{d} \otimes \mathbf{b}) : \mathbf{A} \\ &= (\mathbf{a} \cdot \mathbf{c}) (\mathbf{b} \cdot \mathbf{A} \mathbf{d}) \\ &= \mathbf{a} \cdot \mathbf{I} (\mathbf{b} \cdot \mathbf{A} \mathbf{d}) \mathbf{c}, \end{aligned} \quad (147)$$

(see Holzapfel (2000)). Hence, applying the property (147)₅ to (146)₂ we find the simple format

$$D_{\Delta \mathbf{u}} \delta W_e^{\text{geo}} = \sum_{I=1}^{n_{\text{node}}} \sum_{K=1}^{n_{\text{node}}} \delta \mathbf{u}_I \cdot \mathbf{I} \int_{\Omega_e} \nabla_{\mathbf{x}} N_I \cdot (\bar{p} \mathbf{I} + \bar{\boldsymbol{\sigma}}_e) \nabla_{\mathbf{x}} N_K dv \Delta \mathbf{u}_K. \quad (148)$$

This equation reads in the convenient matrix notation:

$$D_{\Delta \mathbf{u}} \delta W_e^{\text{geo}} = \sum_{I=1}^{n_{\text{node}}} \sum_{K=1}^{n_{\text{node}}} \delta \mathbf{u}_I^T \mathbf{K}_{IK}^{\text{geo}} \Delta \mathbf{u}_K \quad (149)$$

$$\text{with} \quad \mathbf{K}_{IK}^{\text{geo}} = \mathbf{I} \int_{\Omega_e} (\nabla_{\mathbf{x}} N_I)^T \bar{\boldsymbol{\sigma}}_0 \nabla_{\mathbf{x}} N_K dv, \quad (150)$$

where $\mathbf{K}_{IK}^{\text{geo}}$ is called the *geometrical stiffness matrix*, which defines the *stress* contribution to the tangent matrix. It arises from the finite deformation kinematics of the considered problem and relates nodes I to K within a typical finite element domain. The sub-stiffness matrix $\mathbf{K}_{IK}^{\text{geo}}$ has the size $n_{\text{dof}} \times n_{\text{dof}}$, where n_{dof} denotes the number of degree of freedom associated with the element node (for a 3D continuum problem we have $n_{\text{dof}} = 3$ for each node).

The *initial (Cauchy) stress matrix* (containing stresses at the beginning of each iteration-step) is indicated by $\bar{\boldsymbol{\sigma}}_0$. It is a square matrix with the structure

$$\bar{\boldsymbol{\sigma}}_0 = \begin{bmatrix} \bar{p} & & \\ & \bar{p} & \\ & & \bar{p} \end{bmatrix} + \begin{bmatrix} \bar{\sigma}_{11} & \bar{\sigma}_{12} & \bar{\sigma}_{13} \\ & \bar{\sigma}_{22} & \bar{\sigma}_{23} \\ \text{sym.} & & \bar{\sigma}_{33} \end{bmatrix}_e, \quad (151)$$

where the entries $\bar{\sigma}_{ij}$ are found from the respective constitutive equation to be used.

Let us continue with the constitutive contribution $D_{\Delta \mathbf{u}} \delta W_e^{\text{mat}}$ to the linearization. By means of eqs. (107)₂, (108)₁ and (140)₂, we find from (144) that

$$D_{\Delta \mathbf{u}} \delta W_e^{\text{mat}} = \int_{\Omega_e} \delta \mathbf{e}_h : [(\mathbf{I} \otimes \mathbf{I} - 2\mathbb{I})p_h + \bar{\boldsymbol{\epsilon}}_h] : \Delta \mathbf{e}_h dv \quad (152)$$

$$\begin{aligned} &= \sum_{I=1}^{n_{\text{node}}} \sum_{K=1}^{n_{\text{node}}} \int_{\Omega_e} \frac{1}{2} (\delta \mathbf{u}_I \otimes \nabla_{\mathbf{x}} N_I + \nabla_{\mathbf{x}} N_I \otimes \delta \mathbf{u}_I) : \\ &[(\mathbf{I} \otimes \mathbf{I} - 2\mathbb{I})\bar{p} + \bar{\boldsymbol{\epsilon}}_h] : \frac{1}{2} (\Delta \mathbf{u}_K \otimes \nabla_{\mathbf{x}} N_K + \nabla_{\mathbf{x}} N_K \otimes \Delta \mathbf{u}_K) dv. \end{aligned} \quad (153)$$

This equation reads in matrix notation:

$$D_{\Delta \mathbf{u}} \delta W_e^{\text{mat}} = \sum_{I=1}^{n_{\text{node}}} \sum_{K=1}^{n_{\text{node}}} \delta \mathbf{u}_I^T \mathbf{K}_{IK}^{\text{mat}} \Delta \mathbf{u}_K, \quad (154)$$

$$\text{with} \quad \mathbf{K}_{IK}^{\text{mat}} = \int_{\Omega_e} \mathbf{B}_I^T \mathbf{D} \mathbf{B}_K dv, \quad (155)$$

where $\mathbf{K}_{IK}^{\text{mat}}$ is the *material stiffness matrix* for the element nodes I and K of a typical element. It defines the *constitutive* contribution to the tangent matrix and arises from the resistance of the hyperelastic material to imposed deformation. The (spatial) constitutive matrix is indicated by $\mathbf{D}$. The entries of $\mathbf{D}$ are the components of the fourth-order tensor $(\mathbf{I} \otimes \mathbf{I} - 2\mathbb{I})\bar{p} + \bar{\boldsymbol{\epsilon}}_h$. Taking

advantage of the minor and major symmetries of $\bar{\mathbb{C}}_h$ we deduce that $\mathbf{D} = \mathbf{D}_{\bar{p}} + \mathbf{D}_{\bar{c}}$, with the explicit expressions for the matrices

$$\mathbf{D}_{\bar{p}} = \begin{bmatrix} -\bar{p} & \bar{p} & \bar{p} & & & \\ \bar{p} & -\bar{p} & \bar{p} & & & \\ \bar{p} & \bar{p} & -\bar{p} & & & \\ & & & -\bar{p} & & \\ & & & & -\bar{p} & \\ & & & & & -\bar{p} \end{bmatrix}, \quad (156)$$

$$\mathbf{D}_{\bar{c}} = \begin{bmatrix} \bar{c}_{1111} & \bar{c}_{1122} & \bar{c}_{1133} & \bar{c}_{1112} & \bar{c}_{1123} & \bar{c}_{1113} \\ & \bar{c}_{2222} & \bar{c}_{2233} & \bar{c}_{2212} & \bar{c}_{2223} & \bar{c}_{2213} \\ & & \bar{c}_{3333} & \bar{c}_{3312} & \bar{c}_{3323} & \bar{c}_{3313} \\ & & & \bar{c}_{1212} & \bar{c}_{1223} & \bar{c}_{1213} \\ & \text{sym.} & & & \bar{c}_{2323} & \bar{c}_{2313} \\ & & & & & \bar{c}_{1313} \end{bmatrix}. \quad (157)$$

Let us finally specify the pressure part $D_{\Delta \mathbf{u}} \delta W_e^{\text{STP}}$, which is a crucial contribution to the linearized quantity $D_{\Delta \mathbf{u}} \delta \tilde{W}_e^{\text{int}}$. For this specification we require the interpolation of the increment of the volumetric variables. Hence, by analogy with (108)₂ and (108)₄ we write

$$\Delta p_h = \Delta \bar{p}, \quad \Delta \tilde{J}_h = \Delta \bar{\tilde{J}}. \quad (158)$$

By means of (107)₁, the property $\mathbf{I} : \text{sym}(\text{grad} \delta \mathbf{u}_h) = \text{div} \delta \mathbf{u}_h$, and (158)₁ we find from (145) that

$$D_{\Delta \mathbf{u}} \delta W_e^{\text{STP}} = \int_{\Omega_e} \Delta p_h \mathbf{I} : \delta \mathbf{e}_h dv = \int_{\Omega_e} \Delta p_h \text{div} \delta \mathbf{u}_h dv = \Delta \bar{p} v_e \overline{\text{div}} \delta \mathbf{u}_h, \quad (159)$$

where, for convenience, the definition

$$\overline{\text{div}}(\bullet) = \frac{1}{v_e} \int_{\Omega_e} \text{div}(\bullet) dv \quad (160)$$

of the average divergence over the current volume v_e of an element is to be introduced.

Before examining the pressure change $\Delta \bar{p}$ occurring in (159)₃ it is first necessary to determine the directional derivative of $\bar{\tilde{J}}$ in the direction of the increment $\Delta \mathbf{u}$ of the displacement field. Therefore, by means of (158)₂, the linearization of (117) leads to $\Delta \bar{\tilde{J}} = (1/V_e) \int_{\Omega_{0e}} \Delta J_h dV$. With the incremental property $\Delta J_h = J_h \text{div} \Delta \mathbf{u}_h$ we find that the change of the volumetric variable is given by

$$\Delta \bar{\tilde{J}} = \frac{1}{V_e} \int_{\Omega_e} \text{div} \Delta \mathbf{u}_h dv = \bar{\tilde{J}} \overline{\text{div}} \Delta \mathbf{u}_h, \quad (161)$$

where the operator $\overline{\text{div}}(\bullet)$, as defined in eq. (160), is to be used.

By recalling relation (118)₂ and using (161)₂, we find an explicit form of the pressure change, namely

$$\Delta \bar{p} = \frac{\partial \bar{p}}{\partial \bar{J}} \Delta \bar{J} = \frac{d^2 U(\bar{J})}{d \bar{J}^2} \Delta \bar{J} = \frac{d^2 U(\bar{J})}{d \bar{J}^2} \bar{J} \overline{\text{div}} \Delta \mathbf{u}_h \Big|_{\bar{J} = v_e/V_e}. \quad (162)$$

Substituting this result into (159)₃ gives

$$D_{\Delta \mathbf{u}} \delta W_e^{\text{STP}} = \kappa \overline{\text{div}} \Delta \mathbf{u}_h \overline{\text{div}} \delta \mathbf{u}_h, \quad \kappa = \bar{J} \frac{d^2 U(\bar{J})}{d \bar{J}^2} v_e, \quad (163)$$

where $\bar{J} = v_e/V_e$, as shown in (117)₂, and, for notational simplicity, κ has been introduced as an abbreviation. It is important to emphasize that the variables $\Delta \bar{p}$ and $\Delta \bar{J}$ for the pressure and volumetric quantities are eliminated at the element level.

The aim is now to express eq. (163)₁ in the same mathematical structure as (148). By using the definition (160) and the interpolation (105), a straightforward algebraic manipulation gives

$$D_{\Delta \mathbf{u}} \delta W_e^{\text{STP}} = \kappa \frac{1}{v_e} \int_{\Omega_e} \left(\sum_{I=1}^{n_{\text{node}}} \delta \mathbf{u}_I \cdot \nabla_{\mathbf{x}} N_I \right) dv \frac{1}{v_e} \int_{\Omega_e} \left(\sum_{K=1}^{n_{\text{node}}} \Delta \mathbf{u}_K \cdot \nabla_{\mathbf{x}} N_K \right) dv \quad (164)$$

$$= \kappa \sum_{I=1}^{n_{\text{node}}} \sum_{K=1}^{n_{\text{node}}} \delta \mathbf{u}_I \cdot (\overline{\nabla}_{\mathbf{x}} N_I \otimes \overline{\nabla}_{\mathbf{x}} N_K) \Delta \mathbf{u}_K, \quad (165)$$

where the definition

$$\overline{\nabla}_{\mathbf{x}}(\bullet) = \frac{1}{v_e} \int_{\Omega_e} \nabla_{\mathbf{x}}(\bullet) dv \quad (166)$$

of the spatial average gradient of the scalar function $(\bullet)$ over the current volume v_e of an element is to be used.

The pressure part (165)₂, which is given in tensorial form, is recast in the matrix notation as

$$D_{\Delta \mathbf{u}} \delta W_e^{\text{STP}} = \sum_{I=1}^{n_{\text{node}}} \sum_{K=1}^{n_{\text{node}}} \delta \mathbf{u}_I^T \mathbf{K}_{IK}^{\text{STP}} \Delta \mathbf{u}_K, \quad (167)$$

$$\text{with} \quad \mathbf{K}_{IK}^{\text{STP}} = \kappa (\overline{\nabla}_{\mathbf{x}} N_I)^T \overline{\nabla}_{\mathbf{x}} N_K, \quad (168)$$

where $\mathbf{K}_{IK}^{\text{STP}}$ is the ‘pressure’ *stiffness matrix*, which defines the important *pressure* contribution to the tangent matrix.

The total stiffness matrix for a typical element, denoted by $\mathbf{K}_e$, is composed of the three sub-stiffness matrices given through eqs. (150), (155) and (168), which is summed over the nodes of the elements. Thus,

$$\mathbf{K}_e = \sum_{I=1}^{n_{\text{node}}} \sum_{K=1}^{n_{\text{node}}} \int_{\Omega_e} [I(\nabla_{\mathbf{x}} N_I)^T \bar{\sigma}_0 \nabla_{\mathbf{x}} N_K + \mathbf{B}_I^T \mathbf{D} \mathbf{B}_K] dv + \kappa (\overline{\nabla}_{\mathbf{x}} N_I)^T \overline{\nabla}_{\mathbf{x}} N_K. \quad (169)$$

The size of the element stiffness matrix $\mathbf{K}_e$ is then $(n_{\text{node}} \cdot n_{\text{dof}}) \times (n_{\text{node}} \cdot n_{\text{dof}})$, which, for an 8-node brick element, is $(8 \cdot 3) \times (8 \cdot 3) = 24 \times 24$.

Note that motion-dependent loads of a body cause additional external virtual work (for pressure boundary loading see, for example, Holzapfel (2000), p. 383). The external virtual work done by the loads along the virtual displacement need to be considered within *nonlinear* finite element analyses in the form of an additional, in general *non-symmetric*, contribution to the element stiffness matrix $\mathbf{K}_e$ (Schweizerhof and Ramm (1984)).

The construction of the global (tangent) stiffness matrix, denoted by $\mathbf{K}$, follows the standard assembly procedure of element stiffness matrices. We write

$$\mathbf{K}(\mathbf{u}) = \mathbf{A} \sum_{e=1}^{n_e} \mathbf{K}_e(\mathbf{u}_e), \quad (170)$$

with $\mathbf{K}_e$ given in (169). Since both volumetric variables $\bar{J}$ and $\bar{p}$ are eliminated at the element level we now have a (generalized) displacement model with only the global variable $\mathbf{u}$ unknown. Hence, the implementation of the global stiffness matrix $\mathbf{K}$, which is symmetric for hyperelasticity, and the residual $\mathbf{R}$ (compare with (129)₂) is the same as for the standard displacement-based method in which only the displacement field is discretized. It is known that the stiffness matrix $\mathbf{K}$ is the Jacobian matrix of the internal nodal forces according to

$$\mathbf{K}(\mathbf{u}) = \frac{\partial \mathbf{F}^{\text{int}}(\mathbf{u})}{\partial \mathbf{u}}. \quad (171)$$

Note that the derived $Q1/P0$ -element does not satisfy the *Babuška-Brezzi*, or *LBB*, *stability condition* within the geometrical linear theory and instabilities may occur (Brezzi and Fortin (1991)). For mathematical details of the LBB-condition the reader is referred to the book by Oden and Carey (1984).

4.4 Solution Methods of the Spatially Discrete Equations

So far we have considered the discretization of the continuous problem *in space*. To complete our study of finite deformation elastodynamics, we will apply a finite difference scheme to achieve also discretization *in time*.

The task is now to find the displacement $\mathbf{u}(t)$ satisfying the nonlinear initial boundary-value problem in the form of the spatially discrete equations (129)₂ and the given initial data (130) over some closed time interval $t \in [0, T]$ of interest, where $0 = t_0 < \dots < t_{M+1} = T$. Subsequently, we consider the time discretization (partition)

$$[0, T] = \bigcup_{n=0}^M [t_n, t_{n+1}], \quad (172)$$

where n denotes an index on the closed time sub-interval $[t_n, t_{n+1}]$, and $M + 1$ is the number of the sub-intervals. We now concentrate attention on a typical sub-interval, with

$$\Delta t = t_{n+1} - t_n \quad (173)$$

characterizing the associated time increment. For tracing a nonlinear elastodynamic problem not all time sub-intervals need to have the same width.

Assume now that up to a certain time t_n the algorithmic approximations for the solution $\mathbf{u}_n$, the velocity $\dot{\mathbf{u}}_n$, and the acceleration $\ddot{\mathbf{u}}_n$ are specified uniquely by the given motion at t_n . The aim is now to advance the solution to time $t_{n+1} = \Delta t + t_n$ and to *update* these quantities to $\mathbf{u}_{n+1}$, $\dot{\mathbf{u}}_{n+1}$, and $\ddot{\mathbf{u}}_{n+1}$.

In order to solve (129)₂ and (130) for a typical time sub-interval we may use the most-widely used (global) time-stepping algorithm by Newmark (1959); see also Hughes (2000), Chapter 9. This algorithm consists of the equations

$$\left. \begin{aligned} M\ddot{\mathbf{u}}_{n+1} + \mathbf{F}^{\text{int}}(\mathbf{u}_{n+1}) &= \mathbf{F}^{\text{ext}}(t_{n+1}), \\ \mathbf{u}_{n+1} &= \mathbf{u}_n + \Delta t \dot{\mathbf{u}}_n + \frac{\Delta t^2}{2} [(1 - 2\beta)\ddot{\mathbf{u}}_n + 2\beta\ddot{\mathbf{u}}_{n+1}], \\ \dot{\mathbf{u}}_{n+1} &= \dot{\mathbf{u}}_n + \Delta t [(1 - \gamma)\ddot{\mathbf{u}}_n + \gamma\ddot{\mathbf{u}}_{n+1}], \end{aligned} \right\} \quad (174)$$

where β and γ are (free) algorithmic parameters. Different specific choices of these parameters result into different time-stepping algorithms. For comparative studies see, for example, Goudreau and Taylor (1972). The Newmark method is *second-order accurate* – the truncation error is of order $(\Delta t)^2$. The set of nonlinear algebraic equations (174) is solved in each time increment. An iterative solution strategy is based on the efficient predictor/multicorrector algorithms described in Hughes et al. (1979).

Other transient time-stepping algorithms such as the α -method by Hilber et al. (1977) may be considered. The method controls (damps) the amount of high frequency dissipation, while the predictor and corrector formulas of Newmark's method are retained. A variant of the method is found in Chung and Hulbert (1993). For another modification of Newmark's method see, for example, Wood et al. (1981).

Explicit methods. An *explicit* finite element integrator is the *central-difference method* widely used in structural dynamics. The algorithm is deduced from the Newmark formulas (174) by using the central difference as a template, i.e. $\beta = 0$ and $\gamma = 1/2$. It has the form (Wood (1990))

$$\left. \begin{aligned} \ddot{\mathbf{u}}_{n+1} &= \mathbf{M}^{-1} [\mathbf{F}^{\text{ext}}(t_{n+1}) - \mathbf{F}^{\text{int}}(\mathbf{u}_{n+1})], \\ \mathbf{u}_{n+1} &= \mathbf{u}_n + \Delta t \dot{\mathbf{u}}_n + \frac{\Delta t^2}{2} \ddot{\mathbf{u}}_n, \\ \dot{\mathbf{u}}_{n+1} &= \dot{\mathbf{u}}_n + \frac{\Delta t}{2} (\ddot{\mathbf{u}}_n + \ddot{\mathbf{u}}_{n+1}). \end{aligned} \right\} \quad (175)$$

As can be seen from the decoupled eq. (175)₂ the displacements at time t_{n+1} can be computed explicitly from the three matrices $\{\mathbf{u}_n, \dot{\mathbf{u}}_n, \ddot{\mathbf{u}}_n\}$, which are known at t_n . Since $\mathbf{F}^{\text{ext}}$ at t_{n+1} is given, accelerations at t_{n+1} follow from the time discrete equation of motion (175)₁, and, finally, the velocities may be computed from eq. (175)₃. Hence, the explicit method is simple to implement since the required solutions at t_{n+1} only depend on the quantities at t_n . For the case that the mass matrix $\mathbf{M}$ is diagonal (mass lumping), the inverse $\mathbf{M}^{-1}$ of $\mathbf{M}$ is trivial to compute and the integrator turns out to be particularly efficient.

The disadvantage of explicit finite element methods is the fact that they are only stable when Δt is less than a critical value. Hence, explicit methods are particularly suited to compute structural responses in the *high* frequency domain, for which small time steps are often required to capture the associated physical process. To resolve vibrations in the low frequency responses, this method may require an inefficiently large number of time steps due to the stability limit. Hence, for this type of problem explicit integrators are the methods of choice.

Implicit methods and aspects of nonlinear equation solving. One example of an *implicit* finite element integrator is the *trapezoidal rule (average acceleration method)*. This method is widely used for structural dynamics applications and is deduced from (174) by taking $\beta = 1/4$ and $\gamma = 1/2$. Thus,

$$\left. \begin{aligned} M\ddot{\mathbf{u}}_{n+1} + \mathbf{F}^{\text{int}}(\mathbf{u}_{n+1}) &= \mathbf{F}^{\text{ext}}(t_{n+1}), \\ \mathbf{u}_{n+1} &= \mathbf{u}_n + \Delta t \dot{\mathbf{u}}_n + \frac{\Delta t^2}{4}(\ddot{\mathbf{u}}_n + \ddot{\mathbf{u}}_{n+1}), \\ \dot{\mathbf{u}}_{n+1} &= \dot{\mathbf{u}}_n + \frac{\Delta t}{2}(\ddot{\mathbf{u}}_n + \ddot{\mathbf{u}}_{n+1}). \end{aligned} \right\} \quad (176)$$

In practice, $\ddot{\mathbf{u}}_{n+1}$ is eliminated from the time discrete equation of motion (176)₁ by means of (176)₂. This results in a highly nonlinear and coupled algebraic equation for $\mathbf{u}_{n+1}$, given by

$$\frac{4}{\Delta t^2} M \mathbf{u}_{n+1} + \mathbf{F}^{\text{int}}(\mathbf{u}_{n+1}) = \mathbf{F}^{\text{ext}}(t_{n+1}) + M \left[\ddot{\mathbf{u}}_n + \frac{4}{\Delta t^2} (\mathbf{u}_n + \Delta t \dot{\mathbf{u}}_n) \right]. \quad (177)$$

By writing eq. (177) in the structure of (130) we obtain the nonlinear discrete equations of motion (using the trapezoidal rule as a template) in the form

$$\mathbf{R}(\mathbf{u}_{n+1}) = \mathbf{O}, \quad (178)$$

with

$$\begin{aligned} \mathbf{R}(\mathbf{u}_{n+1}) &= \mathbf{F}^{\text{ext}}(t_{n+1}) - \mathbf{F}^{\text{int}}(\mathbf{u}_{n+1}) - \frac{4}{\Delta t^2} M \mathbf{u}_{n+1} \\ &\quad + M \left[\ddot{\mathbf{u}}_n + \frac{4}{\Delta t^2} (\mathbf{u}_n + \Delta t \dot{\mathbf{u}}_n) \right], \end{aligned} \quad (179)$$

where $\mathbf{R}(\mathbf{u}_{n+1})$ is the column matrix of the residual forces, and the unknowns $\mathbf{u}_{n+1}$ are a set of nonlinear *functions*.

The task is now to find the unknown quantities at t_{n+1} , in particular the column matrix $\mathbf{u}_{n+1}$, by solving the set (178) of nonlinear *algebraic* equations using iterative solution techniques. A wide variety of such solving strategies exist in computational mechanics.

One representative example of a solution method for nonlinear algebraic equations is the *Newton-Raphson method*. Although this method is well-established and in common use, the method is insufficiently robust and is associated with high computational cost. It is an equilibrium iteration, which exhibits quadratic rates of convergence once we are close enough to the solution point (local convergence), pre-supposing that ‘correct’ linearization of the nonlinear equations is provided. There are many other iterative solution schemes available, such as the *modified Newton method*, which shows only linear rates of convergence.

Now, within the context of the present volume, we will review briefly the application of the Newton-Raphson solution algorithm to the implicit finite element method. The linearization of (178) may be written as

$$\mathbf{R}(\mathbf{u}_{n+1}^{i+1}) \approx \mathbf{R}(\mathbf{u}_{n+1}^i) + \left(\frac{\partial \mathbf{R}(\mathbf{u})}{\partial \mathbf{u}} \right) \bigg|_{\mathbf{u}_{n+1}^i} \Delta \mathbf{u} = \mathbf{O}, \quad (180)$$

where $\mathbf{u}_{n+1}^i$ is the displacement at iteration i and $\mathbf{u}_{n+1}^{i+1} = \mathbf{u}_{n+1}^i + \Delta \mathbf{u}$ is the new displacement at iteration $i + 1$. Note that the last term in the linear equation (180) is the directional derivative of $\mathbf{R}$ with respect to the displacement estimate $\mathbf{u}_{n+1}^i$ in the general direction of the increment $\Delta \mathbf{u}$. Hence, with the discrete equation (180)₂, the general Newton-Raphson iteration scheme becomes

$$\left(\frac{\partial \mathbf{R}}{\partial \mathbf{u}} \right) \bigg|_{\mathbf{u}_{n+1}^i} \Delta \mathbf{u} = -\mathbf{R}(\mathbf{u}_{n+1}^i), \quad \mathbf{u}_{n+1}^{i+1} = \mathbf{u}_{n+1}^i + \Delta \mathbf{u}, \quad (181)$$

which is to be solved for an increment $\Delta \mathbf{u}$ in the solution followed by the displacement update (181)₂. Here we iterate within each time step until, for example, the magnitude $\|\mathbf{R}(\mathbf{u}_{n+1}^i)\|$ of the residual (i.e. the *Euclidean norm* calculated from the current iterated value), is less than a pre-assigned tolerance that ensures an accurate approximate solution to the nonlinear algebraic problem. One simple convergence test is based on the comparison of $\|\mathbf{R}(\mathbf{u}_{n+1}^i)\|$ with a fraction of the initial residual $\|\mathbf{R}(\mathbf{u}_{n+1}^0)\|$ such that $\|\mathbf{R}(\mathbf{u}_{n+1}^i)\| < \epsilon \|\mathbf{R}(\mathbf{u}_{n+1}^0)\|$ is satisfied, where ϵ is a pre-assigned, non-dimensional tolerance parameter. Another measure of accuracy is, for example, the *energy norm* determined by the expression $\|\Delta \mathbf{u}^T \mathbf{R}(\mathbf{u}_{n+1}^i)\|$ (for more details see, for example, Belytschko et al. (2000)). If the convergence test is satisfied the time sub-interval is updated and the Newton-Raphson iteration starts with the next step to advance the solution.

By using this concept and by taking $\mathbf{u}_{n+1}^i$ instead of $\mathbf{u}_{n+1}$ in relation (179), we obtain from (181)₁ the linear set of equations for the displacement increment $\Delta \mathbf{u}$, i.e. the iterative driver

$$\left[\mathbf{K}(\mathbf{u}_{n+1}^i) + \frac{4}{\Delta t^2} \mathbf{M} \right] \Delta \mathbf{u} = \mathbf{R}(\mathbf{u}_{n+1}^i), \quad (182)$$

where the residual at iteration i is given as

$$\begin{aligned} \mathbf{R}(\mathbf{u}_{n+1}^i) &= \mathbf{F}^{\text{ext}}(t_{n+1}) - \mathbf{F}^{\text{int}}(\mathbf{u}_{n+1}^i) - \frac{4}{\Delta t^2} \mathbf{M} \mathbf{u}_{n+1}^i \\ &\quad + \mathbf{M} \left[\ddot{\mathbf{u}}_n + \frac{4}{\Delta t^2} (\mathbf{u}_n + \Delta t \dot{\mathbf{u}}_n) \right], \end{aligned} \quad (183)$$

and the stiffness matrix evaluated at the current value $\mathbf{u}_{n+1}^i$, i.e. at *each* iteration i and time t_{n+1} , is of the form

$$\mathbf{K}(\mathbf{u}_{n+1}^i) = \left(\frac{\partial \mathbf{F}^{\text{int}}}{\partial \mathbf{u}} \right) \bigg|_{\mathbf{u}_{n+1}^i}. \quad (184)$$

The stiffness matrix (184) is the time discrete version of (171). The displacement is then updated via (181)₂. Note that the deviation of the approximate solution from the exact solution

decreases quadratically at each iteration when the linearization (184) is used. In practice, the notion ‘Newton-Raphson method’ often refers to an algorithm in which $\mathbf{K}$ is formed at *each* iteration. So, for large-scale computations, the solution of the nonlinear algebraic problem may be associated with high computational effort due to the formulation and factorization of $\mathbf{K}$ at every iteration. Alternatively, modified Newton methods may be applied in which $\mathbf{K}$ is computed only once (at the first iteration) and then kept fixed during the remaining iterations of the time step, or even the entire solution strategy. This widely used method requires more iterations to converge and does not show the superior convergence properties of the Newton-Raphson method, but the computational cost per iteration is less. It is used mainly for weakly nonlinear problems.

For the quasi-static case the inertia terms are negligible, and hence eqs. (182) and (183) may be reduced to the specific forms

$$\mathbf{K}(\mathbf{u}_{n+1}^i) \Delta \mathbf{u} = \mathbf{R}(\mathbf{u}_{n+1}^i), \quad \mathbf{R}(\mathbf{u}_{n+1}^i) = \mathbf{F}^{\text{ext}}(t_{n+1}) - \mathbf{F}^{\text{int}}(\mathbf{u}_{n+1}^i), \quad (185)$$

where the stiffness matrix $\mathbf{K}(\mathbf{u}_{n+1}^i)$ is given through eq. (184). The Newton-Raphson iteration boils down to the solution of the sequence of linearized problems (185), so that equilibrium is restored at t_{n+1} , which results in satisfaction of (178).

The advantage of implicit finite element methods over explicit methods is the fact that for linear transient problems and suitable implicit integrators the time steps Δt can be chosen arbitrarily in order to obtain *unconditionally* stable solutions. However, for large time steps the accuracy of the solution decreases. Although no general proof of unconditional stability for nonlinear systems exists, we are taught by computational experience that time steps within implicit methods can be chosen much larger than within explicit methods while retaining stability. Hence, implicit integrators are particularly suited for computing structural responses in the *low* frequency domain, for which larger time steps are desirable.

Line search method. The Newton-Raphson method converges slowly if the starting iteration is substantially far from the (exact) solution (Luenberger (1984)), so larger time steps reduce the robustness of the Newton-Raphson method. In addition, the modified Newton-method may also lead to poor convergence rates due to the fact that the affiliated stiffness matrix is only a rough approximation to the consistent one, as evaluated in (184). Consequently, the displacement updates may not be effective any more in the sense that the updated values are further away from the desired solution. Since Newton-type methods have an unfortunate tendency to wander off if the initial guess is not sufficiently close to the root, it is therefore necessary to enhance these methods by ‘post-processing’ strategies to preclude divergence and to improve the convergence rate. The enhancements should be performed in an economical way. The widely used *line search method*, discussed briefly here, has proven to be effective in controlling the solution and preventing divergence of the nonlinear iteration through Newton-type methods.

Basically, the line search method interprets the increment $\Delta \mathbf{u}$, found by the Newton method, as a ‘*search direction*’. It is scaled by a (scalar) *search parameter*, denoted by s . Rather than using the update formula (181)₂, the new displacement is now computed by the formula

$$\mathbf{u}_{n+1}^{i+1}(s) = \mathbf{u}_{n+1}^i + s \Delta \mathbf{u}, \quad (186)$$

where $\mathbf{u}_{n+1}^{i+1}$ and $\mathbf{u}_{n+1}^i$ denote the new and old solutions. For nonlinear dynamic (transient) problems the direction $\Delta \mathbf{u}$ is determined by solving (182), with the use of eqs. (183) and (184), while

for nonlinear quasi-static problems $\Delta \mathbf{u}$ is given through eqs. (185) and (184). The aim is now to find the search parameter s such that some norm on the residual matrix is minimized with respect to the non-scaled (standard) update, i.e. the full Newton step $s = 1$. Note that the line search method becomes less effective in improving the convergence rate of Newton-Raphson iterations the closer $\mathbf{u}$ approaches the solution point.

Now we define a scalar-valued *nonlinear* function $g(s)$ such that the search parameter s satisfies the condition

$$g(s) = \Delta \mathbf{u}^T \mathbf{R}(\mathbf{u}_{n+1}^{i+1}(s)) = 0. \quad (187)$$

This scalar equation for s admits a general representation of the residual $\mathbf{R}$ evaluated at $\mathbf{u}_{n+1}^{i+1} = \mathbf{u}_{n+1}^i + s\Delta \mathbf{u}$. For nonlinear dynamic problems the residual is given through eqs. (183) and (186) using the trapezoidal rule as a template, while for quasi-static problems the residual is given by (185). The task is now to design a line search algorithm that selects s in such a way that at the end of each iteration the search direction $\Delta \mathbf{u}$ is *orthogonal* to the matrix $\mathbf{R}(\mathbf{u}_{n+1}^{i+1})$ forming the residual forces.

The condition for s in the form of the definition (187) has to be satisfied to some tolerance, i.e. $g(s) \approx 0$. There is no explicit expression for the search parameter s in the presence of nonlinear problems. Hence, a solution procedure is required for solving the nonlinear scalar problem $g(s) = 0$ for the single scalar unknown s . One possibility is to use again a Newton-type method, however, the solution procedure aims to be efficient so as to keep the computational costs of the line search algorithm as low as possible. In practice, the condition $g(s) = 0$ is regarded as too stringent and it would not be efficient (or not even necessary) to apply an ‘exact’ line search. One accepts convergence of the line search process with a relatively large pre-assigned tolerance.

A line search algorithm may return values of $s > 1$, which in practice may lead to poor convergence rates for nonlinear problems. Hence, the root should be expected in the interval between 0 and 1. By substituting $s = 0$ and $s = 1$ into (187) we may deduce the condition $g(1) \cdot g(0) < 0$. If this condition is satisfied then s is determined by some sort of root finder. The search parameter s is then accepted if the condition

$$|g(s)| \leq \epsilon |g(0)|, \quad (188)$$

holds, where ϵ is a user-specified (line-search) tolerance parameter. A value of $\epsilon = 0.5$ is proposed by Matthies and Strang (1979). For alternative convergence criteria the reader is referred to the books by Luenberger (1984) and Bazaraa et al. (1993). A widely used line search algorithm is based on residual measures that are interpolated by a quadratic function of s (see, for example, Bonet and Wood (1997)).

Another enhancement for Newton-type methods is the *quasi-Newton method* (also called the *secant method*) with the BFGS-update (named after its creators: **B**ROYDEN, **F**LETCHER, **G**OLD-FARB and **S**HANNO). For more details see, for example, Matthies and Strang (1979), Dennis and Schnabel (1983) and Luenberger (1984).

Finally, we wish to emphasize that Newton’s methods are not appropriate for tracing nonlinear responses near limit points (one example is the quasi-static computation of the snap buckling phenomena of structures). The stiffness matrix approaches singularity in the neighborhood of limit points and the solution diverges. So-called *arc length methods* are well-suited to constrain the iterative solution. The description of this topic is beyond the scope of this article, the reader

may consult the survey article by Riks (1984). For more details of the implementation techniques of solution algorithms within a finite element program for both dynamic and static nonlinear problems the reader is referred to the books by Crisfield (1991), Crisfield (1997), Bathe (1996), Zienkiewicz and Taylor (2000), Belytschko et al. (2000) and Wriggers (2001) among many others.

5 Representative Numerical Examples

The following numerical examples are supposed to be representative in the sense that they aim (i) to show the performance of the structural arterial models outlined in Sections 2 and 3, and (ii) to document finite element results that are in good qualitative agreement with experimental data. For this purpose we consider an arterial segment with no pathological changes of the innermost layer. Hence, we approximate the arterial segment by two separate thick-walled fiber-reinforced circular layers, i.e. the media and the adventitia, which behave incompressibly.

The first example is concerned with a finite element analysis of the mechanical behavior of an artery during clamping. The three-dimensional deformation and the stress distributions in the arterial wall during the clamping procedure are presented. The remaining two numerical examples are concerned with the investigation of the characteristic viscoelastic (relaxation and creep) behavior of a healthy young artery under various (static and dynamic) boundary loadings.

All three-dimensional nonlinear finite element analyses have been performed with the mean dilatation $Q1/P0$ -element and on a HP-J7000 workstation under the UNIX operating system. A detailed account on the underlying three-field variational principle, the finite element discretization and the consistent linearization procedure was provided in Sections 4.1-4.3. The proposed structural models have been implemented in Version 7.3 of the multi-purpose finite element analysis program FEAP, originally developed by *R.L. Taylor* and documented by Taylor (2000).

5.1 Elastic Behavior of an Artery During Clamping

Arterial clamps are used to compress arteries during surgery so that blood flow is arrested. On the basis of the elastic arterial model, as outlined in Section 2, a three-dimensional finite element model for arterial clamping is reviewed (see Gasser et al. (2002) and references therein). The numerical model provides a tool for studying the layer-specific mechanical response of clamped arteries subject to an arbitrarily chosen clamp design.

In Section 2.4 the numerical example of the deformation behavior of a carotid artery during combined bending, inflation, extension and torsion has demonstrated that it is essential to incorporate residual stresses (and strains) in the load-free configuration. In order to consider residual stresses associated with the load-free configuration Ω_0 , we introduce opened-up (reference) configurations for the media and adventitia. Each arterial layer in the reference configuration is assumed to be unstressed (the residual stresses are entirely removed by leaving all other properties of the material unchanged) and taken to correspond to an open sector of a circular cylindrical tube with opening angle α , wall thickness H , inner radius R_i and length L , as indicated in Figure 8. We assume that the media occupies $2/3$ of the arterial wall thickness and that the collagen fibers are helically wound. Their orientations at a reference point $\mathbf{X}$, characterized by the angle φ , are different for each layer. Specific geometrical data for each arterial layer are summarized in Figure 8. The arterial clamp is idealized as a pair of (rigid) cylinders with a radius of 3.0 (mm).

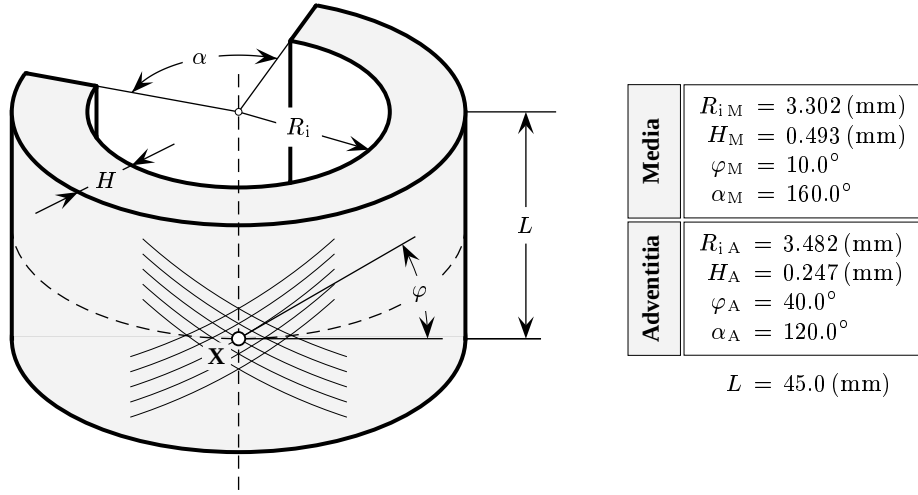


Figure 8. Opened-up (stress-free) configuration of an arterial layer and associated geometrical data for the media and adventitia.

The (elastic) material parameters $c_i, k_{1i}, k_{2i}, i = M, A$, which are involved in the strain-energy functions (38) and (39), were fitted to the experimental data of a human left anterior descending coronary artery (LAD), as given in Carmines et al. (1991). For purposes which were made clear on p. 18 we set $c_M = 10c_A$. The penalty parameters $\kappa_i, i = M, A$, see eq. (80), were chosen to be 10^4 (kPa). The resulting values are summarized in Table 1.

Media	Adventitia
$c_M = 27.0 \text{ (kPa)}$	$c_A = 2.7 \text{ (kPa)}$
$k_{1M} = 0.64 \text{ (kPa)}$	$k_{1A} = 5.1 \text{ (kPa)}$
$k_{2M} = 3.54 \text{ (-)}$	$k_{2A} = 15.4 \text{ (-)}$
$\kappa_M = 10^4 \text{ (kPa)}$	$\kappa_A = 10^4 \text{ (kPa)}$

Table 1. Elastic material parameters $c_i, k_{1i}, k_{2i}, i = M, A$, for the media M and adventitia A, and penalty parameters κ_i .

Finite element model. Due to the symmetry of the considered geometry and the loading condition of the problem, only a wedge of the artery with 90° is discretized by 1200 $Q1/P0$ -elements — 750 for the media and 450 for the adventitia. We used eight finite elements across the wall thickness with the same radial dimension within a layer (five for the media and three for the adventitia). We discretized the circumferential and axial directions of one arterial layer by 10 and 15 finite elements, respectively, while we considered a refined mesh around the clamp region. The nodes at the media/adventitia interface are linked together. The number of elements used turns out to be sufficient to achieve accurate numerical results. The common symmetrical boundary conditions for the structure are applied. During the whole computation the top and bottom faces of the artery are modeled as remaining planar.

In order to handle the contact problem (the interaction of the boundary of the clamp with the boundary of the artery) we use a *point to surface* strategy. The cylindrical surface of the arterial clamp plays the role of a slave surface modeled by 121 *slave points*, which are nodes. The *master surface* is associated with the (outer) boundary of the adventitia, at which contact is expected. It is modeled by master points which are interpolated from 80 four-node (contact) facets.

Finally, frictionless contact is assumed between the arterial boundary and the boundary of the clamp. The control of the penetration between the clamp and the artery is performed by means of the *penalty method*. The discrete (contact) problem is summarized in Table 2.

Arterial wall (wedge of 90°)			Arterial clamp	
$Q1/P0$ -elements	Media	750	Slave nodes	121
	Adventitia	450		
4-node (contact) facets	Outer boundary of the adventitia (<i>master surface</i>)	80		

Table 2. Summary of the discrete contact problem (wall/clamp interaction).

Loading process. The loading process of the two-layer structure is separated into three steps:

1. The *stress-free* arterial segments, with geometrical data given in Figure 8 and parameters as summarized in Table 1, are applied to an initial (pure) bending deformation (for more details see Section 2.2). This deformation process generates the unloaded but *stressed* circular cylindrical shape of the arterial segment, for which the opening angle $\alpha = 0.0^\circ$. Then, the nodes at the media/adventitia interface (i.e. the basal lamina externa) are linked together. Finally, we end up with homogeneous stress and strain states in the circumferential and axial directions, which we now use as the reference for subsequent deformation processes.
2. The circular cylindrical two-layer structure is then inflated up to an internal pressure $p_i = 13.33$ (kPa) (100 mmHg, which is the regular mean arterial pressure), and axially stretched up to an assumed value of $\lambda_z = 1.1$. These loads are considered to be physiological, producing the physiological state of deformation.
3. The interaction of the boundary of the clamp with the boundary of the artery is finally computed as a *displacement-driven* problem of the rigid cylinder (clamp). The arterial clamp is placed perpendicular to the axis of the artery (for the geometrical set-up of the clamping process see Figure 9(a)).

Deformed states of the clamped artery. The deformed states of the artery before and after pinching off with a clamp are illustrated in Figure 9(a)(b) (the physiological configuration is characterized by state **(A)**, while deformed state **(F)** shows the nearly occluded artery). In order to clearly demonstrate the deformations the figure takes advantage of the symmetries by showing one half of the artery. As can be seen from Figure 9(b), the region that is affected by the clamping mechanism is relatively small. Figure 9(c) illustrates different deformed states of the artery, which occur during the clamping process. State **(A)** represents again the physiological configuration (before contact with the clamp), while states **(B)**-**(F)** show the decrease of the arterial lumen due to clamping.

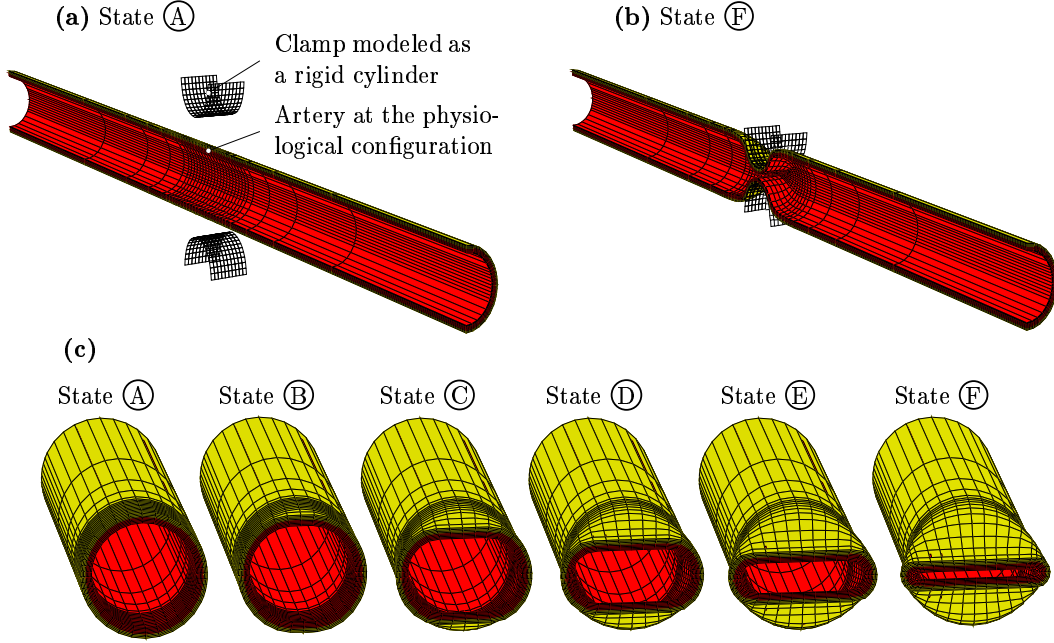


Figure 9. Deformed states of the artery (media and adventitia) at different displacements of the clamp. (a) Artery at the physiological configuration before pinching off with a clamp. (b) Deformed state of the artery, for which the maximum clamp displacement has been applied (full contact between the clamp and the artery). (c) Different deformed states.

Evolution of stresses in the artery during clamping. Now we study the evolution of circumferential and axial stresses at a representative point of the artery during clamping.

Three-dimensional finite element analyses showed that in the neighborhood of the clamp, there is a tendency for *circumferential* (Cauchy) stresses in the wall to decrease and *axial* stresses in the media to increase. To demonstrate the local evolution (change) of stresses in the artery during clamping, we pick a representative point P , which is located at the inner surface of the media as indicated in Figure 10.

In order to show the (local) stress evolution at P we introduce, for convenience, the two (dimensionless) scalar quantities

$$s_\theta = \frac{\sigma_\theta}{\sigma_{\theta \text{ phys}}}, \quad s_z = \frac{\sigma_z}{\sigma_{z \text{ phys}}}, \quad (189)$$

which determine the normalized stresses s_θ in the circumferential direction and the normalized stresses s_z in the axial direction. In eq. (189), σ_θ and σ_z denote the Cauchy stresses in the circumferential and axial directions, while $\sigma_{\theta \text{ phys}}$ and $\sigma_{z \text{ phys}}$ denote the associated stress quantities in the physiological state (A), at the same point P .

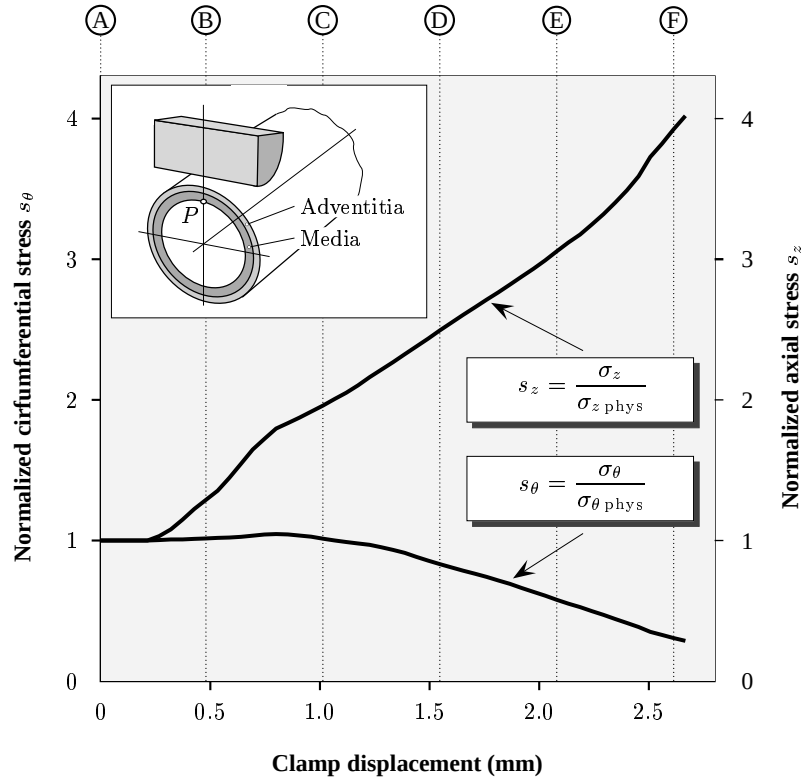


Figure 10. Evolutions of the normalized circumferential and axial stresses (s_θ, s_z) with the clamp displacement at a certain point P located at the inner surface of the media. Note the high increase of s_z with respect to the physiological configuration.

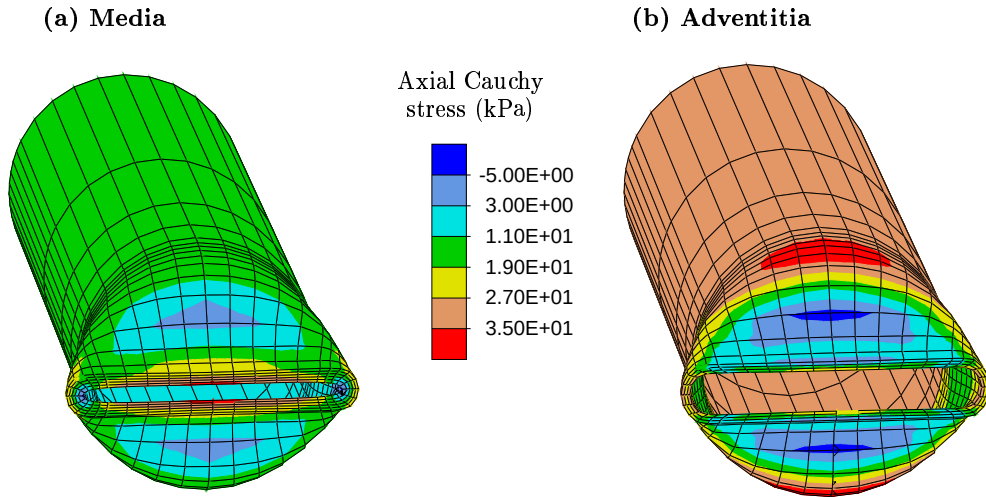


Figure 11. Distribution of the axial Cauchy stress in (a) the media and (b) the adventitia at the final state of the clamping process.

In Figure 10 the (local) evolution of the normalized stresses s_θ , s_z with the clamp displacement are plotted. As can be seen, the axial stress s_z increases significantly, while the circumferential stress s_θ decreases with progressive clamp displacement. Remarkably, at the final state ⑥ of the clamping process, with an occlusion of 92.5% referred to the lumen diameter at the physiological state ① (the maximum dimensions of the lumen are 0.378 (mm) and 5.983 (mm)), the value of the normalized axial stress s_z at the considered point P is about four times higher than that at the associated state ①. Since the media is relatively vulnerable in regard to axial extension, this may be associated with medial damage and damage of the attached endothelial cells. This is not an obvious effect, is not described in the literature, and cannot be investigated experimentally. The relevance of stress analyses is underlined by the fact that even non-destructive loading that exceeds the physiological limit leads to biological responses (Jackiewicz et al. (1996)) that might be clinically adverse.

Figure 11 illustrates the distribution of the axial Cauchy stress in the media and the adventitia. As can be seen, the maximum values of the stresses in the media and adventitia are similar; however, their locations are different.

Since the media is not designed to carry high axial loads (this is a task of the adventitia), the clamping process may cause injury in the media. In view of the high circumferential stresses at the inner surface of the media one may conclude that the intima is also highly deformed and damaged (compare with the experimental observations of, for example, Slayback et al. (1976) and Margovsky et al. (1997); see also the work by Harvey and Gough (1981) grading the damage of the arterial wall). It is important to note that the axial stress increase in the media due to clamping is strongly affected by the shape of the clamp jaws.

For the distribution of circumferential stresses in the arterial layers at the load-free and physiological configurations the reader is referred to Gasser et al. (2002).

5.2 Viscoelastic Behavior of an Artery under Various Boundary Loadings

Two representative numerical examples are investigated in order to demonstrate the reliability and efficiency of the viscoelastic arterial model outlined in Section 3. On the basis of a healthy and young arterial segment, which is under various (static and dynamic) boundary loadings, we aim to point out some of the peculiar viscous effects of arterial deformations.

Elastic and viscoelastic material parameters. For both examples we take the *elastic* material parameters for the media and adventitia, and the penalty parameters, to be as summarized in Table 1 (for a description of these parameters see p. 53).

Based on the statements of Section 1.1 that *viscous* effects are attributed mainly to the mechanical behavior of smooth muscle cells, we associate the time-dependent response with the media, which is the heterogeneous arterial layer containing a very large muscle content. For our study we use experimental *in vitro* data for a human LAD (Gow and Hadfield (1979)), since a human LAD has already been used for the identification of the elastic material parameters. As mentioned in Section 3.2, the dissipation of arterial soft tissues in cyclic loading is relatively insensitive over a wide frequency spectrum. In order to replicate this characteristic dissipative response we choose a set of five Maxwell elements ($m = 5$) and a set of five relaxation times $\tau_1, \dots, \tau_5$ that cover a time domain of four decades. The free-energy factors $\beta_1, \dots, \beta_5$ are determined in such a way that the dissipation is (approximately) constant between frequencies

$f = 0.01 \text{ (s}^{-1}\text{)}$ and $f = 100.0 \text{ (s}^{-1}\text{)}$. This frequency domain is sufficient for the investigation of physiological loading. Note that a viscoelastic model based on a single relaxation time, such as can be found in the literature, would never be able to describe the dissipation response of arterial soft tissues under cyclic loading.

A detailed identification process for quantifying the viscoelastic material parameters involved in the constitutive model, as outlined in Section 3, is presented in Holzapfel et al. (2002a), Section 5.1. The complete set of viscoelastic material parameters for the media, which is taken for both numerical examples, is summarized in Table 3.

Media
$\beta_1^\infty = 0.353 \text{ (-)}$ $\tau_1 = 0.001 \text{ (s)}$
$\beta_2^\infty = 0.286 \text{ (-)}$ $\tau_2 = 0.010 \text{ (s)}$
$\beta_3^\infty = 0.298 \text{ (-)}$ $\tau_3 = 0.100 \text{ (s)}$
$\beta_4^\infty = 0.285 \text{ (-)}$ $\tau_4 = 1.000 \text{ (s)}$
$\beta_5^\infty = 0.348 \text{ (-)}$ $\tau_5 = 10.00 \text{ (s)}$

Table 3. Viscoelastic material parameters for the media.

Finite element model. The three-dimensional finite element analyses are based on 8-node $Q1/P0$ -elements. Due to the axi-symmetry of the problems, only a sector of the structure (a wedge of any angle) is discretized by eight finite elements (five for the media and three for the adventitia) and analyzed. The nodes at the media/adventitia interface are linked together and common symmetrical boundary conditions for the structure are applied. During the whole computation the top and bottom faces of the artery are modeled as remaining planar.

(i) Axial relaxation and creep test. In the first example we compare the axial relaxation and creep responses under combined bending, inflation and axial extension of the segment, with (one-dimensional) experimental data of Tanaka and Fung (1974).

We consider the opened-up geometry of a circular cylindrical tube (Figure 8). Bending deformations are applied to the open medial and adventitial sectors to give a closed two-layer thick-walled tube, which is then stretched to $\lambda_z = 1.1$ in the axial direction (simple tension) and fixed subsequently at this elongation. These two processes are performed rapidly so that no time remains for relaxation. Hence, we assume that the relaxation process starts at $t = 0^+$ (for computational reasons we have chosen the deformation period to be 10^{-7} (s)).

Now the axial reaction force, say F_z , is computed as a function of time and normalized with the axial reaction force at $t = 0^+$. Figure 12 shows the relaxation of the normalized axial reaction force, which is denoted by the ratio $k_F = F_z(\lambda_z, t)/F_z(\lambda_z, 0^+)$, versus time t at axial stretch $\lambda_z = 1.1$. The (linear) decrease of k_F is indicated by triangles, while

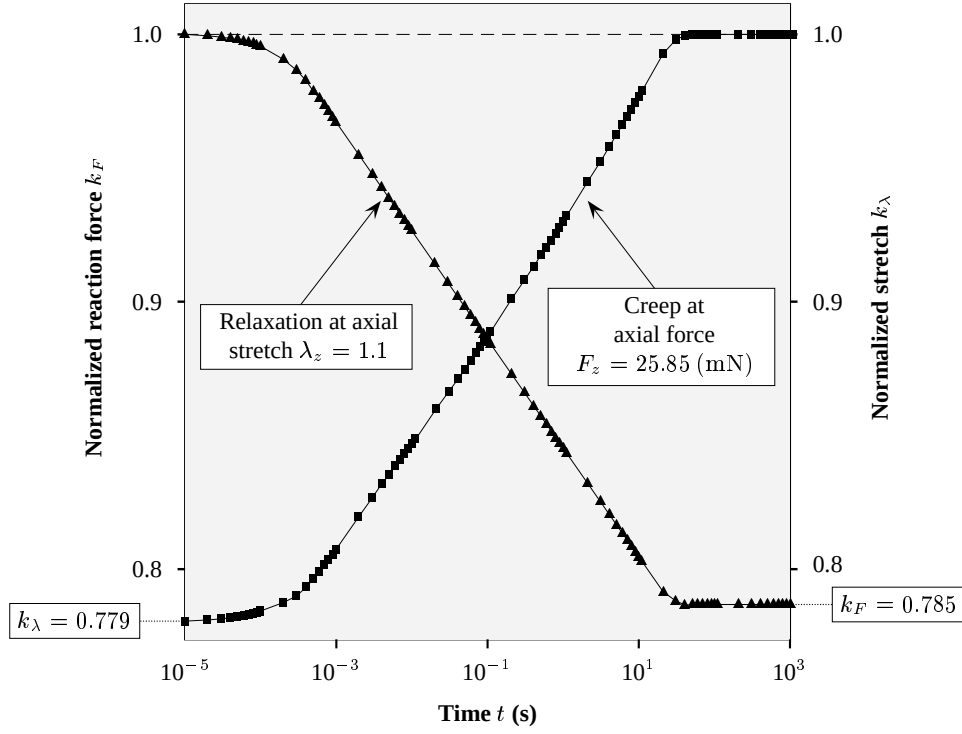


Figure 12. Axial relaxation and creep responses of an arterial segment. Triangles show the relaxation of the normalized axial reaction force $k_F = F_z(\lambda_z, t)/F_z(\lambda_z, 0^+)$ versus time t at axial stretch $\lambda_z = 1.1$. Squares show the creep of the normalized axial stretch $k_\lambda = \lambda_z(F_z, t)/\lambda_z(F_z, 0^+)$ versus time t at axial force $F_z = 25.85$ (mN).

thermodynamic equilibrium (no change in the values of the state variables at any particle of the system) is reached at $k_F = 0.785$ after some time.

In order to compute the arterial response due to creep, we apply an axial force of $F_z = 25.85$ (mN) on the closed tube, which is then kept fixed during the creeping process. By analogy with the above, the viscoelastic process starts at $t = 0^+$. The value of F_z is chosen so that the axial stretch λ_z of the arterial segment is 1.1 at $t = 0^+$, which gives the same initial conditions as in the relaxation test. Now the axial stretch λ_z is computed as a function of time and normalized with $\lambda_z = 1.1$. Figure 12 shows the creep of the normalized axial stretch, which is denoted by the ratio $k_\lambda = \lambda_z(F_z, t)/\lambda_z(F_z, 0^+) = \lambda_z/1.1$, versus time t at axial force $F_z = 25.85$ (mN). The (linear) decrease of k_λ is indicated by squares, while thermodynamic equilibrium is reached at $k_\lambda = 0.779$. In comparison with experimental data (Tanaka and Fung (1974)) the results given in Figure 12 represent qualitatively the typical viscoelastic behavior of arterial segments.

- (ii) **Dynamic inflation test.** The second example investigates the dynamic inflation of the arterial segment of the muscular type under different dynamical loading characteristics. The

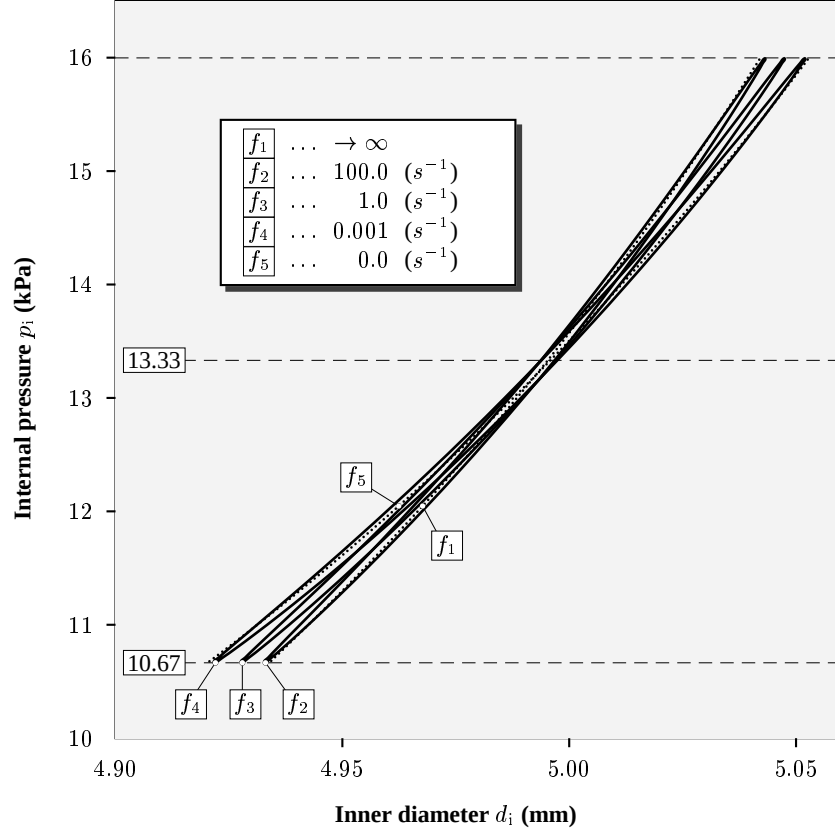


Figure 13. Evolution of the inner diameter d_i of an arterial segment subjected to static and dynamic internal pressure p_i . The solid lines show numerical results based on sinusoidal pressure loads with the frequencies $f_2 = 100.0 \text{ (s}^{-1}\text{)}$, $f_3 = 1.0 \text{ (s}^{-1}\text{)}$ and $f_4 = 0.01 \text{ (s}^{-1}\text{)}$. The dotted curves show numerical results based on the limiting cases $f_1 = 0.0 \text{ (s}^{-1}\text{)}$ and $f_5 \rightarrow \infty$ of the frequencies, representing the stiffest and softest (elastic) responses, respectively.

inner diameter response is investigated in regard to changes in internal pressure and frequency. Sufficiently slow and fast processes, for which viscous effects are absent, are studied as limiting cases.

To achieve a closed two-layer thick-walled tube (pure) bending deformations are applied to the open medial and adventitial sectors. The closed tube is then inflated up to the internal pressure $p_i = 13.33 \text{ (kPa)}$, axially stretched up to $\lambda_z = 1.1$ (simple tension) and then *fixed* at this elongation. We consider a time period of 10^6 (s) in order to compute the equilibrium state of the arterial segment, which we use as the reference state.

Starting from this state we subject the arterial segment to a sinusoidal pressure load $p_i(t)$ varying between 10.67 (kPa) and 16.0 (kPa) and representing the change of blood pressure between the diastolic and systolic phases. We study load cycles with the specific frequencies of $f_2 = 100.0 \text{ (s}^{-1}\text{)}$, $f_3 = 1.0 \text{ (s}^{-1}\text{)}$ and $f_4 = 0.001 \text{ (s}^{-1}\text{)}$. The evolution of the inner

diameter d_i with internal pressure p_i for each of the given frequencies is plotted in Figure 13, and indicated by solid lines. The hysteresis loops demonstrate the typical insensitivity to the frequency over several decades.

As can be seen, the loading and unloading cycles involve about the same dissipation, which is represented by the area between the curves (a measure of the non-recoverable energy). In addition, the deformation responses are stiffer at higher frequencies, a fact which has already been reported by several researchers (see, for example, Learoyd and Taylor (1966) and Fung (1971)) and which could not be modeled using pseudo-elasticity. Note that the hysteresis is quite small, which is due to the small amplitudes of the pressure loads.

We also investigate sufficiently slow processes, $f_1 = 0.0 \text{ (s}^{-1}\text{)}$ (enough time remains for the arterial segment to adjust itself internally), and sufficiently fast processes, $f_5 \rightarrow \infty$. For computational reasons we have chosen the frequencies to be $f_1 = 10^{-5} \text{ (s}^{-1}\text{)}$ and $f_5 = 10^5 \text{ (s}^{-1}\text{)}$. These two limiting cases describe (reversible) processes during which the arterial segment is in equilibrium at all times and are associated with stiffest and softest responses, respectively. Viscous effects do not arise and the dissipation is zero. The numerical results are plotted in Figure 13 and indicated as dotted curves.

6 Layer-specific FE-Modeling of Balloon Angioplasty using MRI and Mechanical Testing

A detailed understanding of the mechanical procedure of balloon angioplasty requires three-dimensional modeling and efficient numerical simulations. Following Holzapfel et al. (2002b), this section aims to expose a fully three-dimensional model for eight distinct arterial components associated with specific mechanical responses.

The 3D geometrical model is based on *in vitro* MRI of a human stenotic *post-mortem* artery and is represented by NURBS surfaces (Non-Uniform Rational B-Splines). The NURBS approach is proven to be efficient for representing complex geometric information processed by computers. Mechanical tests of the corresponding vascular tissues provide the fundamental basis for the formulation of large strain constitutive laws, which model the typical mechanical characteristics of arterial components under supra-physiological loadings occurring during balloon angioplasty.

The 3D finite element realization considers the balloon-artery interaction and accounts for vessel-specific axial *in situ* pre-stretches. 3D stress states of the investigated artery during balloon expansion are presented. The proposed approach provides a tool that has the potential (i) to improve procedural protocols and the design of interventional instruments on a lesion-specific basis, and (ii) to determine post-angioplasty mechanical environments, which may be correlated with restenosis responses.

Simplified models, which do not consider either axial *in situ* pre-stretch, three-dimensional geometries or material anisotropy, may lead to significant deviations of the stress states compared with the reference model (see the comparative study in Holzapfel et al. (2002b)).

6.1 Introduction and Motivation

Balloon angioplasty and stenting are purely *mechanical* procedures, which aim to dilate stenotic or occluded arteries in order to restore blood flow. A more detailed understanding of the un-

derlying mechanics of angioplasty and an optimization of the interventional protocols may lead to significant improvements of the outcome. Surprisingly, only a small number of physical and computational models concerned with balloon angioplasty have been published, although this interventional treatment is of great and steadily growing medical, economical and scientific interest (American Heart Association (2000)). Basically, a suitable model has to be based on realistic experimental data, appropriate geometrical and physical modeling, and on efficient numerical realization using the finite element method. In particular, the model must operate in both the physiological and the supra-physiological loading domain.

Interventionists are interested in methods that allow determination of appropriate procedural parameters such as diameter ratio (balloon to artery), balloon length, inflation pressure etc., as well as optimal instruments such as stents with specific designs. Moreover, models that allow the prediction of the primary success of interventional treatments on a lesion-specific basis may improve patient selection, i.e. the selection of patients for which balloon angioplasty is the optimal therapeutical choice. Additionally, for biomedical engineers, appropriate models may be helpful in order to develop novel designs of interventional instruments such as balloons and stents and to prove their performance on a computational basis. From a computational point of view these types of problems are of particular interest since they involve a number of challenging fields in computational mechanics, such as nonlinear continuum and contact mechanics and the numerical simulation of nonlinear material behavior at large strains.

Existing models do not, unfortunately, consider essential mechanical characteristics of arterial stenoses undergoing large deformations at therapeutic loadings, which are far beyond the physiological domain (for a survey see Holzapfel et al. (2002b)).

6.2 Geometrical, Physical and Finite Element Modeling

MR image based geometrical modeling. An external iliac artery (male, 68 years) was excised from a corpse during autopsy within 24 hrs after death. For subsequent mechanical testing only fresh tissues were used. Axial *in situ* pre-stretch, defined as *in situ* length/*ex situ* length denoted by λ_{is} , was 1.04. A straight segment of the artery (20.0 (mm)) with an eccentric stenosis was scanned by means of high-resolution magnetic resonance imaging (cross-sections with an in-plane resolution of 0.3 (mm) and an axial resolution of 1.0 (mm)). For each scanned cross-section, the borders of the arterial components were traced manually by a series of points. The traced cross-sections were combined along the arterial axis, which resulted in a set of 3D point clouds representing the boundary surfaces of the components. Finally, each of the boundary surfaces was represented as a NURBS spatial model (Hoschek and Lasser (1993) and Piegel and Tiller (1997)), which was fitted to the associated point cloud.

After scanning, the artery was cut through transversely into two halves. One half was used for corresponding histological analyses to allow material characterization, while the other half was dissected anatomically into its major components, which are non-diseased intima I-nos¹, collagenous cap I-fl (fibrotic part at the luminal border), fibrotic intima at the medial border I-fm, calcification I-c, lipid pool I-lp, non-diseased media M-nos, diseased fibrotic media M-f and adventitia A (Holzapfel et al. (2000b)). From these components, stripes with axial and

¹ The abbreviation 'nos' is frequently used in histopathology and stands for **not otherwise specified**. In the context of the present study it means 'no appreciable disease', or, more precisely 'non-atherosclerotic'.

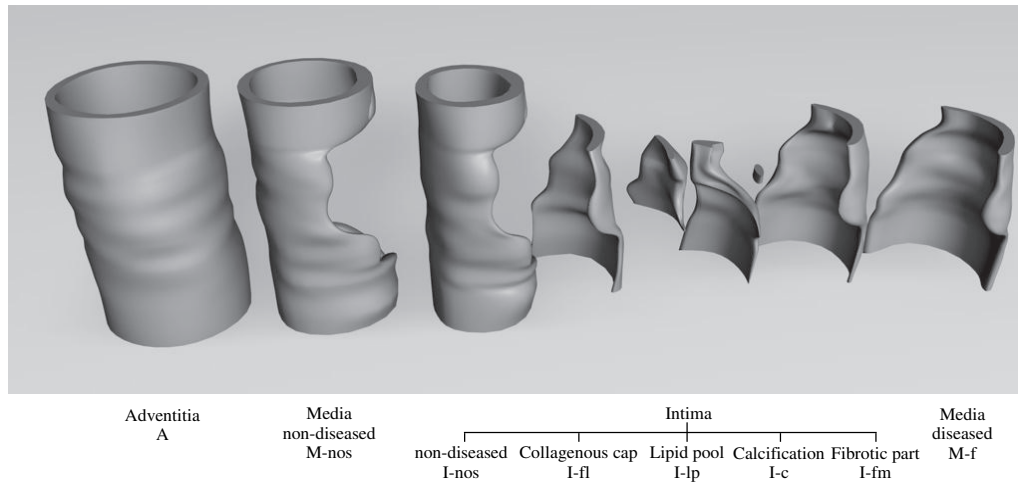


Figure 14. 3D reconstruction of eight distinct components of a human stenotic artery distinguished by two major regions: the *non-diseased wall* (consists of I-nos, M-nos and the associated part of A), and the *eccentric plaque region* (consists of I-fl, I-lp, I-c, I-fm, M-f and the associated part of A).

circumferential orientations were cut out and underwent cyclic uniaxial extension tests with continuous recording of tensile force, stripe width and gauge length at a constant crosshead speed of 1.0 mm/min. For most specimens preconditioning was achieved by executing five successive loading cycles. Typical specimen geometries can be estimated from Figure 14. The calcification and the lipid pool were excluded from the tests and considered as rigid body and incompressible (very soft) solid, respectively. A description of the procedure and the experimental results obtained can be found in Holzapfel et al. (2000b).

Based on the classification scheme described above, a three-dimensional geometrical model of the considered stenosis was generated using interpolation between the 21 serial high-resolution magnetic resonance slices. Figure 14 shows the three-dimensional reconstruction of the eight distinct tissues of the human iliac artery. As seen from the figure, two major regions can be distinguished: the *non-diseased wall* which consists of the non-diseased intima I-nos, the non-diseased media M-nos and the associated part of the adventitia A, and the *eccentric plaque region*, which consists of the collagenous cap I-fl, the lipid pool I-lp, the calcification I-c, the fibrotic intima at the medial border I-fm, the diseased fibrotic media M-f and the associated part of the adventitia A. Note that in contrast to young laboratory animals, the non-diseased intima I-nos exhibits considerable thickness (and mechanical strength). This is caused by intimal hyperplasia, which is defined as proliferation of intimal cells and increase of extracellular matrix without any lipid deposits. Arteries with diffuse intimal hyperplasia are non-diseased by definition (Stary et al. (1992)). For human arteries, thickness ratios of intima/media from about 0.1 to 1.0 or more are documented (see the review article by Stary et al. (1992)). Non-published measurements performed in the authors' laboratory showed an average thickness ratio of intima/media/adventitia of 13/56/31 for aged non-diseased iliac arteries Schulze-Bauer et al. (2001).

Physical modeling. The resulting data sets, which contain geometrical information, histological characterizations and mechanical behaviors serve as a fundamental basis for the generation of the specific physical models.

The investigated arterial components are assumed to be (nearly) incompressible fiber-reinforced composites, which exhibit strongly nonlinear anisotropic responses and which undergo large strains (Holzapfel (2001)). They show a pronounced stiffening behavior in the large strain domain due to collagen fiber recruitment (see Holzapfel (2001) and Humphrey (2002)). Specific parallel arrangements of the collagen fibers in form of fiber families (Rhodin (1980)) lead to anisotropic material responses. For the arterial tissues investigated we observed that most tissue types responded with abrupt failure when exposed to loads that are far beyond the physiological range, while the non-diseased media responded with non-recoverable deformations before tissue failure occurred (for the stress-strain response of the non-diseased media, see Holzapfel et al. (2000b)). The physiological range of tissue stresses is assumed to be in the order of 50 – 100 kPa at the mean arterial pressure of 100 mmHg and axial *in situ* stretch. This is justified by experimental data published by Cox (1976) for circumferential wall stresses in dog iliac arteries, by Fung et al. (1979) for wall stresses of rabbit iliac arteries in both directions, and by unpublished data of human aged iliac arteries performed in the authors' laboratory (Schulze-Bauer et al. (2001)).

The non-recoverable deformations may come mainly from damage mechanisms and also from plastic-type phenomena. In order to trace a practicable approach and due to the lack of appropriate data about microstructural processes associated with non-recoverable deformations, we decided to apply the theory of plasticity. For the associated continuum basis and algorithmic formulation the reader is referred to Gasser and Holzapfel (2001). The calcification I-c and the lipid pool I-lp are modeled as a rigid body and an incompressible solid (neo-Hookean material), see eq. (35), with a very low (stress-like) material parameter $c = 0.1$ (kPa). All other arterial components involved are based on the constitutive description as presented in Section 2.3. The fiber orientations used for the constitutive models are determined by means of histology and the image processing tools described in Section 1.2. It turned out that the arterial components investigated exhibit a symmetric arrangement of the two fiber families.

Particular physical models of the arterial components were obtained by fitting the general constitutive formulations to the corresponding experimental data using the *Levenberg-Marquardt* algorithm. Best fit constitutive parameters characterizing the elastic and inelastic responses of different arterial tissues, and the associated fiber direction angles used for the subsequent numerical simulation are adopted from Holzapfel et al. (2000b) (see also Holzapfel et al. (2002b)). All material models involved have been implemented in the finite element program ABAQUS V5.8 (Hibbitt et al. (1998)).

Numerical modeling. Based on this particular approach the incompressibility constraint is enforced by means of an augmented Lagrangian method (Simo and Taylor (1991)), which provides superior numerical performance compared with standard penalty methods (Holzapfel (2000)).

Finite element discretization. The use of NURBS provides smooth surfaces and a suitable basis for mesh adaption procedures that allow mesh refinement with respect to the (original) reference geometry. Based on the geometrical model, a mesh generator creates finite element meshes with a user-defined degree of refinement. For convenience, throughout the entire dis-

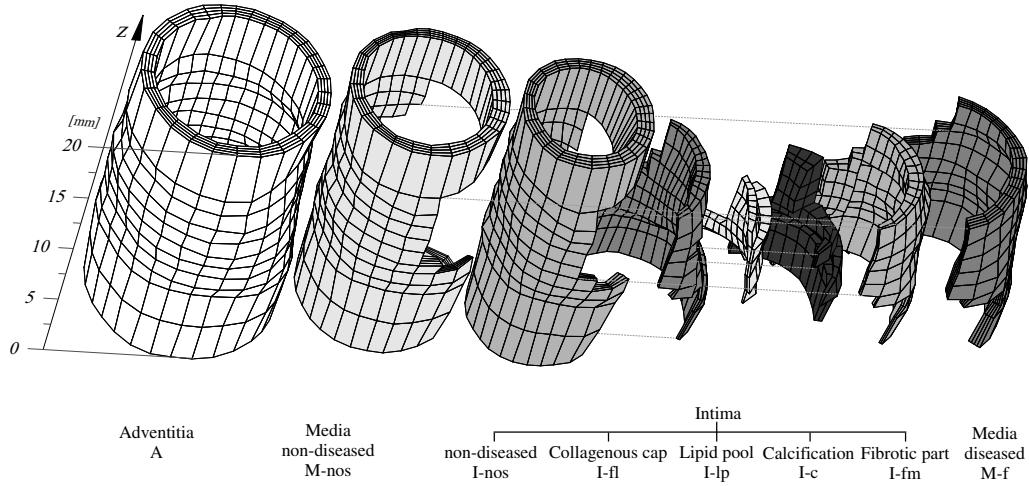


Figure 15. Exploded view of the 3D discretization of a human stenotic artery.

crete structure, the numbers of finite elements in the radial, circumferential and axial directions are constant. This allows easy access to the node numbers, as required for the prescription of boundary conditions and contact surfaces. To capture the incompressibility constraint the robust $Q1/P0$ -element, as described in Sections 4.1-4.3, has been used.

Figure 15 shows the mesh obtained in an exploded view, where the specific discretizations of the individual arterial tissues can be seen. In the radial direction the intima, media and adventitia consist of four elements each. The whole arterial model is generated with 3828 isoparametric 8-node brick elements. Each finite element is associated with one out of eight distinguished material responses. Thus, there is an abrupt change of material properties across element boundaries, which separate different arterial components. Because of the lack of according experimental data, no interpolation for a smooth transition of material properties was applied. Hence, stress analysis seems to be more reliable within an arterial component than at its boundaries.

The (fully inflated) balloon is modeled as a rigid cylinder-shaped structure of hexahedral finite elements with a diameter of 10.0 (mm) and a length chosen to be equal to the stretched vessel segment. The assumption of a rigid cylinder is justified by the fact that fully inflated angioplasty balloons behave as non-compliant cigar-shaped structures.

Boundary conditions and loading process. Regarding boundary conditions the artery is fixed axially at one end, while the other end is exposed to displacements according to the axial *in situ* pre-stretch. Additional boundary conditions are prescribed in order to avoid rigid-body motions of the artery and the angioplasty balloon. During the experimental preparations a significant axial retraction of the adventitia was documented after dissection from the remaining arterial tissues (for a general discussion see Schulze-Bauer et al. (2002)). Therefore, we included the observed adventitial axial pre-stretch in the numerical model, which was measured as 1.2. Each of the discrete arterial components is tied to its adjacent components, i.e. we just prescribe displacement continuities across the boundaries of neighboring components. This is motivated

by the fact that the ‘adherence materials’ that interconnect the arterial components have small thicknesses compared with the dimensions of the components (for the case of healthy arteries these materials are called internal and external elastic membranes). Hence, even if these materials are highly deformable, only minor movements of the components against each other are possible. Moreover, for diseased tissues these materials are rather stiff due to the high content of collagen fibers. Cracking of these layers, which may occur during angioplasty procedures, is not implemented in the proposed numerical model.

We consider three consecutive states of loading: (i) intraluminal (physiological) pressure of 13.3 kPa (100 mmHg, which is the regular mean arterial pressure) before balloon angioplasty, (ii) full expansion of the angioplasty balloon ($d = 10.0$ mm diameter) in addition to the applied intraluminal pressure, and (iii) intraluminal pressure of 13.3 kPa after balloon angioplasty. Note that, during the whole angioplasty process a surface (pressure) load of 13.3 kPa is applied to the inner boundary surface of the artery. Both the balloon inflation and the stent expansion are modeled as displacement-driven processes. At the beginning of the unloading process the prescribed displacements are then replaced by equivalent reaction forces at each node, which are driven to zero.

Contact problem. For the modeling of the contact problem, which occurs between the boundary of the arterial wall and the angioplasty balloon, we use the *master-slave approach* (slave nodes are constrained not to penetrate the master surface). The contact constraint is applied by means of the penalty method. Additionally, the *dual* contact problem (interchange master and slave surface) is solved in order to avoid problems with the contact algorithms associated with normal vector computations on the contact surfaces. Finally, frictionless contact is assumed between the arterial boundary and the boundary of the angioplasty balloon.

6.3 Numerical Results and Clinical Implications

The aim of this section is to report three-dimensional stress results in the arterial components, induced by balloon angioplasty, in the human stenotic artery investigated and to conclude with clinical implications. All finite element analyses have been performed with the finite element program ABAQUS V5.8 and on an UNIX-based HP-J7000 workstation.

Numerical simulation of balloon angioplasty. Figure 16 illustrates exploded views of the artery at full expansion of the angioplasty balloon (10.0 (mm) diameter) showing circumferential and axial Cauchy stresses in (kPa). Note that for the purpose of visualization a smoothing algorithm was developed. Mean values of Gauss point stresses were computed for each element in circumferential and axial directions. The stress value of a particular node is recovered by averaging the mean values of all involved elements. Then the internal stresses are obtained by a linear interpolation in the same manner as for the displacements. Generally, the global stress distribution is significantly higher (about $2 - 3\times$) in the circumferential than in the axial direction, as expected for the radial balloon expansion. In the non-diseased wall elastic tissue limits are exceeded, which causes non-recoverable deformations. They occur in the entire media M-nos and in parts of the intima I-nos, indicated in Figure 16 by hatched regions. In the clinical context the phenomenon of non-recoverable deformations, occurring predominantly in the media, is termed ‘controlled vessel injury’. This is known to be an essential mechanism for the luminal gain of the angioplasty procedure (Castaneda-Zuniga (1985)). For the non-diseased intima I-nos this may be

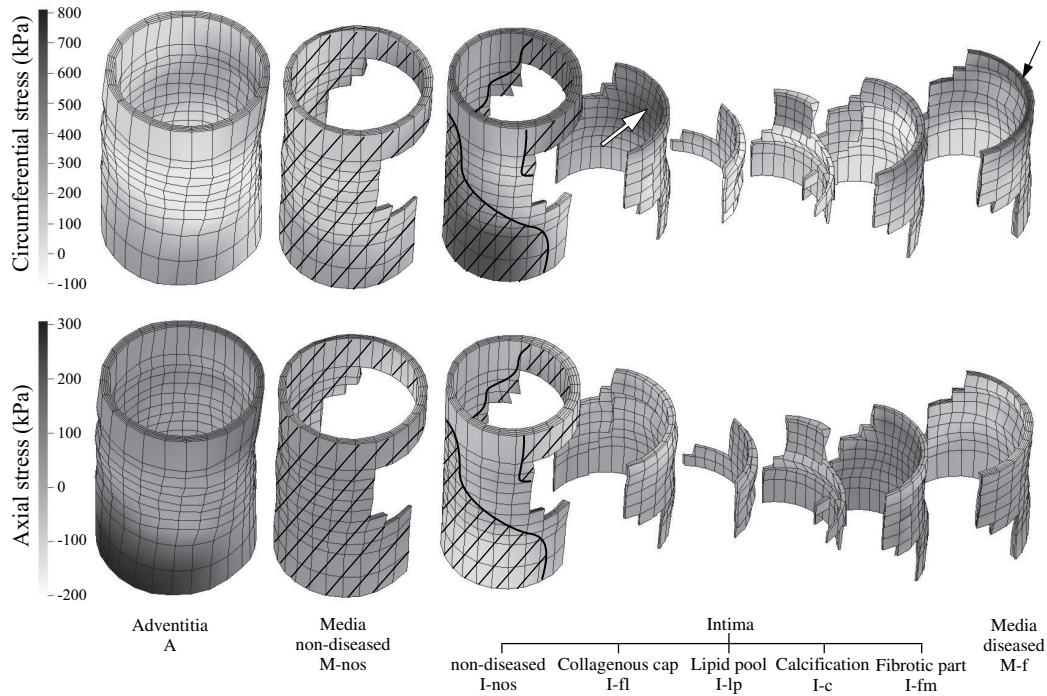


Figure 16. Exploded views of a human stenotic artery at full expansion of the angioplasty balloon (10.0 (mm)) showing circumferential and axial Cauchy stresses.

interpreted as the occurrence of intimal fissures (compare with section ‘Methods’). Interestingly, although the adventitia is modeled as a homogeneous material and has a rather regular geometry, it exhibits a complex three-dimensional stress distribution, which strongly motivates the use of a 3D model. The high-stress region of the adventitia at $z = 0$ (mm) is an effect of the rigid calcification opposite to it and may also be caused by the prescribed boundary conditions.

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The (eccentric) plaque region shows a sandwich-like stress pattern in the circumferential direction. Here the (liquid-like) lipid pool and the calcification are low-stress regions. They are

embraced by two stiff structures, i.e. the collagenous cap I-fl at the inner side, and the fibrotic part of the intima I-fm and the diseased media M-f at the outer side. These components undergo high stresses, which may lead to tissue failure. In particular, circumferential stress concentrations (> 700 (kPa)) are seen at the center of the collagenous cap (indicated by the thick arrow in Figure 16), which is a typical location for plaque rupture, as documented in pathological studies (Richardson et al. (1989)). According to our experimental data the ultimate tensile stress of the collagenous cap was not reached in the simulation performed. Rupture of the collagenous cap is associated with the risk of subsequent thrombus formation in real-life procedures. In addition, tissue failure may also occur in the diseased media M-f, which may initiate layer dissection (indicated by a thin arrow in Figure 16). This observation has not been recognized in previous mechanical studies. Such an effect cannot be observed by means of standard clinical imaging modalities. Nevertheless, in real-life interventions, it may cause, for example, intramural hemorrhage, which subsequently may lead to dissecting aneurysm formation. For this specific morphology no stress concentrations are seen at the plaque shoulders, which are the lateral transition zones between the atherosclerotic plaque and the non-diseased intima. However, tissue failure at this location is a frequent clinical finding (The et al. (1992) and Mario et al. (1995)) and may lead to dissection (delamination) of the plaque from the underlying arterial wall.

The proposed numerical model offers the opportunity to quantify the angioplasty-induced changes of the mechanical environment on a component-specific basis. One quantity that characterizes the changes in the mechanical environment is the local stress difference, i.e. post-angioplasty stress minus pre-angioplasty stress. Figure 17 shows the circumferential stress differences at $z = 6.0$ (mm) for the previous simulation, whereas the stress differences are plotted onto the post-angioplasty configuration at 13.3 (kPa). Additionally, the luminal shape before angioplasty at 13.3 (kPa) is indicated by a dashed curve in order to demonstrate the luminal gain.

There are significant local and component-specific changes of the mechanical environment indicated by a complex pattern of stress differences. In real-life interventions the stress differences that depend crucially on the interventional protocol used, may initiate component-specific biological responses, which ultimately may lead to a renarrowing of the dilated artery. This pathological process is called restenosis and is the major shortcoming of angioplasty (occurrence up to 40 % within six months – Fischman et al. (1994) and Macaya et al. (1996)). Arterial tissues may be regarded as regulatory systems that control their physical and chemical environments. In other words these tissues compare actual values of these quantities with set point values and react on deviations with a specific biological response such as growth, remodeling, atrophy etc. in order to restore the set point values. In our case the stress differences may be seen as deviations of the actual stress state from the (tissue-specific) referential stress state. The subsequent biological reaction may lead to restenosis and thus, cause therapeutic failure. The proposed model allows to investigate the driving forces of this process, i.e. the stress differences, on a component-specific basis. For the underlying mechanics of arterial remodeling the reader is referred to the works by Rodriguez et al. (1994), Taber (1995), Section 7.2, Hayashi et al. (1996), Taber (1998), Rachev et al. (2000), and references therein.

Special merits of the methodology. One aim of this section was to introduce and describe the numerical methodologies of a model, which yields the most sophisticated simulations of interventional procedures to date. For this type of studies the proposed model will provide a suitable computational basis in the future. Based on a large number of investigated stenoses the approach

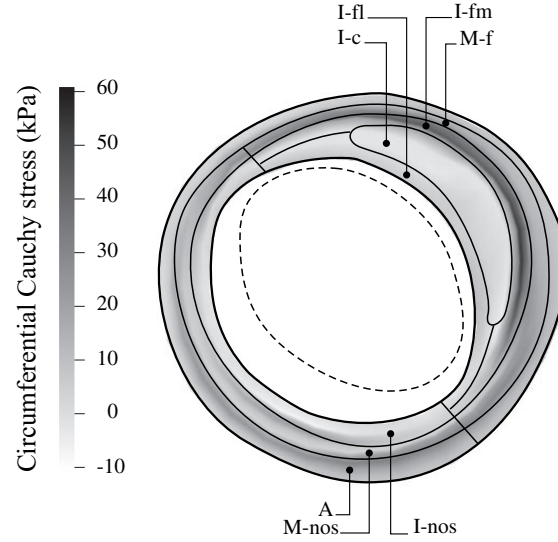


Figure 17. Differences in circumferential stresses (post-angioplasty stresses minus pre-angioplasty stresses at 13.3 (kPa)) for a cross-section ($z = 6.0$ (mm)) after a full expansion of the angioplasty balloon with a diameter of $d = 10.0$ (mm). Stresses are plotted onto the post-angioplasty configuration at 13.3 (kPa). The luminal shape before angioplasty at 13.3 (kPa) is indicated by a dashed curve.

may provide a ‘virtual workbench’ for the design and evaluation of interventional instruments, such as stents. Moreover, the proposed model approach bears the potential for the determination of lesion-type specific procedures, i.e. the definition of optimal procedural protocols for certain plaque geometries and compositions with respect to certain target quantities such as the maximum luminal gain, intimal injury, medial overstretch etc. Another potential application evolves from the fact that for the first time it is possible to investigate the change in the mechanical environment caused by an interventional procedure on a layer-specific basis. This may allow a better understanding of the corresponding biological response, and hence could be a potential contributor for restenosis research.

Morphological data for such investigations may come from high-resolution magnetic resonance imaging of the arterial wall and plaque, which holds great promise for characterizing atherosclerotic plaques and is, therefore, of huge interest for the medical community. MRI has shown to discriminate between various tissue types involved in atherosclerosis *in vitro* (Toussaint et al. (1995), Martin et al. (1995), Shinnar et al. (1999) and Rogers et al. (2000)). Moreover, the challenging task of *in vivo* wall imaging yielded promising results for carotid arteries (Yuan et al. (1995), Toussaint et al. (1996) Luk-Pat et al. (1999)), for the aorta (Fayad et al. (2000b)), and even for coronary arteries (Fayad et al. (2000a)). Thus, it can be expected that in the near future MR wall and plaque imaging will become clinical reality so that morphological input for models as the proposed one are available. Clearly, the combination of mechanical analysis and

morphological data has an enormous potential to significantly improve current diagnostics and interventional procedures.

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